Manual

Goods Receipt Inspection Planning

WEP-PPW 8.1

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Copyright

Contents

1 Goods Receipt Inspection Planning - Overview

Purpose

This component allows creating inspection plans based on master data defined earlier plus generating and managing inspection requirements which, in combination with the associated inspection steps, form the basis for the collection of inspection data.

Implementation Considerations

Since creating inspection plans is the basis for later collection of inspection data, this component is required to collect measured values and attributive inspection decisions.

Integration

This component mainly serves the components:

- Quality data entry / info functions (AIP)
- Additional inspection planning / inspection steps
- Family inspection planning
- Reports of production control inspects and
- Failure mode analysis / action tracking

Features

The following functions are available:

- Maintenance functions to collect and process relevant master data (article, company, failures, actions, characteristics etc.)
- Maintenance functions to create and modify inspection plans
- Versioning inspection plans including managing historical data
- Inspection plan release and activation using special rights
- Allocating or modifying inspection plan characteristics of various characteristic types (variable, attributive)
- Assigning operations and inspection stations so as to structure the inspection steps on the basis of the inspection plans.
- Defining of tolerance limits, unit, sampling scheme, control charts to be used, activating automatic failure generation etc.
- Allocating any document (drawings, pictures, videos of any format and internal notes) to characteristics, inspection plans, inspection plan characteristics, inspection requirements or inspection step characteristics in a document list

- Generating inspection requirements or steps based on inspection plans created earlier

2 Summary

2.1 Notes on the Document

This document describes the “article” application of the Manufacturing Operation Center (MOC). General information on how to use MOC can be found in the document entitled “moc_cc.pdf“.

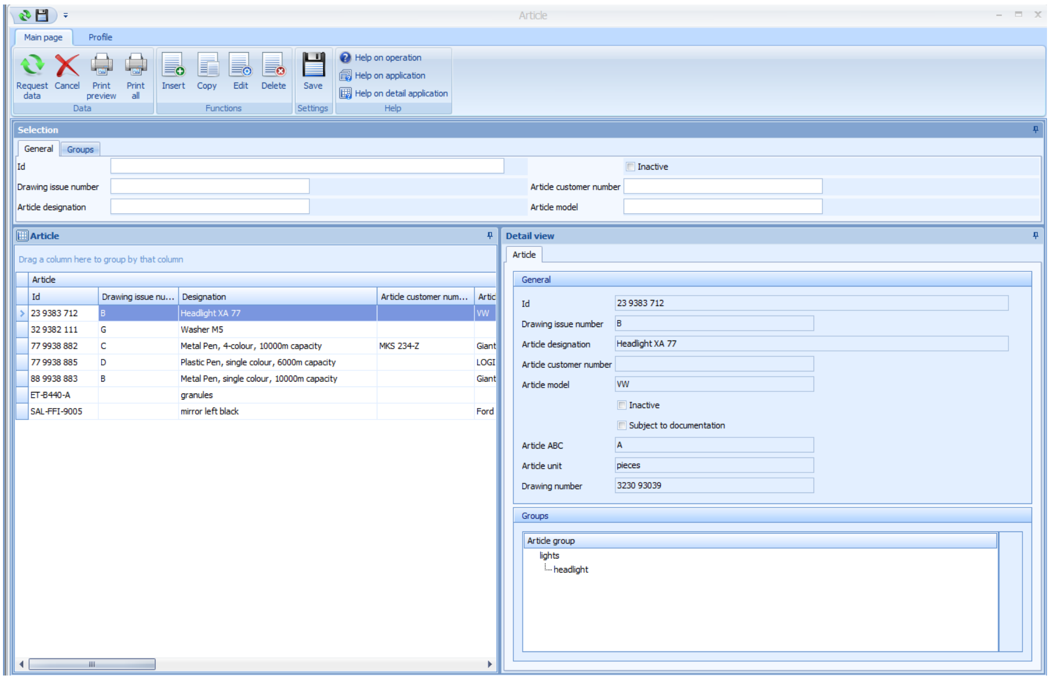
3 Article

The article catalog has been designed to edit/keep articles. Article data is a global catalog that is used in many CAQ modules and in PDV (Process Data Collection). Provided that there is an interface to a higher-level system (e.g. ERP system), articles may be created automatically via this interface. As soon as a new article is created or changed, e.g. in the ERP system, the article data record is automatically created or changed in the HYDRA-CAQ article catalog based on the defined information.

3.1 Starting the Function

Menu	Master data → Quality management → Article Master data → Process data processing → Article
Transaction code	atc
Function authorization	atc

3.2 Default Application Layout



3.3 Toolbar

The toolbar provides the different functions available for this application and possibly links to other applications. The functions included in the “general” tab of the toolbar refer to all detail applications. In addition to the standard functions, such as help, request data, save application settings, and print preview, the other tabs also include specific functions that are specially tailored to the respective detail application. The following sections describe the individual application functions.

“Data” Category

Request data

The information to be displayed within the application is requested on the basis of the entered selection criteria. This process might take some time depending on the dataset from which the system filters data and on the selection result to be transferred and displayed.

Cancel

The query sent by clicking the “request data” button can be canceled using this function.

Print preview

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.

Save

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking “Yes”.

“Functions” Category

Add

Adds a new article.

Copy

Copies the selected article.

Edit

Edits an already existing article

Delete

Deletes the selected or several selected articles.

“Help” Category

Help on operation

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled “moc_cc.pdf”. It describes how to use MOC in general and applies for all applications.

Help on application

This function opens the manual corresponding to the respective application from which the help was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.

Help on detail application

This function opens the application manual at the section where the respective detailed application is described.

3.4 Selection parameters

The application provides the following selection criteria:

"General" tab

- Article no.:
  Article number

- Drawing issue number:
  Drawing issue number of the article, often also referred to as index

- Designation:
  Article name

- Inactive:
  Inactive, active articles. The checkbox is not activated by default.

- Customer article no.:
  Customer article number

- Article model:
  Article model

"Groups" tab

- Group:
  The article group tree can be opened using the button if an article group is to be filtered. There is a function to accept and cancel the activity.

3.5 "Article" Detail Application

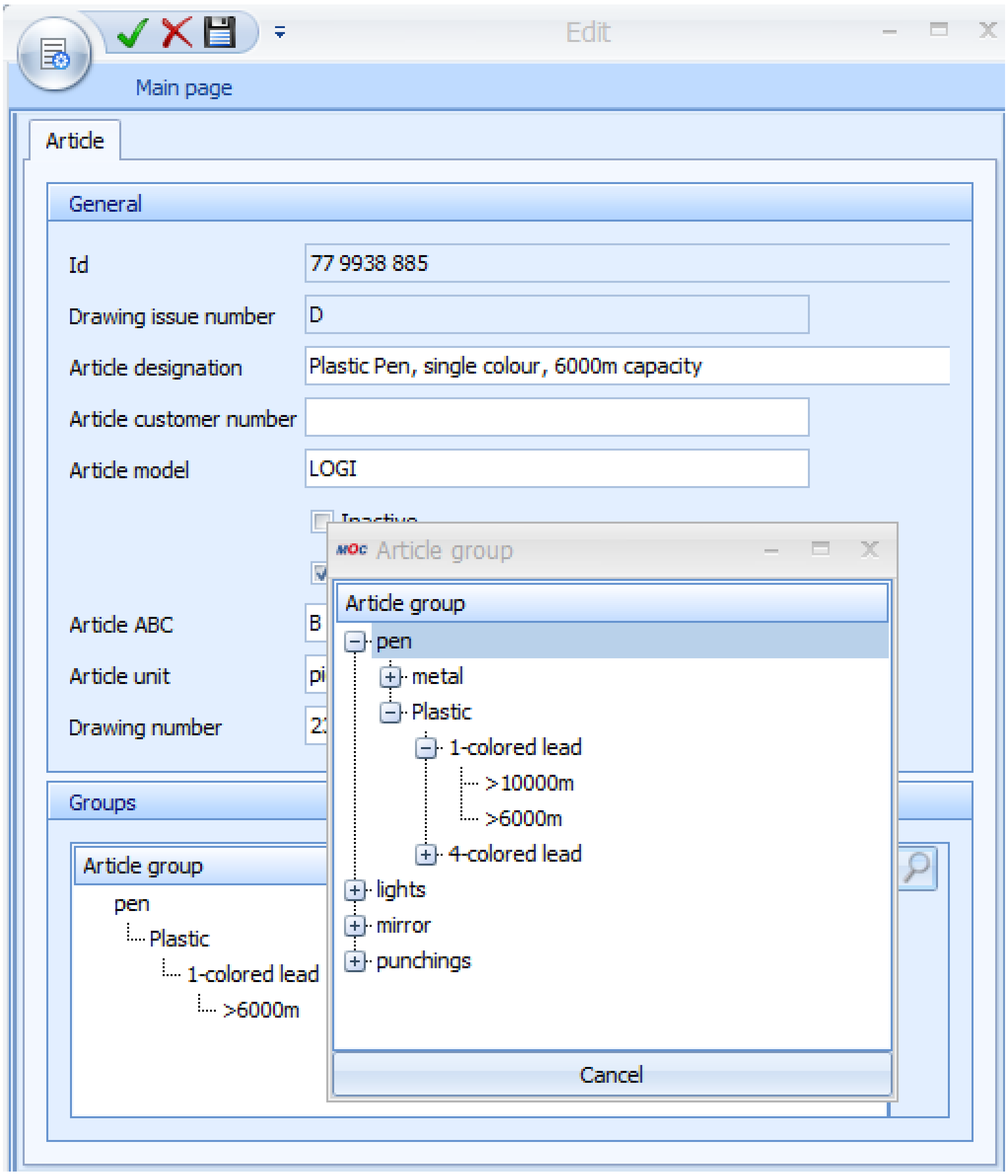
The article number as well as the drawing issue number uniquely identify articles in all areas of HYDRA-CAQ referring to the article catalog. The drawing issue number, also referred to as article index, may be very important, in particular, for inspection planning and when inspection orders are generated. Thus, it is possible to create, for example, an inspection plan for article 12938 with the drawing issue number A and B within inspection planning, whereas there are different inspection specifications for each drawing issue number. Unless the drawing issue number is indicated and thus may be part of the inspection plan, the system that generates the inspection requirements, must deliver this drawing issue number.

Consequently, the “article no.” and “drawing issue number” fields are key fields, i.e. when a new article is saved, it is first checked whether an article with this key information already exists.

By distinguishing between active and inactive articles, it may be defined whether or not the articles are available in certain selection lists. Thus, no inspection plan can be created for an inactive article. However, inactive articles may be evaluated at any time. Moreover, inactive articles can also be reactivated at any time.

Furthermore, an article can be defined as being subject to documentation. In addition the dialog provides the fields customer article number, article model, article ABC, drawing number as well as the possibility to assign units. The catalog of units is accessed to assign units (dimensions).

The corresponding group needs to be assigned, provided that article groups are to be evaluated or family inspection plans are to be used at a later point in time. Groups are assigned by opening the group tree using the magnifier button. Using the hierarchic tree entries the required group can be selected in the group tree and accepted by double clicking.



The assigned group including the hierarchical group structure then appears in the “groups” field of the editing dialog of articles.

When articles are displayed in a list, the group hierarchy is represented by the columns “group 1 to group 5”.

Groups are maintained in the “article groups” application and are described in the document entitled “MOC_Groups.pdf “.

4 Summary

4.1 General notes on the document

This document describes the “Groups“, e.g. article groups, application of the Manufacturing Operation Center (MOC). For general information on how to use MOC, please refer to the “moc_cc.pdf“ document.

5 Groups

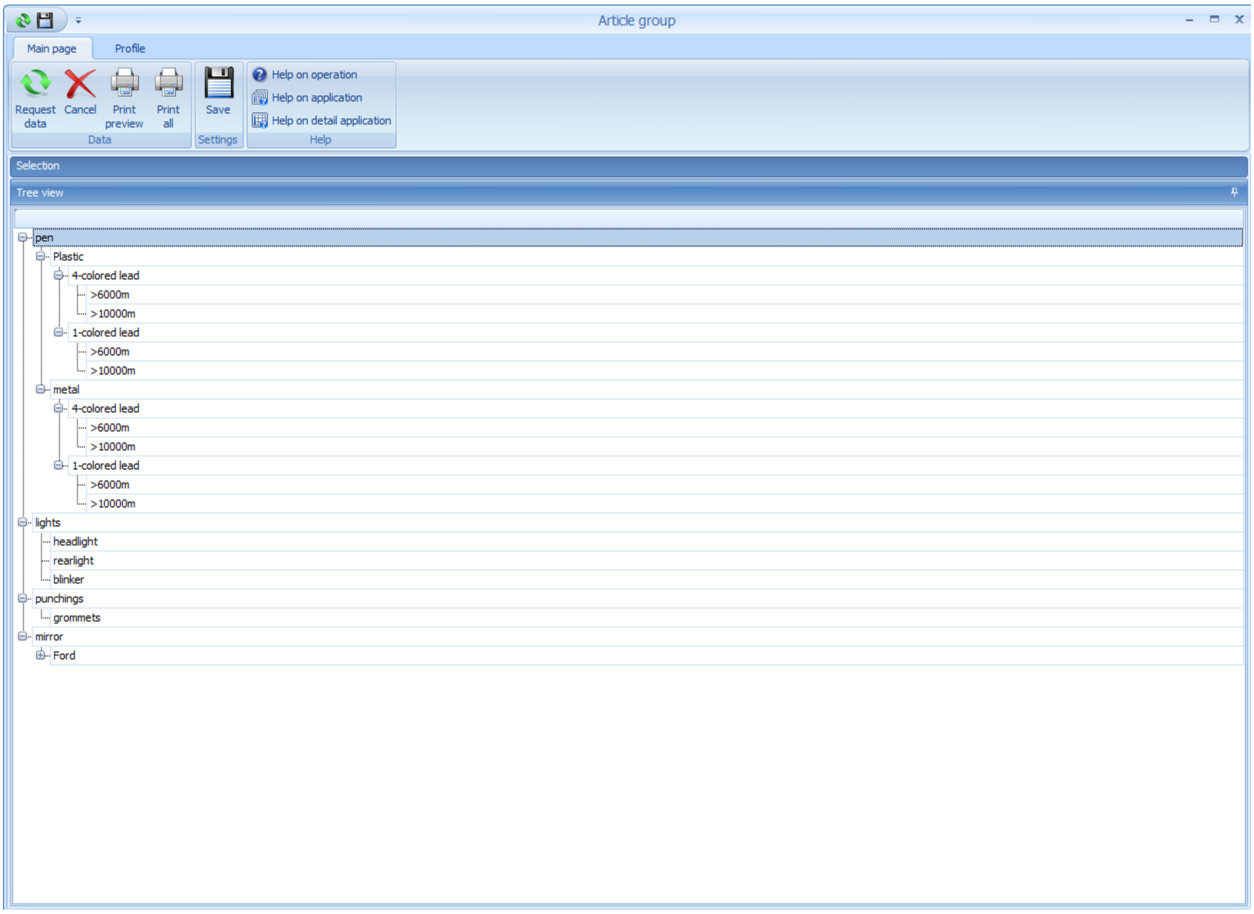
The group catalogs have been designed to create and edit groups for the different applications. The created groups may be assigned to master data of the corresponding application. Consequently, article groups may be created, for example, and assigned to the articles. In this case, it is also possible to create inspection plans on the basis of article groups.

Basically, the creation of groups is also reasonable for failure mode analyses.

5.1 Starting the function

Menu	Master data → Quality management → Article groups Master data → Process data processing → Article groups Master data → Quality management → Measure groups Master data → Quality management → Failure type groups Master data → Quality management → Failure location groups Master data → Quality management → Failure cause groups Master data → Quality management → Causer groups Master data → Quality management → Cost type groups
Transaction code	atcgr → Article groups measgr → Measure groups ftypgr → Failure type groups flocgr → Failure location groups fcaus → Failure cause groups origr → Causer groups costgr → Cost type groups
Function authorization	atcgr -→ Article groups measgr.* → Measure groups ftypgr.* → Failure type groups flocgr.* → Failure location groups fcaugr.* → Failure cause groups origr.* → Causer groups costgr.* → Cost type groups

5.2 Default Application Layout



This figure of the article group catalog is exemplary for all groups.

5.3 Toolbar

The toolbar contains the function calls that are available for this application and possibly links to other applications. The functions placed on the “general” tab of the toolbar refer to all detail applications. In addition to the standard functions such as “help”, “request data”, “save application settings” and “print preview”, the other tabs include specific functions that are tailored to the corresponding detail application. The individual functions of the application are listed in the paragraphs that follow.

"Data" category

Request data

The information to be displayed within the application is requested according to the entered selection criteria. This process might take some time depending on the data set from which the system filters data and on the selection result to be transferred and displayed.

Cancel

This function cancels the query sent by clicking the “request data“ button.

Print preview

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.

Save

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking “Yes”.

"Functions" category

There are no special functions for this detail application. Groups are created, changed and deleted using the context menu of the right mouse button within the tree structure view.

"Help" category

Help on operation

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled “moc-cc.pdf”. It describes how to use MOC in general and applies for all applications.

Help on application

This function opens the manual that describes the application from which the help function was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.

Help on detail application

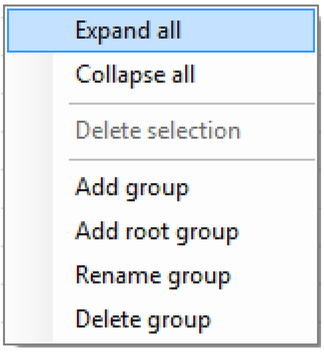
This function opens the application manual at the section where the respective detailed application is described.

5.4 Selection parameters

There are no selection parameters. A specific group can be found by using the "fast selection" function. To use the "fast selection" function just open the group tree structure on the 1st level, select the first entry and enter the first letter of the group in question. Consequently, the first group starting with this letter that is found is selected. The "fast selection" function also integrates subordinate groups that are not opened. If the requested term is included in a group that is not opened, it will be opened automatically.

5.5 "Groups" Detail Application

A group may be created, changed or deleted by opening the context menu of the right mouse button within the display area of the tree structure.



It is altogether possible to define groups up to the fifth hierarchy level. The "add root group" function has to be selected in the context menu of the group tree to create a new main group (1st level). The menu entry "add group" generates a sub-group (level 2 to 5). A designation, which is directly entered in the list view, has to be assigned for this new group. To be able to save the new group, click above or below this new entry within the group tree. Then a confirmation prompt appears asking whether or not the new group is to be saved. Provided that this question is affirmed ("yes"), the entry is saved. The same applies for renaming of groups. Regardless of which hierarchy level is concerned, the entry has to be selected to be able to edit the group designation. Changes can directly be entered in the corresponding line of the tree view. Click above or below the entry to be changed to be able to save the modification. No confirmation prompt appears when it comes to renaming.

An entry is also deleted by selecting a group entry and executing the "delete group" function. Only the group that is at the bottom of the group tree can be deleted.

The "expand all" context menu entry opens all groups up to the lowest hierarchy level. The "collapse all" context menu option closes all entries up to the first level.

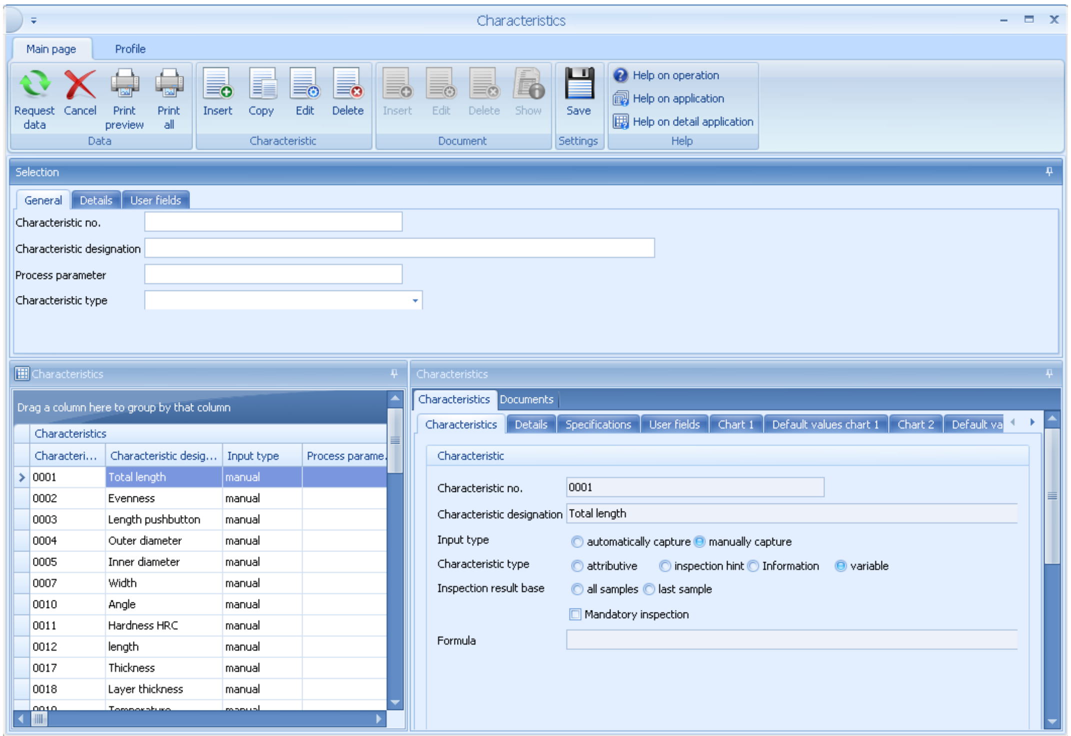
The "delete selection" function cannot be used in the maintenance of groups dialog. This function is enabled, for example, in the maintenance of articles application if an article group is selected and this selection is to be removed/deleted.

6 Characteristic Master Data

Summary

Menu	Master Data → Quality Management → Characteristics
Transaction code	chrq
Function authorization	chrq

Utilization



The characteristics catalog has been designed to define characteristics and, as a result, to predefine characteristic data of inspection plans. For this reason, it is directed to people involved in inspection planning.

The characteristics catalog is one of the most important basic catalogs for without this catalog no inspection plan could be created. As it has been designed to predefine characteristics data for inspection plans it provides extensive input possibilities. Generally, only this data, which has not to be modified at a later point in time when assigning it in the inspection plan, should be entered in the characteristics catalog. The definition of limit values, for example, is usually not reasonable as they are only known when the inspection plan is created. For a relation to a specific article is only established with the assignment to an inspection plan. If you keep this always in mind it becomes obvious which information needs to be defined. For this reason, it should be considered thoroughly whether the characteristic "outer diameter" is to be created only once and the detailed information is to be defined later in inspection planning or whether it would be better to create several characteristics for the "outer diameter" specifying the limit values already. In the majority of cases, it is better to only enter a few general characteristics. The required evaluations/reports also play a role in this. If a new "outer diameter characteristic" is created for nearly every tolerance modification it is almost only “applicable” for one article and comprehensive evaluations are not possible when failures are analyzed later"

It is important that each detail defined here can be modified in the inspection plan or that details, which have not been stated, can be added.

The configurations made in the characteristic master data are not final. Characteristic master data has rather been designed as a template for inspection planning. All settings of the characteristic master data can be supplemented and modified in inspection planning.

Integration

The characteristic catalog is a global catalog that is used in many QM applications. Please find below some possible fields of application that refer to the characteristic catalog.

- Inspection planning for production, goods receipt, goods issue, initial samples and calibration
- Inspection requirements for production, goods receipt, goods issue, initial samples and calibration
- Failure analysis in complaints management
- Several reports/evaluations

Prerequisite

There are no special requirements.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
  Characteristics number

- Characteristic designation:
  Designation of the characteristic – Please note: wildcards may be used "*"

- Characteristic type:
  Inspection type: attributive, inspection chart, variable

"Details" tab:

- Gages

  Selects a gage

- Gage designation:
  Selects a gage designation

"User fields" tab:

- If user fields are created they may be selected

If several selection criteria are used overlapping results are displayed in the characteristic master data.

In addition, the column filter allows for the content of each individual column to be filtered.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

"Characteristics" tab

Characteristic no.

Unique number of the characteristic

Characteristic designation

Designation of the characteristic

Input type

Automatic or manual collection type. This field controls the release of HYDRA-PDV fields (for automatic collection). If the automatic collection function is selected, the characteristic type is restricted to the "variable" option.

Characteristic type

Decides whether the inspection is based on the collection of measured values (variable) or on the indication of the number of identified errors (attributive). When it comes to the attributive inspection, the decision is often based on "pass" or "fail". The inspection chart and the information characteristic are further characteristic types. The information characteristic has only been designed to display a document while the inspection is running. Subject to the input type, the lower area of the dialog provides corresponding sampling schemes.

Inspection result base

Decides whether all samples or only the last sample are used to determine the inspection result (pass/fail).

Mandatory inspection

If this checkbox is activated, at least one measured value has to be entered for this characteristic, before an inspection order can be completed with this characteristic.

Calculate characteristic

Identifies a characteristic to be calculated

The authorization “iriscp.formula“ is required to show this field in inspection plan characteristics.

Formula:

Please also see the section entitled "formula calculation"

The authorization “iriscp.formula“ is required to show this field in inspection plan characteristics.

"Details" tab

"Gage" group

Gage

Defines whether a gage or gage group is to be assigned to the characteristic:

Assignment of the gage to be used (or even gage group).

The resource management may be accessed using the resource type "PRM". The authorization “iriscp.gage“ is required to show this field in inspection plan characteristics.

Gage designation

Shows the gage designation

"Properties" group

Certificate printing

The corresponding definition determines whether this characteristic is to be printed (show selection or print always) or not (print never) when certificates are printed out at a later stage (e.g. acceptance, inspection certificate). If the "display selection" option is chosen, a list of the characteristics that are assigned to this flag is displayed prior to printing. These characteristics have been predefined for printing in the list. However, this selection may be removed. Finally, all selected characteristics and the characteristics assigned to the "print always" option are printed out on the certificate. Characteristics assigned to the "print always" flag do not appear in a selected list, as they are printed in any case. However, it is important to note that these certificate flags only affect certificate forms.

Error weighting

For information purposes, a rating can be made here if the inspection result for this characteristic is "fail".

"Inspect" group

Analysis selection catalog

Using this selection function, an analysis selection catalog may be selected. This catalog restricts the selection options when failures are entered (failure types, failure locations, etc.). (But nevertheless, all available failures may be entered by directly entering their number).

Designation of analysis selection

Shows the designations of analysis selection catalogs

"Specifications" tab

Once the "specifications" tab has been selected, the sample scheme and dimensions may be entered. In this context, it has to be considered that (as already mentioned) the definition of tolerance limits in the master data of characteristics is only reasonable if certain conditions are met. The same applies to the definition or calculation of action and warning limits. The different possibilities are described anyway.

"Sampling scheme" group

Sampling scheme

The following sampling schemes are available:

- 100% inspection
- k value inspection
- Batch inspection
- n-c inspection
- SPC inspection

The sampling scheme determines the course of the inspection. With an n-c inspection with the parameters 5-0, 5 items are checked and 0 failures may be detected.

The section entitled "Sampling scheme" describes this process in more detail.

Sample size/assumable sample size

Specification of the sample size or assumable sample size depending on the sampling scheme, see section "sampling schemes".

Acceptance quantity

Acceptance quantity for the n-c inspection, please also see section "sampling schemes".

Interval type

Input for SPC or n-c inspections: time, pieces, once, none. Please also see the section "sampling schemes".

Interval value

Specifies the interval subject to the interval unit.

The authorization ”iriscp.interval“ is required to show this field in inspection plan characteristics.

Interval unit

For n-c or SPC inspections, e.g. minutes, hours.

The authorization ”iriscp.interval“ is required to show this field in inspection plan characteristics.

With output batch change

Causes an inspection to become due when output batches change.

The authorization ”iriscp.interval“ is required to show this field in inspection plan characteristics.

With machine status change

Causes an inspection to become due when machines change.

The authorization ”iriscp.interval“ is required to show this field in inspection plan characteristics.

Source status

Specifies the source statuses, separated by commas, (specific, non-productive machine statuses), which are to cause an inspection to become due when switching into a productive machine status.

The authorization ”iriscp.interval“ is required to show this field in inspection plan characteristics.

With the change of shifts

Causes an inspection to become due when shifts change.

The authorization ”iriscp.interval“ is required to show this field in inspection plan characteristics.

"Construction measures" group

Unit

Pieces, meter, kg, etc. Unit of the characteristic. Units are assigned by accessing the unit catalog.

Decimal places

Number of decimal places. Leading zeros in front of the comma are not displayed in the specification fields. The number of decimal places that is defined within system settings is indicated by default.

Size

Plausibility and tolerance limits can be entered as absolute, relative or percentage values. Please note that relative or percentage lower limits (lower tolerance limit, lower process limits) have to be indicated with a negative algebraic sign.

Norm

Computation of tolerances on the basis of specific norms (e.g. ISO metric fits). Subject to the selected norm, further information is requested (e.g. engineering fit). Tolerance limits are computed automatically on the basis of these specifications.

Engineering fit

Computation of tolerance limits on the basis of a specific norm and fit. The selected fit depends on the selected norm.

UPL

Specifies the upper plausibility limit

UTL

Specifies the upper tolerance limit (upper specification limit)

Target value

Indicates the target value

LTL

Specifies the lower tolerance limit (lower specification limit)

LPL

Defines the lower plausibility limit

Generate error (UTL)/(LTL)

If the checkbox "generate error" is enabled it is defined whether a violated limit value automatically (in the background) results in the failure type "limit value violation" (AUTO:TG> or AUTO:TG<) to be defined when measured values are recorded. This option is not available for attributive characteristics, as the specification is only used for information purposes in this case.

"User fields" tab

In case user fields are defined for characteristics, they are displayed and may be edited here.

"Chart 1/chart 2" tab

The control charts to be used can be defined in the "chart 1/chart 2" tab. These control charts are later available in the measurement recording for terminals (AIP). Altogether two different control charts may be defined, whereas the action limits, warning limits and the mean value may be defined for each control chart if variable characteristics are in use. There are two different possibilities to define these limit values. They are either entered manually or determined automatically on the basis of specific default values that are defined in the "default values" tab "chart 1/2". For further information on control charts, please see the sections entitled "control chart for variable characteristics" and "control charts for attributive characteristics".

Chart 1 / chart 2

Defines the control chart that is to be displayed in the measurement recording dialog at the terminal. Action limits may also be defined on the basis of the control chart type.

Upper AL

Specifies the upper action limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

Upper WL

Defines the upper warning limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

MV (Mean value)

Specifies a mean value, e.g. as basis for the automatic calculation of limits by the system.

Lower WL

Defines the lower warning limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

Lower AL

Specifies the lower action limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

Generate trend error

The option "generate trend error" has to be activated to be able to generate an automatic error if a trend exists (e.g. seven values in a row are descending or ascending, the number of values is defined while the system is customized). To determine a trend, the samples of an inspection step characteristic are taken into account, sorted by the sample number, regardless on which machine data has been recorded.

Generate error (UWL) / (LWL)

If the checkbox "generate error (UWL) / (LWL)" is enabled it is defined whether a violated limit value automatically (in the background) results in the failure type "limit value violation" (AUTO:WG> or AUTO:WG<) to be defined when measured values are recorded. A violation of the limit value refers in this case to the defined control chart. If, for example, an xq chart is defined an automatic error will only be generated, provided that the corresponding xq value of the control chart, not the single value, is beyond the warning limits.

Generate error (UAL) / (LAL)

If the checkbox "generate error (UAL) / (LAL)" is enabled it is defined whether a violated limit value automatically (in the background) results in the failure type "limit value violation" (AUTO:EG> or AUTO:EG<) to be defined when measured values are recorded. A violation of the limit value refers in this case to the defined control chart. If, for example, an xq chart is defined an automatic error will only be generated, provided that the corresponding xq value of the control chart, not the single value, is beyond the action limits.

"Default values chart 1 / default values chart 2" tab

For further information on control charts, please see the sections entitled "control charts for variable characteristics" and "control charts" for attributive characteristics.

"Default for calculating limit values" tab

Calculation type

Default values to compute limit values: Cpk, Sigma, sQuer/an, RQuer/dn, relative deviation of XQuer, percentage deviation of XQuer

Cpk

Default value of cpk

Sigma

Default value or computed sigma value

RQuer /sQuer

Default value for RQuer/sQuer

"Non-action probability" group

Action limits (non-action probability)

Selects the action probability (only visible with the computation type: cpk, Sigma, RQuer /sQuer)

Warning limits (non-action probability)

Selects the action probability (only visible with the computation type: cpk, Sigma, RQuer /sQuer)

"Deviation from XQuer specification" group

rel. AL

Direct entry of the action limits (only visible if the calculation types "rel. deviation/deviation in percent" are used)

rel. WL

Direct entry of the warning limits (only visible if the calculation types "rel. deviation/deviation in percent" are used).

"Confidence interval" group

Confidence interval

One-sided or two-sided. It may be chosen between "on-sided" and "two-sided" for the control charts R and s.

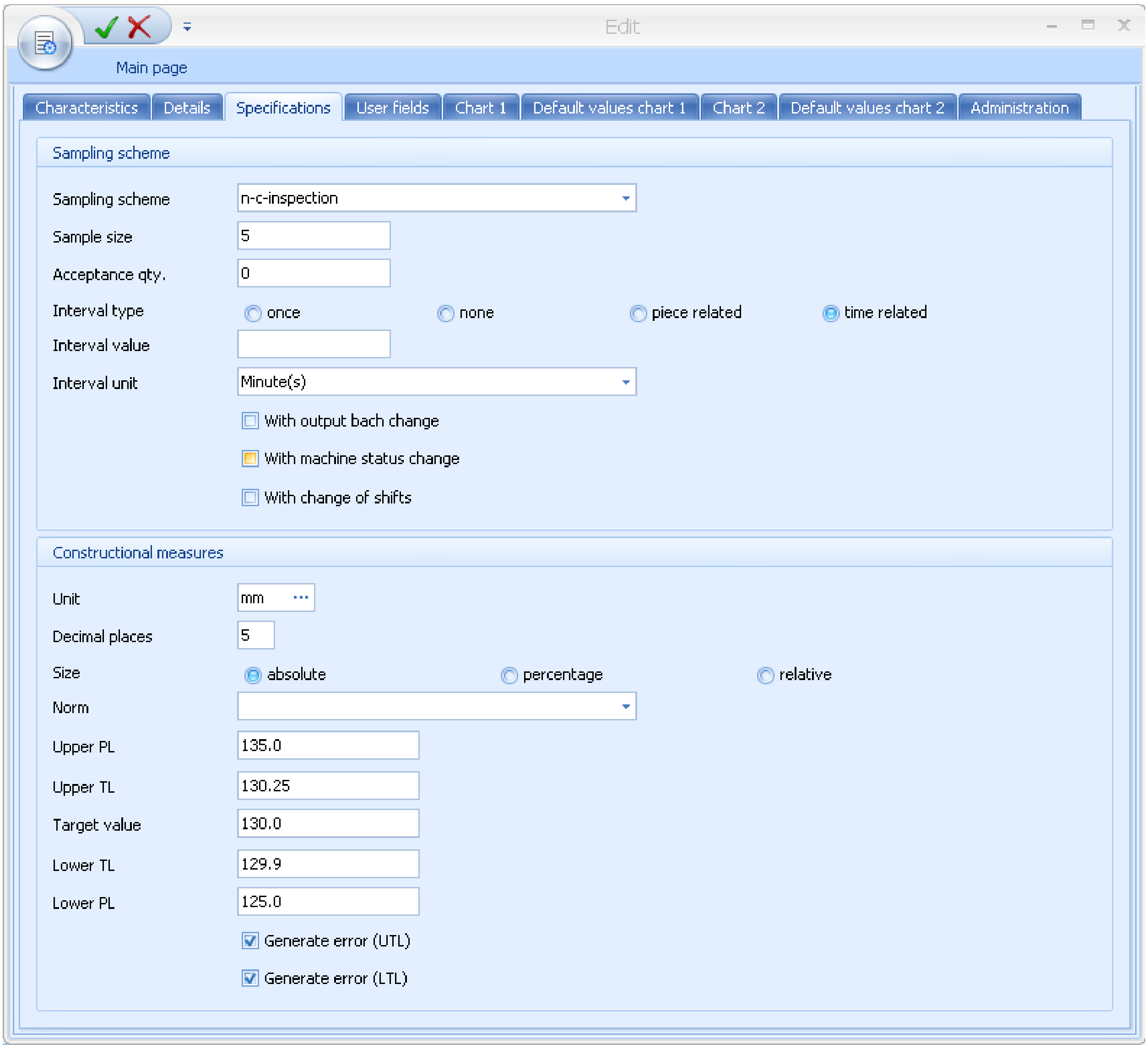
"XQuer" group

XQ

Target value, tolerance center, mean value of xq chart, input (only visible and can only be selected with an xq control chart)

Editing functions

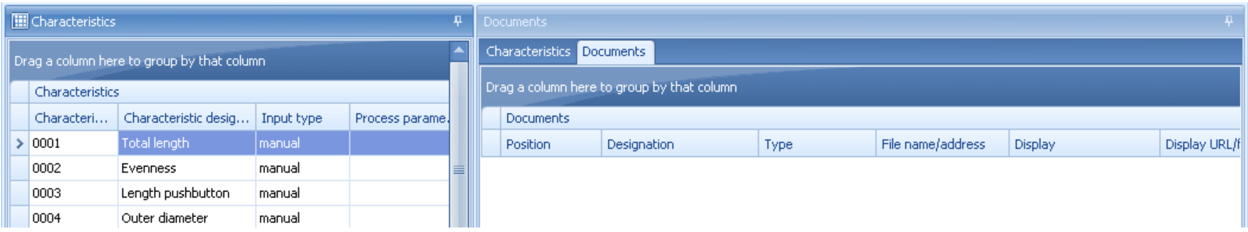
The below screenshot is one example for how an editing dialog could look like. The design and alignment of fields may deviate from that of this screenshot.



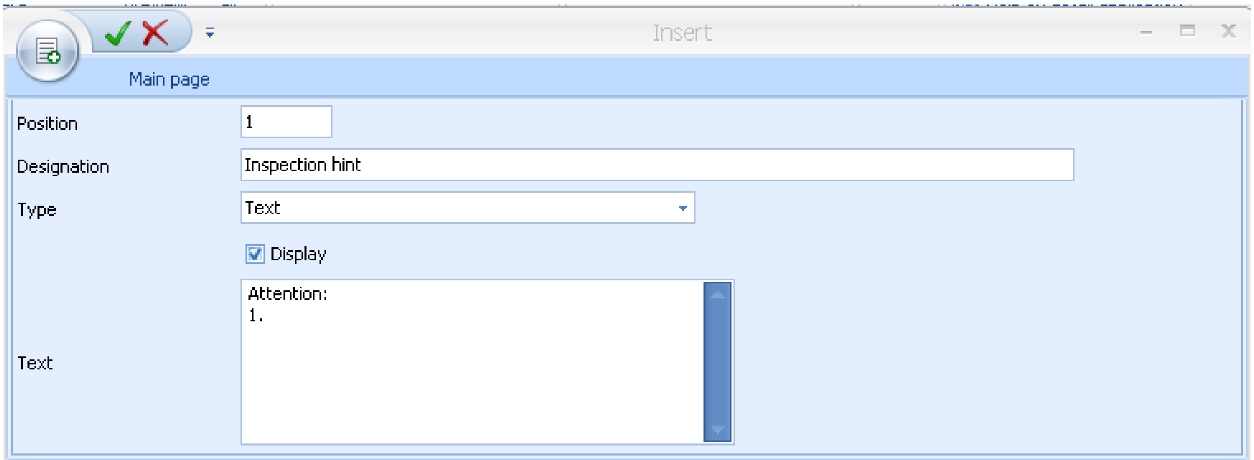
Toolbar

There are no further specific function keys in addition to the standard features.

"Documents" detail application



Provided that the "documents" tab has been activated, as many documents as required may be assigned to each characteristic. If this tab is activated corresponding editing buttons are enabled in the toolbar to edit the documents.



When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

"URL" and "text" are provided as file types. The file name including the path may be entered for the "file" type. The file type "URL" allows for the Internet or intranet to be accessed. The third file type "text" allows for the text to be entered directly.

Please note:

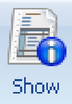
The different variants of the file type “URL” supported by the shop floor client are described in the relevant manual dealing with the shop floor client. It might, for example, be the case that “https” URL entries can be displayed on MOC but not on the AIP shop floor client.

A designation may be assigned to each defined document. Moreover, it can be determined in which order the documents are to be listed. This is made by the "position" field (numeric input). The specifications have to be unique within this list. In addition, the "display with inspection" checkbox determines whether or not the document can be displayed during the inspection process.

In this case as well it has to be reflected thoroughly whether documents are to be assigned without a precise reference to an article. Normally, document assignment depends on the article.

“Documents“ toolbar

In addition to the standard features, the “Documents” detail application also provides a button to view the documents.



Show documents

With this button it is possible to open and show a document, which is attached by a link. In this case, however, a program needs to be installed that is able to open the linked file type, e.g. Acrobat Reader for PDF documents.

6.1 Sampling scheme

The user can choose from five sampling schemes in a specified list. Subject to the selected sampling scheme, some additional information has to be defined. Depending on how they are used in the different inspection plan areas (e.g. production, goods receipt, goods issue) only a subset of these sampling schemes is available.

Sampling scheme n-c inspection: The sample size is entered in the "sample size" field (= n) and the maximum number of admissible non-conforming units is entered in the "acceptance quantity" (=c) field. The figure "c" is defined as acceptance number. This means: if n = 50 und c = 1 the characteristic, and thus the item, is only rated as "fail" if two non-confirming units occur, provided that the sample size amounts to 50.

Sampling scheme 100% inspection: The sampling scheme "100% inspection" is generally only used in the goods receipt and goods issue. In this case, the sample size is computed from the actual quantity of the inspection requirement and corresponds to it.

Sampling scheme SPC inspection: The sampling scheme "SPC inspection" nearly corresponds to the "n-c" inspection plan. The only difference is that the acceptance limit "c" is not used in this case.

Sampling scheme batch inspection: The sampling scheme "batch inspection" only applies to the areas "goods receipt" and "goods issue" in the standard configuration. The percentage specifying how much percent of the batch is to be checked is entered here. Later in the inspection order characteristic the sample size is calculated from the actual quantity of the inspection requirement and multiplied by the specified percentage.

Provided that action limits are to be calculated, the expected (assumable) sample size has also to be entered here.

Sampling scheme k-value inspection: With the k value inspection the entered k value is checked against the computed k value and if this value is violated the sample is rated "fail".

6.2 Control charts for variable characteristics

The variable characteristics provide the charts xq, s and R.

The production dispersion is used in many computations of the statistical quality assurance. One example is the calculation of capability indices and action limits of a quality control chart. Vice versa, it is possible to estimate the dispersion of production, provided that a process capability index is indicated, and the action limits are calculated on this basis.

The specifications for the calculation of limit values can be found in the tab "default values chart 1" or "default values chart 2", where values to estimate the production dispersion can be entered. The action and warning limits can be calculated on the basis of these specifications. However, it is also possible to enter the production dispersion directly. This program provides altogether three computation options.

At first, the specifications are described on the basis of the xq and s chart. The differences with the R chart are explained in more detail in the sections that follow.

There is often a specification for the process capability index cpk. This specification is reasonable, as respecting the process capability index cpk implies that items can be produced within the range of tolerance. On the basis of the default cpk value, the system computes internally an estimated value for Sigma, which is entered to the right of the "Sigma" option for information purposes. This computed basic estimated value has in turn been designed to calculate the limit values of the xq/s chart. The computation is performed, once further data has been entered using the "calculate" button. The calculation method "cpk" is set by default. In addition, there are also the calculation methods "sigma" and sQuer/an".

This cpk value of 1,33 ensures that 99.725% of the characteristic values are within the tolerance. However, it also often required that 99.994% of the characteristic values are within the tolerance limit, which corresponds to a cpk value of 1.67.

The calculation method sq/an means that an estimate of the standard deviation is calculated from the quotient of the medium standard deviation and a correction factor an. This correction factor depends on the sample size, which is expressed by the index n. The values for a_n are defined in the system and are requested automatically. This estimate of the standard deviation is best in case that there is no specification of the process capability index and the production dispersion is unknown and thus the specification of the sq-value has still to be corrected by a correction factor. Moreover, it is the most efficient one under the given conditions. The sq-value has also to be specified for a subsequent calculation of limit values. This can be entered in the field on the right next to the sq/an option. If the “calculate” button is clicked in the preceding dialog the estimate sq/a_n is calculated from the specification of sq and it is entered on the right next to the sigma option for information. This estimate is then the basis for the calculation of action and warning limits of the xq/s chart

The third calculation method requires a sigma value being specified. In this case it is assumed that sigma is known and consequently the correction factor is not required. To the right of the “sigma” option the sigma value is entered. In contrast to the previous method sq/a_n is replaced by sigma. In cases of doubt and since in the majority of cases sigma is not really known, the calculation method based on the specification of sq including the automatic determination of the estimate sq/a_n should be performed for variances.

If the “relative deviation from Xq” or the “deviation from Xq in percent” is selected as “specification for calculating limit values” the input option for “action probability in %” disappears and the “deviation from target value” can be entered instead. Limit values are then calculated on the basis of these values and the default calculation of Xq (target value, tolerance center, mean value of XQ chart, input).

Further details have to be made in order to determine action and warning limits of the Xq chart. An Xq value has to be specified. The system offers the possibilities to equate the Xq value with the tolerance center or the target value or to specify a value manually. If the process is supposed to be aligned to the mean value the tolerance center should be preferred as Xq value.

The action probability must be entered in per cent in order to calculate action and warning limits of the Xq/S-chart. For this purpose, it has to be defined beforehand whether the calculation is to be made on the basis of one-sided or two-sided limit values. One of the two options has to be selected.

Having defined the “one-sided“ or “two-sided“ option, the action probability has to be entered in per cent. All possible and reasonable action probabilities are defined in the system and need only to be selected from the list. The computation of the Xq chart is based on normal distribution. The value 99,725 is to be selected if, e.g., 99,725% of characteristic values are supposed to lie within the action limits. As the specification of a sigma area is commonly used by some users, the corresponding sigma area is displayed along with the respective action probability for information.

If warning limits are also to be calculated an action probability has also to be entered here. Please take into account that the probability value of the warning limit has to be less than the action limit value.

No sigma area is displayed in the selection list since the distribution of chi² is considered for the calculation of limit values of the s-chart. Apart from that, the entry is the same as for the Xq chart.

The calculation of limits is triggered as soon as the specifications made are saved, provided that the "calculate" checkbox has been clicked before.

As already mentioned, the user is also able to enter the limit values directly without any defaults.

If the R-chart is selected instead of the Xq or s-chart, the option sq/a_n is replaced by Rq/d_n in the specifications for the calculation. The calculation of the estimate of sigma is made on the basis of the default mean range R divided by the correction factor d_n. This correction factor in turn depends on the sample size n. The corresponding values are defined in the system and are selected automatically. Apart from that, the rest is the same as for the Xq or s-chart. Please note that limit values are not calculated on the basis of a chi² distribution for the R-chart but on the basis of a table defined in the system, which is based on standardized ranges.

6.3 Control charts for attributive characteristics

The p- and u-charts are available for attributive characteristics.

p designates the share of non-conforming units in the sample and u designates the failures per unit in the sample. It is important for the p-chart that each item is either described as error-free or faulty. If an item shows several failures it is referred to as faulty only once.

In contrast to variable characteristics there are no lower limit values. Furthermore, it is normally not necessary to state the values UTL, LTL and target value.

A pq or uq value has to be entered in per cent to calculate specifications automatically. This can be done in the default values dialog.

The calculation of limits is triggered as soon as the specifications made are saved, provided that the "calculate" checkbox has been clicked before.

Calculation is respectively based on normal distribution. The value 99,725 has to be selected if, e.g. 99,725% of the characteristic values are supposed to lie below the upper action limit. As the specification of a sigma area is commonly used by some users the corresponding sigma area is displayed with the respective action probability for information.

6.4 Computation of formulas

On the basis of an entered formula, measured values can be calculated automatically by way of measured values or statistic values of other characteristics that have been checked before.

The first part of the formula specifies where the calculation of formulas is supposed to take place. The following types are available:

- V – Calculation on the level of single values (Value).
  A single value is generated for the calculated characteristic for each single value of the characteristics involved.

- S - Calculation on the level of samples (Sample).
  Exactly one single value is generated for the calculated characteristic for each sample of the characteristics involved.

- C - Calculation on the level of characteristics (Criteria).
  Exactly one single value is generated for the calculated characteristic (with respect to the overall statistic of all characteristics involved)

The actual formula follows this identifier. The following operators, functions and constants are supported in this context:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates natural logarithms
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates common logarithms
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
x + y	Addition
x – y	Subtraction
x / y	Division
x * y	Multiplication
x ** y	Calculates x raised to the power of y
Absolute term (constant)	
pi	3.141592654
e	2.718281828

If constant, numeric values are used in formulas it has to be taken into account that no separators are used for thousands digits. If these constants are floating-point numbers it has to be considered that they use a dot as decimal separator instead of a comma.

The following syntax applies for the variables identifying the single values or statistic values of the order characteristics involved [x:y:z]. The x parameter identifies the statistic value to be used. The available values are listed as follows. Please consider possible restrictions regarding the calculation level.

- `X` – Single value
  (is only available for calculations on the level of single values)

- `AVG` – Mean value
  (is only available for calculations on the level of samples or characteristics)

- `MIN` – Minimum
  (is only available for calculations on the level of samples or characteristics)

- `MAX` – Maximum
  (is only available for calculations on the level of samples or characteristics)

- `SUMX` – Sum of single values
  (is only available for calculations on the level of samples or characteristics)

- `R` – Range
  (is only available for calculations on the level of samples or characteristics)

- `S` – Standard deviation
  (is only available for calculations on the level of samples or characteristics)

- `N` – Sample size
  (is only available for calculations on the level of samples or characteristics)

- `M` – Number of samples
  (is only available for calculations on the level of characteristics)

The y parameter describes how the corresponding characteristic is supposed to be identified. The following possibilities are available:

- `SENO` – identification via the MS (maintenance sequence/operation sequence) of the characteristic (serial number)

- INCR – Identification via the characteristic number (inspection criteria)

  If the characteristic number is not unique within the inspection requirement it is not predictable which one of the applicable characteristics is used at the time of calculation.

The parameter z identifies the characteristic on the basis of the field content determined by parameter y. Either the MS/OP number or the characteristic number of the calculation source is written in this field. If the characteristic number has a blank it should be replaced by an underscore within the formula.

Example 1:

A new characteristic is computed from the single values of the characteristic assigned to the number "LENGTH"/"LAENGE" divided by 2.5. A corresponding single value is supposed to be calculated for each single value of the source characteristic (calculation on the level of single values).

→ Formula: `V: [X:INCR:LAENGE] / 2.5`

Example 2:

The characteristic "surface" results from the product of the characteristics with the characteristic number “LENGTH”/”LAENGE” and “WIDTH TOTAL”/”BREITE GES”. A single value of the characteristic "surface" is supposed to be calculated for each single value of both source characteristics (calculation on the level of single values).

→ Formula: `V: [X:INCR:LAENGE] * [X:INCR:BREITE_GES]`

Example 3:

The characteristic "maximum margin width" results from the subtraction of the minimum of the characteristic "inside diameter" (MS 10) from the maximum of the characteristic "outside diameter" (MS 20). A single value of the characteristic "maximum margin width" is supposed to be calculated for each sample of both source characteristics (calculation on the basis of samples).

→ Formula: `S: [MAX:SENO:20] - [MIN:SENO:10]`

The escalation CPAUMW.CALCULATED_CRITERIAS_GET_VARIABLE_VALUE is triggered if unknown variables are used within a formula (faulty parameters x and/or y). However, the escalation management module has to be licensed in this case.

When it comes to the calculation of formulas, it is allowed to compute new calculated characteristics from calculated characteristics that have already been computed. However, this nesting may not have more than 10 references one below the other. Furthermore, double concatenations must not be created (Example: characteristic A is calculated from characteristic B and characteristic C; characteristic C is calculated from characteristic A).

Tool numbers, machine numbers or cavity numbers, etc. are not defined for the single values computed.

The same number needs to be assigned to all source samples of the calculation to take over a corresponding number (batch number, sample number, serial number, etc.). If there is no number that is assigned to all source samples a number will not be assigned to the calculated sample. Provided that several numbers are found that are assigned to all source samples only the first number that has been found is assigned to the calculated sample. This function only applies to numbers, which have been assigned on sample level.

7 Specification List

1.1 Summary

Menu	Master data → Quality management → Specification list
Transaction code	sclq
Function authorization	sclq

Utilization

The specification list has been designed to create specifications, e.g. within the framework of family inspection planning.

To be able to use the specifications of the specification list for the generation of an inspection step with inspection step characteristics, the property "from list" has to be set in the corresponding inspection plan characteristics. In this case, only active entries of the specification list are used. A specification list entry is searched on the basis of the following key fields, when generating an inspection step.

- Area
- Article number, drawing issue number
- Resource number
- Characteristic no.
- Operation number and operation designation
- Customer no.
- Supplier number

The order of searching the specification list may be configured while customizing the system.

The specification list does not replace inspection planning. It rather is a supplement to family inspection plans.

Integration

This function is a fundamental components of family/group inspection planning, as the inspection specifications that vary with each item in the article group of a group inspection plan are defined in the specification list.

Prerequisite

An inspection plan for the article group including corresponding characteristics referring to the specification list has to exist to be able to use specification lists (configuration within inspection plan characteristics: "from list").

Selection criteria

The application provides the following selection criteria:

Area

List of available CAQ areas

Specification no.

Unique specification number

Version no.

Unique version within the specification. Only active, provided that the version control function has been activated within system settings.

Active

Active or inactive entries are filtered.

Special case

Special cases/normal cases or both are filtered.

Article number

Article number of the specification list entry; can be selected from the catalog for article master data.

Article designation

Article designation

Customer number

Customer number can be selected from the company catalog

Customer name

Customer name

Supplier number

Supplier number can be selected from the company catalog

Supplier designation

Supplier name

Operation

Operation number

Workplace

Workstation, e.g. machine

Note: The definition of machine related specification list entries isn´t supported from the standard. This field can be used for customer specific implementations.

Characteristic no.

Characteristics number

Characteristic designation

Characteristic designation

Characteristic type

The characteristic type is filtered: variable, attributive, inspection chart

If several selection criteria are used overlapping results are displayed in the specification list entries.

Field descriptions

"Characteristic key" tab

"Characteristic" group

Area

Specifies the area for which the specification list entries are to apply.

Specification no.

Number of the specification

Version no.

Version number of the specification

Active

Checkbox. Shows whether the entry is active or inactive.

Workplace

Indicates the workplace.

Note:: The definition of machine related specification list entries isn´t supported from the standard. This field can be used for customer specific implementation in order to define machine related inspections for the same article. This might be required if e.g. machines of different types are used.

Machine designation

Designation of the workplace/machine

Resource

A resource (e.g. tool) may be indicated.

Designation

Name of the resource

Characteristic no.

Unique number of the characteristic

Characteristic designation

Name of the selected characteristic number

Operation

Number of the operation

Operation designation

Designation of the operation

"Properties" group

Characteristic type

Variable, attributive or inspection chart

Input type

Specifies whether data is collected manually or automatically.

Special case

Indicates whether this characteristic does no longer need to be checked after x "pass" inspections in a row.

Number of pass inspections in a row

The number of "pass" inspections in a row that is required for the characteristic to be no longer required to be checked.

No characteristic

Specifies whether this characteristic is not required to be checked for the combination of the indicated key fields.

"Article" group

Article number

Article number for this specification list entry

Article designation

Designation of the selected article number

Drawing issue number

Drawing issue number of the selected article number

"Companies" group

Customer number:

Customer no.

Customer name

Customer name of the selected customer number

Supplier number

Supplier no.

Supplier name

Supplier name of the selected supplier number

The key fields of the "characteristic key" tab can no longer be changed if a list entry is changed (edited).

"Specifications" tab

Go to

The fields included in the "specifications" tab correspond to those of the characteristic master data and are described in the documentation MOC_CharacteristicsQm.

"Chart 1/2" tab

Go to

The fields included in the "chart 1/2" tab correspond to those of the characteristic master data and are described in the documentation MOC_CharacteristicsQm.

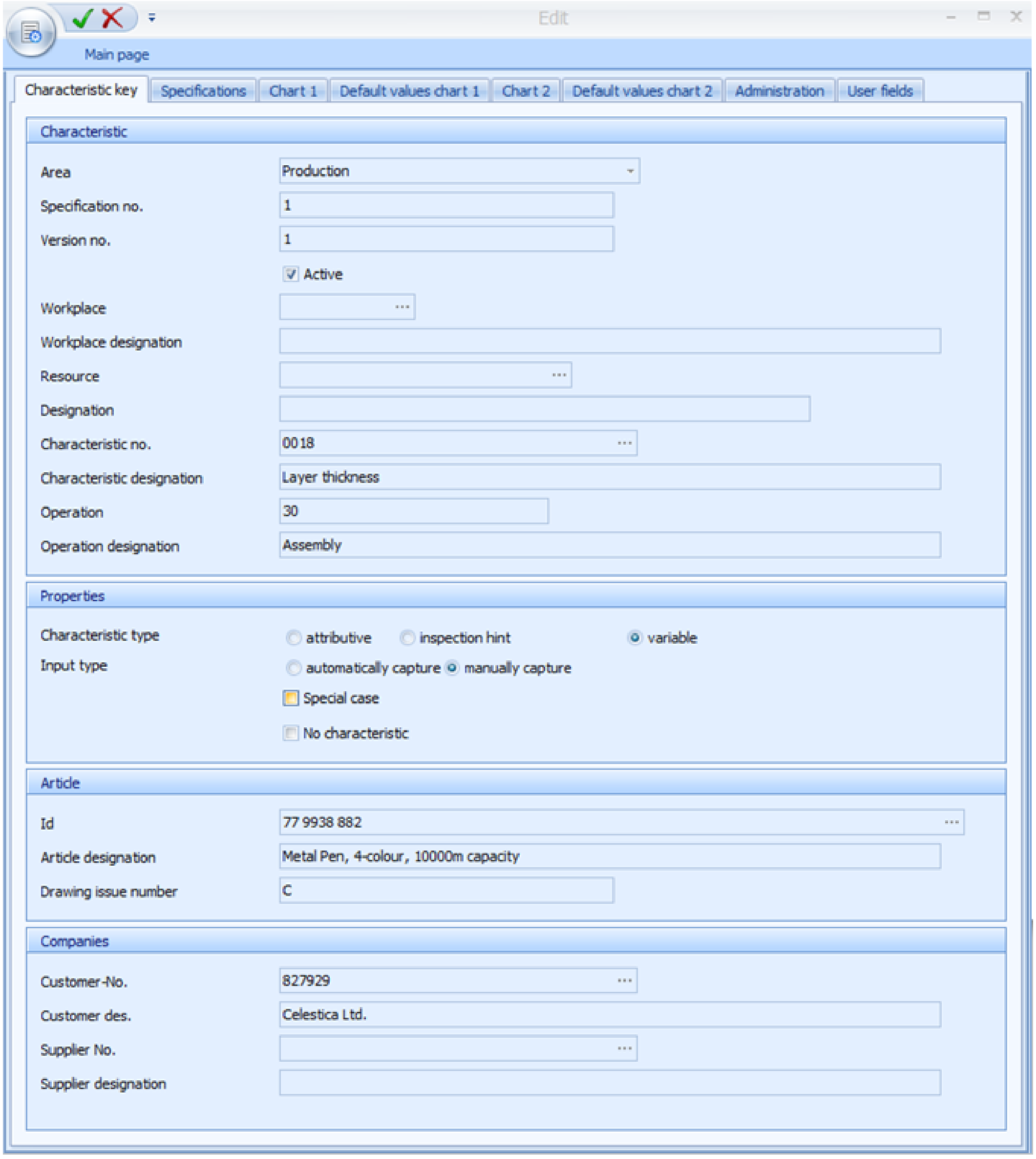
Default values chart 1/2

Go to

The fields included in the "default values chart 1/2" tab correspond to those of the characteristic master data and are described in the documentation MOC_CharacteristicsQm.

Editing functions

The following dialog opens to edit a data record:



Toolbar

The below-mentioned additional functions are available besides the standard functions.

Activate

Function authorization: sclq.active

Activates a specification list entry. A previously released version is automatically deactivated.

Deactivate

Function authorization: sclq.release

Deactivates a specification list entry. The specification list entry is no longer used.

8 Configuration of Workplaces and Resources

Summary

Menu	Master data → Resources → Resource configuration Master data → Workplaces/ Machines → Workplace configuration
Transaction code	res
Function authorization	mdres

The resource configuration is the central function for resource management in MES.

Usage

The master data for both workplaces and machines as well as for other resources (tools, DNC resources, etc.) are managed here. The resources are roughly classified according to resource type. This type is also connected with corresponding functions and applications that open other functional components of the MES especially for the respective type.

Integration

This application can be used to view the resource information of all of the resource types present in HYDRA. However, the data record maintenance depends on the resource type. In this way, depending on the resource type, not all fields can be maintained and not all resources can be created or deleted.

Based on the resource type, other applications are present in the MES that are specially customized for these types. For example, the application package of machine data collection is based on resources of type "Machine".

In addition to the resource configuration, the resource overview application is present, which does not permit data maintenance, but does enable administration operations for daily handling of resources such as the stock transfer of a resource.

Requirement

Before a workplace or machine is created, a year model/ shift calendar must be created. To use the various resource types in a meaningful way, the advanced licenses for these types must also be available.

Selection criteria

The following selection criteria are available in the application:

Resource from ... to ...

This selection criterion refers to the resource. Wildcards (placeholders *) can be used.

Short designation

Short designation of resource. Only relevant for resources of type MNR.

Resource type

Type of resource.

Workplaces and machines always have resource type MNR, while in the context of customizing other resources can be given individual resource types. Predefined resource types include:

DNC	NC-/ DNC program
DOC	Document
ENE	Energy meter
ENT	Extraction device
ENT	Extraction device
MNR	Workplace/ Machine
PAC	Packaging, transportation container
PRM	Inspection and measuring equipment
PER	Production staff / general
PRU	Set up staff
TEM	Tempering equipment
VOR	Device
WNR	Tool

Using the predefined resource types is recommended.

The detail resource information is adapted depending on the resource selected in the table overview.

Designation

Designation of the resource.

Group

Workplace/ Machine group of the resource. Only relevant for resources of type MNR.

Cost center

Cost center of resource.

Short name

Short name of the resource

Resource family

Family to which the resource is assigned.

Responsibility area

Responsibility area to which the resource is assigned.

Storage location

Master storage location of resource.

MD user fields

MD user fields 1- 6 of the resource. If a resource family is selected in the selection panel, the field names are displayed based on the assigned user field definition.

Field descriptions

This detail application includes three main tabs:

- Resource configuration
- Resource list
- Resource attributes

Main tab "resource configuration"

The configurations and master data of resources are defined here.

General tab

Resource type

Resource type of the resource. When the HYDRA system is delivered, some resource types are predefined. Further resource types can be created within the scope of HYDRA customizing.

Resource

The number of the resource or workplace to be entered is input in this field.

The maximum number of places of this number is as follows, depending on the resource type:

- Resources of type MNR: maximum of 8 places
- Resources of type <> MNR: maximum of 20 places

Permitted characters include ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890/_.-+#. Spaces and other special characters are not allowed. For technical reasons, * (asterisk) and % (percent) can be input, but are nonetheless not permitted because they are not valid characters. When the input field is exited, lower case letters are automatically transformed into CAPITAL LETTERS.

Notes regarding workplaces/ machines (resource type MNR):

For technical reasons, for resources of type MNR there is no check for the maximum number of places. For this reason it must be ensured that the length of the resource number (= workplace/ machine number) has a maximum of 8 digits.

If the option "Numeric machine number" (HYDRA basic settings) is activated for use on DOS-based terminals, it must be ensured that the resource number (= workplace/ machine number) includes only numerical digits and that the length of the number is exactly 8 places. If necessary, when creating the workplace/ machine, zeros must be added to the beginning of the number to extend it to 8 digits.

Short designation

Short designation of resource. This field can only be used with workplaces/ machines (resources of type MNR).

Designation

This field is used to assign a short, unique designation for each resource. This designation is displayed in reports and overviews and at the terminal and it is useful for orientation.

Responsibility area

Responsibility areas are used such that in various evaluations, the user is only shown the data to which he/she has access according to his/her responsibility area authorization.

The responsibility area can also remain empty. In this case, the resource is always displayed regardless of the user's assigned responsibility authorizations.

Cost center

The cost center to which the resource belongs is entered in this field.

Inventory number, engraving number, drawing number, manufacturer, owner

Additional information that functions as a comment.

Acquisition date, acquisition costs

Additional information that functions as a comment.
The currency is configured across the system in the HYDRA basic settings.

Storage location

Location at which the resource is stored when it is not being used (home storage location).

In connection with the material and production logistics (MPL), the material buffer specified in this field is used during input batch logon for reposting the logged on input batch(es) from the previous material buffer to this material buffer (upstream of the machine).

Delivery date, start-up date, guaranty date

Additional information that functions as a comment. These fields are only available with the licensing of the gage management.

External designation, resource type designation, usage, purchase order number

Additional information that functions as a comment. These fields are only available with the licensing of the gage management.

Supplier and party in charge including detail fields

Additional information that functions as a comment. These fields are only available with the licensing of the gage management.

Workplace configuration tab

This tab is only available if a resource of the type "MNR" is selected.

Workplace master data

Workplace category

N Machine

P Workplace

Definition as machine or workplace. As regards processing, both of these categories are identical if only BDE or MDE and PDV are in use.

J Machining center

The "Machining center" category and its functionality are described in detail in the BDE-BEA product documentation.

L Line (MDE-SFL only)

A Aggregate (MDE-SFL only)

The categories "Aggregate" and "Line" and their functions are described in detail in the MDE-SFL product documentation.

Q CAQ inspection station

Workplace is defined purely as a CAQ inspection station without affecting the BDE or MDE statistics.

R Reel-based manufacturing (MPL-RF only)

S Cutting unit (MPL-RF only)

The categories "Reel-based manufacturing" and "Cutting unit" and their functions are described in detail in the MPL-RF product documentation.

D Parallel output batches (MPL only, starting with MPL 7.2)

Parallel output batches can be produced on the machine for an operation with a batch management requirement.

C Packing station (MPL-PAL only, starting with MPL 7.2)

Specific posting functions are used on the machine for mapping a packing station. The functions are described in detail in the MPL-PAL product documentation.

M Melting aggregate

This option defines a machine as melting aggregate in terms of composition.

Workplace type

E Single workplace

G Group workplace

Please note

Terminals can be assigned to both group and single workplaces. However, in this case it must be noted that the terminal is set to operation mode "BDE" or the option **Processing** is set to "BDE processing" when assigning workplaces to terminals.

External workplace

This field identifies external workplaces. Currently, it only functions as a comment.

Locked

If this identifier is set, the machine/ workplace has been (logically) deleted. The following modifications are no longer permitted in this case:

- Order postings on the terminal
- Order posting on the console/MOC (e.g. using the Order overview function)
- Modifications in event maintenance

Furthermore, the machine/ workplace is no longer displayed in the *Machine overview* or in the graphic planning board of the HYDRA shop floor scheduling module (HLS).

Company

This field is used to differentiate the individual machines/ workplaces. Within the system, it also functions as report/evaluation option to some extent.

Group

Assignment of the workplace/ machine to a logical group. In the context of the planning this has to do with a capacity group in which the primary capacities are summarized.

If a new workplace is created, it is automatically assigned to a group of the same name (menu BDE: Master data > Workplaces/machines > Groups), which is defined as a capacity group. If there is no capacity group yet, then it is automatically created and then the workplace is automatically assigned to it.

Category

The category of the machine is entered here. From this category, a validation check can be activated according to the customizing configuration BDE: Master data > Order configuration > Order types, Plausibilities tab, option "Check for specifications in backlog of orders" *(value Category)*.

Year model

A valid year model must be entered here. During entry, times to be posted are compared with this shift model. If no planned year model is stored in the HLS tab, this shift model is also used in the HYDRA shop floor scheduling module (HLS).

Standard rate machine

The arithmetical standard rate of machines can be entered here for calculations. In the HYDRA shop floor scheduling module (HLS) this value is used for some (evaluated) key figures.

Standard labor rate

The arithmetical standard labor rate can be entered here for calculations. In the HYDRA shop floor scheduling (HLS) this value is used for the key figure "Evaluated labor utilization".

Performance level

The performance level of the workplace/ machine can be entered in percent in this field. This value is considered in the HYDRA shop floor planning and in the evaluation of material requirements when calculating the remaining run time.

Incentive wages indicator

Defines the type of calculating incentive wages. Mostly, this flag is used together with the incentive wages based on formulas for customer-specific configurations. In addition, the "incentive wage indicator" can also be used as selection criterion for the determination of wage types within the incentive wage determination.

This field should be left empty if the incentive wages determination is not in use.

The incentive wages indicator G=group piecework has a special meaning. If the workplace/machine has this flag, a premium group needs to be assigned every time an order is logged on . This can be performed either by the "assignment of premium groups" option of the incentive wages determination or, optionally, by an additional field in the terminal dialog for logging orders on. If no assignment is available, the logon of the order will be rejected by issuing a validation error.

Therefore, the incentive wage indicator G = Group Piecework may only be assigned if the group premium conditions are met in the incentive wages determination, as otherwise orders can no longer be logged on!

The meaning of the other incentive wages indicators is specified according to the customer's requirements while customizing the system.

File

It is possible to assign a graphic to every machine/ workplace. Among other uses, the graphic is displayed in the workplace overview or in the AIP. The following image formats are supported: jpg, gif, tif, bmp, ico, emf, wmf.

In the path configuration, the path must be configured using the PATH identification "MOCWPIMG"; the length of the file name for graphics files is limited to 12 places (8.3 notation). Note for Unix installations: Please use lower case letters only for file names.

Maximum capacity (KG)

If a machine is configured as melting aggregate, the maximum capacity in KG can be defined here.

Accuracy class, unit, etc.

Information fields in order to describe the accuracy. These fields are only available if the license for creating gages has been purchased.

Entry

Display 3rd list

A third list can be displayed/ activated in the basic screen on a Windows-based terminal (CTWIN / AIP) using the options displayed here. Depending on the options set, users can switch between the respective lists on the terminal. The following settings are possible. Note here that the display in the lists depends on the module set:

→ Input material (MPL): Logged on input materials/ batches are displayed.

→ Resources (WRM): Logged on resources and tools are displayed.

→ Staff (BDE): Logged on staff is displayed.

→Output material (MPL): Produced output batches are displayed.

Show material/ PRT list when OP is logged on

This option is only relevant in connection with WRM and resources logged on at Windows-based terminals (CTWIN / AIP).

If this option is set, when an OP is logged on, a specific logon dialog is called. This dialog contains a component/ PRT list in which resources are displayed that meet at least one of the following conditions:
- the identifier "Logon on terminal" is set on the resource type;
- the identifier "Log on with OP" is set to "Explicit logon"
- the resource is a so-called "required resource" (identifier on the resource).

Please note: As long as the workplace is relevant for MPL, material components are also displayed in this list.

Sequencing list

This option is used to define which operations are to be displayed in the sequencing list on the terminal. The following settings are possible:

S Basic setting. The value is taken from the option of the same name in the *HYDRA basic settings*.

M Pool of workplaces. Only the operations planned at the workplace are displayed in the sequencing list on the terminal.

G Pool of workplaces and groups. The OPs displayed in the sequencing list on the terminal are those that are either planned at the current workplace or another workplace in the group or are still located in the pool of groups.

K Pool of workplaces and categories. Only those operations that are planned at workplaces of the category are displayed in the sequencing list on the terminal.

H Group control. The OPs displayed in the sequencing list on the terminal are those that are either planned at the current workplace or another workplace in the group.

Number of OPs in sequencing list

The maximum number of OPs that are to be displayed in the sequencing list on the terminal can be stored here.

Compulsory sequence

This option can be used to specify whether or not logging the OPs on in the planned sequence is compulsory. The following parameters are permitted:

N Disabled

J Enabled/active

If the parameter is enabled when the OP is logged on a check is made as to whether or not there is an OP in the order pool for this machine/ workplace that is planned for the same time or previous to this OP in the sequence, but was not yet started (i.e. status = V/prepared). If yes, the OP logon will be rejected.

Please note: If the planning in HYDRA is done using order sequencing (menu ADE: Planning > Order sequencing), a configuration of the sequencing list that is not equal to "M" (pool of workplaces) and an active compulsory sequence can result in a combination that does not make sense.

Dialog control

To meet this requirement, a dialog control that deviates from the standard behavior can be defined at the workplace in the dynamic dialog configuration on the Windows-based terminal (CTWIN / AIP) and then a corresponding reference can be made to it in the dialog.

This configuration is only relevant and can only be used in the context of the HYDRA customizing. Otherwise it has no significance.

Logon of several OPs

If several different operations are to be processed on the machine, this identifier must be selected. Otherwise HYDRA only allows one order/ operation to be logged onto the machine.

Possible values:

J As many OPs as desired can be logged on at the same time.

Please note: On a machine to which a terminal with operation mode MDE is assigned, a maximum of 20 operations can be logged on at the same time. If more than 20 operations must be logged on at the same time, this limitation can be removed after a review by MPDV.

N Only 1 OP can be logged on

1...9 A maximum of n OPs can be logged on

Posting

Quantity posting to person

If selected, this function is used to post the quantity of order interruptions/ logoffs to the person who is logged on for the longest period.

Detailed information about quantity posting to persons can be found here.

Posting on OPs that are not logged on

If selected, a partial confirmation/upload, interruption or logoff for an OP can be performed on this machine, even without a previous logon.

Posting the machine time in connection with operations logged on at the same time

If set, the machine time for OPs logged on at the same time is posted as a proportion on the operations and persons:

J Proportionate posting on OP and person acc. to the number of OPs

N No proportionate posting. If the option is not set, every operation receives the complete machine time.

V According to the default quantity of the OPs. Here it must be ensured that the default quantity (target quantities in primary quantity unit) in the operation > 0.

Z According to the standard time of the OPs (available starting with ADE 7.3). Here it must be ensured that the standard time (processing time) in the operation > 0.

Please note:

This option is also evaluated for group workplaces; in general, the option should not be set for them.

Automatic logoff of personnel when shift ends

This identifier is only relevant if an "X" is set for the identifier of the same name on the order type. The configuration is carried out by MPDV while customizing the system. Otherwise it has no significance.

This identifier is used for the more detailed configuration of the personal data collection at MDE workplaces. Because fully automatic shift ends are generated by the terminals when using the HYDRA MDE, a setting can be made here regarding whether the people logged onto the workplace are automatically logged off at the end of the shift or if they remain logged on.

J Always log off staff when shift ends

N Always save staff when shift ends except for manual logoffs

X Evaluate the person's settings. Now the person will be searched for the corresponding setting

Automatic OP posting when shift ends

This configuration is only relevant in the context of the HYDRA customizing by MPDV and can only be used then. Otherwise it has no significance!

This identifier is only relevant if an "X" is set for the identifier of the same name on the order type.

J Interrupt and log on again at beginning of shift

N Interrupt

MPL

Further information about the HYDRA module MPL can be found in the corresponding MPL documentation.

Batch management

Activates the entry of the batch number for this machine within the posting dialogs on the terminal. Possible values are:

N No batch processing

L Batch tracing (input/ output batches) in the context of HYDRA MPL/MPL-RF

D Throughput batch processing in the context of HYDRA MPL/MPL-RF

J BDE batch management

The following functions are only available in connection with the Material and production logistics module and are supported only on the Windows-based terminals (CTWIN / AIP).

Preceding material buffer

Irrelevant.

→Subsequent material buffer

If a material buffer is specified in this field, the field *Target buffer* in each of the entry dialogs (e.g. output batch change, log operation off) is automatically populated with this value.

If no material buffer was entered in the entry dialog (e.g. deleted from the input field), the output batch is automatically posted to this material buffer.

Automatic generation of batch number

If this identifier is set, a batch number is automatically generated for the output batch to be produced. Otherwise during the operation logon or output batch change, the input of the batch number of the new output batch to be produced is expected.

Please note: If, in the field *Batch management* the setting D (= *Throughput batch recording*) is activated, the value for *Automatic generation of batch number* is automatically set to "J". In this case, manual entry of the batch number is not possible.

Consumption balance

During the OP logoff, an additional dialog (V_BLZ) is opened displaying the material components and their consumption quantities in relation to the current OP logon. In this dialog, the operator has the option to log off input batches that are still running. The option is only enabled, once the consumption balance has been activated for the material type of the output material.

Generate transport order for output batches

This option creates a transport order relating to batches for a generated output batch. The transport is started from the material buffer in which the output batch is produced. The option set here is overridden by the configuration of the relevant option within the material type.

Generate transport order for input material

This option creates an article-related transport order relating to a material component, when an operation is planned for a machine using the shop floor scheduling module. Transportation is started from the output material buffer of the preceding operation. The option set here is overridden by the configuration of the relevant option within the material type.

HLS

Further information about the HYDRA module HLS can be found in the corresponding HLS documentation.

Planning function

This identifier specifies whether or not a workplace or a machine will be displayed and if so, in which MOC planning function.

P	Planning in the graphic planning board of the HYDRA shop floor scheduling (HLS) or in the graphic order sequencing (GAV), i.e. the workplace is planned using the HLS or the graphic order sequencing and for this reason it is visible there, but not in the order sequencing table (AVG). Please note: Whether or not a workplace is displayed in the HLS or the graphic order sequencing depends on other settings as well: - Assignment to a group identified as a "capacity group" - Responsibility area authorization for this workplace - Planning profile

H Only relevant when using the HYDRA shop floor scheduling module (HLS).

Same as P. Furthermore, the workplace is also considered in the automatic (server-based) planning. The activation of the automatic (server-based) planning can only be performed while customizing HYDRA.

T The workplace is only considered in automatic, server-based planning; in the graphic planning board the workplace is not displayed. The activation of the automatic (server-based) planning can only be performed while customizing HYDRA.

A Planning in the order sequencing table (AVG), i.e. the workplace is planned using the AVG module.

N No planning; the workplace is displayed in neither the order sequencing AVG table nor in the graphic order sequencing nor in the HLS module.

Planned year model

A special year model used only for planning in the HYDRA shop floor planning (HLS) can be entered here. It does not affect entry and posting in the ADE/MDE module. If no planned year model is defined, then the BDE year model (*Master data* tab) is used for the planning.

Availability

The available capacity of a workplace/machine can be defined here. The default value for the available capacity is 1000 [per mill].

In the HYDRA shop floor scheduling, the capacity check and automatic assignment assume that each operation has a capacity requirement of 1000 [per mill], i.e. only one order/ operation can run on the workplace/machine at a time. In case of a manual multiple assignment, a dialog informs the user about the double assignment; automatic assignment always assumes a single assignment.

This setting can be used to extend the availability of the workplace such that a multiple assignment is permitted. If the workplace capacity allows, for example, processing of two operations at the same time, the available capacity should be set to 2000 [per mill] in this field.

If nothing is entered in this field or if the value 0 is entered, the system interprets this as the standard value of 1000 [per mill].

Utilization of this function is subject to the corresponding license and requires further customizing services by MPDV, depending on the relevant field of application.

Quantities tab

This tab is only available if a resource of the type "MNR" is selected.

Conversion factors to basic quantity

At the machine or workplace, the quantities can be collected in various quantity types and accounts. In general, the following quantity accounts are supported:

Yield

Scrap

Rework (Windows terminal CTWIN/AIP only)

Open quantity (problem quantity; Windows terminal CTWIN/AIP only)

The following **quantity types** are supported for each quantity account:

Primary quantity

Secondary quantity (Windows terminal CTWIN/AIP only)

Tertiary quantity (Windows terminal CTWIN/AIP only)

Basic quantity (Windows terminal CTWIN/AIP only)

The actual use of several quantity types or accounts depends on the system design. For example, for manual entry of rework quantity, a corresponding input field must be configured in the entry dialog (by customizing).

Automatic quantities can only be entered in the "primary quantity" unit.

Quantity units and conversion factors for base quantity

A quantity unit is defined for each quantity type. The alternative quantity accounts can be entered directly (manually). If this is the case, automatic conversion is not carried out.

If the alternative quantity accounts are not entered manually, a conversion into the alternative accounts is carried out on the server based on the conversion factors or the units configured in the MOC machine master data.

Basis for HYDRA-MDE quantity conversion

The basis that will be used for quantity conversion is set here.

A Use the conversion factors of the OP that is logged on. If no operation is logged on, the quantity conversion from the machine/workplace configuration is used.

M Use conversion factors from the workplace configuration for quantity conversion.

Units and conversion factors for base quantity (P)

Quantity unit (P)

Indicate the quantity unit in which the entry at this machine/ workplace is primarily carried out. In case of automatic entry of quantities, these quantities are generally viewed as primary quantities.

If an automatic quantity conversion into another quantity type is to be carried out, indicate the conversion factors to basic quantity here.

Units and conversion factors for base quantity (S)

Quantity unit (S)

Here indicate the secondary quantity unit in which quantities are to be posted to the workplace/ machine. If an automatic quantity conversion is to be carried out, indicate the conversion factors to basic quantity here.

Units and conversion factors for base quantity (T)

Quantity unit (T)

Here indicate the tertiary quantity unit in which quantities are to be posted to the workplace/ machine. If an automatic quantity conversion is to be carried out, indicate the conversion factors to basic quantity here.

Units and conversion factors for base quantity

Quantity unit (B)

Here indicate the base quantity unit in which quantities are to be posted to the workplace/ machine.

Manual entry of quantities, yield

Manual entry of yield

This option should be set if quantities are to be entered manually, if allocation with another quantity account is to be implemented or if the manual quantities are to be posted as cycles.

On Windows-based terminals this option does not affect the quantity fields displayed in the entry dialogs; modifications to these quantity fields are to be made using dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of yield

Requirement: The option "Manual entry" must be set.

Using this option, manually entered quantities can be offset against other quantity accounts. In this case, the quantity entered is deducted from the account with which the allocation is to be implemented.

Note here that negative values can also occur due to the respective quantity conversion.

Please note

This option may NOT be set for DOS terminals if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of yield as cycles

Requirement: The option "Manual entry" must be set.

If this option is set, manually entered quantities are posted simultaneously as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

Manual entry of quantities, scrap

Manual entry of scrap

This option should be set if quantities are to be entered manually, if allocation with another quantity account is to be implemented or if the manual quantities are to be posted as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the entry dialogs; modifications to these quantity fields are to be made using dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of scrap

Requirement: The option "Manual entry" must be set.

Using this option, manually entered quantities can be offset against other quantity accounts. In this case, the quantity entered is deducted from the account with which the allocation is to be implemented.

Note here that negative values can also occur due to the respective quantity conversion.

Please note

This option may NOT be set for DOS terminals if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of scrap as cycles

Requirement: The option "Manual entry" must be set.

If this option is set, manually entered quantities are posted simultaneously as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

Manual entry of quantities, rework

Manual entry of rework quantity

This option should be set if quantities are to be entered manually, if allocation with another quantity account is to be implemented or if the manual quantities are to be posted as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the entry dialogs; modifications to these quantity fields are to be made using dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of rework

Requirement: The option "Manual entry" must be set.

Using this option, manually entered quantities can be offset against other quantity accounts. In this case, the quantity entered is deducted from the account with which the allocation is to be implemented.

Note here that negative values can also occur due to the respective quantity conversion.

Please note

This option may NOT be set for DOS terminals if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of the rework quantity as cycles

Requirement: The option "Manual entry" must be set.

If this option is set, manually entered quantities are posted simultaneously as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

Manual entry of quantities, open quantity

Manual entry of open quantity

This option should be set if quantities are to be entered manually, if allocation with another quantity account is to be implemented or if the manual quantities are to be posted as cycles.

On Windows terminals this option does not affect the quantity fields displayed in the entry dialogs; modifications to these quantity fields are to be made using dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of open quantity

Requirement: The option "Manual entry" must be set.

Using this option, manually entered quantities can be offset against other quantity accounts. In this case, the quantity entered is deducted from the account with which the allocation is to be implemented.

Note here that negative values can also occur due to the respective quantity conversion.

Please note

This option may NOT be set for DOS terminals if yield is offset against scrap or scrap is offset against yield in the counter configuration.

Posting of open quantity as cycles

Requirement: The option "Manual entry" must be set.

If this option is set, manually entered quantities are posted simultaneously as cycles. Note here that the entered quantity is posted 1:1 as a cycle (partitioning is not considered).

MDE configuration tab

This tab is only available if a resource of the type "MNR" is selected.

Monitoring

Monitoring type

The following types of monitoring can be selected:

Monitoring by operating signal

No monitoring

Cyclical monitoring

If **cyclical** or **operating signal** monitoring was selected, a malfunction can only be entered if a request is made using the terminal ("Assign malfunction"). If **no automatic monitoring** is specified, a new machine status can be specified at any time.

In **cyclical** monitoring, when a counting pulse occurs, an automatic switch is made into the "Production" status. If operating signal was selected and when the operating signal is set, a switch is made into **Production**. If no automatic monitoring was selected, the "Production" status must be assigned manually.

→Entry of malfunction reason required with specified delay time in [s]

(Only with terminal type CT 73x, CT 83x, not with master terminal/DS-100)

If a downtime without a reason was recognized, after the specified delay time the input window for "Change machine status" opens automatically on the terminal. If the terminal goes back into production, the window still remains open.

If a machine status is input now (during production), this input activates a reposting event that reposts the most recently recorded status from "General disturbance" to the new status that was input. If the reposting is correct, the window closes; otherwise, it remains open.

However, if a downtime is recognized again (with or without a reason), the previously noted status can no longer be reposted. The window closes automatically.

If another downtime without a reason is recognized and the delay time has expired, then the input window opens as described above.

If a downtime without a reason is recognized and the machine switches into production before the delay time expires, then the automatic request for a malfunction reason is not carried out.

Important note:

This reposting only affects the HYDRA machine data collection; online correction of the resource performance accounts of the currently running OP is not possible !

Note regarding data maintenance:

All machine status modifications are displayed in the tabular event maintenance of MOC. However, the reposting event is locked and cannot be edited.

In order to perform recalculations correctly now with respect to orders and machines, the original event with the status "NOT ASSIGNED" must be changed to the correct status.

The reposting event does not affect the recalculation!

Minimum malfunction duration

If the **operating signal** monitoring type was selected, the duration that a malfunction must last until it is recognized and logged in as a malfunction is specified in seconds in this field.

Minimum cycle time

If **cyclical** monitoring was selected, a minimum cycle time can be specified in seconds in this field.

From this minimum cycle time and the target cycle stored in the (logged in) operation and compensated with the cycle extension, the terminal determines the maximum value and uses it as the cycle time specification.

If both the minimum cycle time and the target cycle stored in the operation are 0, the cycle time specification is set to 60000 seconds [per 1000 machine cycles].

Cycle extension

If **cyclical** monitoring was selected, the percentage for extending the target cycle time must be input here in a range from 0 to 5000.

The target cycle stored in the (logged in) operation is compensated with this percentage. In this way, a value less than 100 indicates a shortened cycle; a value greater than 100 indicates an extended cycle.

Number of target cycles

If **cyclical** monitoring was selected, here the number of cycles (0 to a maximum of 9) can be indicated according to which the terminal automatically switches from a status not equal to production into production status <u>within the cycle time</u> (prerequisite: no production lock is currently set in the status not equal to production).

With some production processes, there are machine cycles in set up phases. By setting a value greater than 0, the current machine status does not change immediately. Please note: Until a switch is made into production, the quantities entered in this case are neither posted to the yield nor the scrap quantities.

Cycles to be evaluated

This entry should be populated with 0.

Note regarding data maintenance:

All machine status modifications are displayed in the event maintenance dialog. However, the reposting event is locked and cannot be edited.

In order to perform recalculations correctly now with respect to orders and machines, the original event with the status "NOT ASSIGNED" must be changed to the correct status.

The reposting event does not affect the recalculation!

Administration

Posting during prod. lock

This setting can be used to set the type of posting of the counting pulses collected during the so-called production lock. This configuration takes effect with all counters configured as "Yield".

Posting as scrap

If the counting pulses were configured on the counter, they are offset against the partitioning/ pulse factor and posted as scrap. However, any defined allocations with another quantity account are not carried out in this case.

Posting as yield parts

Posting of the counting pulses as yield

No posting

No quantities are posted during the production lock.

→→→Pulse factor specific to machines

The pulse factor is used, for example, if lengths (e.g. from a wheel) are to be collected.

For machines for which a discrete or integral number of quantities (e.g. pieces) are entered per pulse, the value should always be set to 0. In this case, the pulse factor is not used for the evaluation. That means that the posting of the number of cycles corresponds with the actual pulses transferred via the MSS (machine interface).

The signals transferred from the machine (counting pulses) are collected by the MSS. The quantity is calculated and posted as follows according to the configured number of pulses:

Quantity for the machine = pulse * partitioning for the machine/ pulse factor for the machine

Quantity for the operation = pulse * partitioning for the operation/ pulse factor for the operation

Please note: The pulse factor will be evaluated as a fraction. In this way, in the quantity calculation the pulse is included as a denominator while the partitioning is considered a numerator.

Pulses that occur during a malfunction or a production lock (configuration of *Posting during prod. lock* > scrap) are interpreted as scrap. The scrap quantities are also calculated using the formula given above.

Partitioning specific to machines

The partitioning specific to the machine is indicated here. This is included in the quantity calculation in a multiplication with the partitioning entered in the operation. If this is not desired, the value 1 must be input here.

Extended weekend automatic

If this option is selected and with the corresponding configuration the status that was available before status 999 was activated will be assigned at shift start.

Please note:

To set the option, the workplace must already be assigned to a terminal.

Detailed information about the automatic activation of status 999 can be found in the chapter *Day types* .

Wait. period short-term disturb.

To improve the overview, e.g. in machine history, a short-term malfunction status can be configured per machine/ workplace. In the context of machine monitoring this status serves as a "container" for any unconfirmed statuses that existed only for a certain (short) period of time.

If a downtime is automatically recognized on the terminal and the machine automatically goes back into production, a check is made regarding whether or not this disturbance is shorter than the time period configured here for short-term malfunctions.

If this is the case, the malfunction, which does not yet have a reason, is given the status (reason) configured on the machine as the status for "short-term malfunctions".

Inputs/ outputs

Machine lock/ Target quantity reached/ Machine downtime/ Free I/O

The terminal outputs used for production statuses are indicated in these fields

Machine lock output	is set if the status "not assigned" occurs or a status in which the option "machine lock" is set.
Target quantity reached output	is set if the target quantity of the OP is reached.
Machine downtime output	is set if a status not equal to P**roduction** was recognized for the machine**.** When changing into production, the output is immediately reset to 0.
Free I/O	Free input/ output for customer-specific customization.

These statuses can be used for connecting a monitoring light or a horn, for example.

The assignment of an output by entering the corresponding number in one of the fields specifies which relay is interconnected by the terminal when the predefined status occurs. If "0" is entered, no action occurs. Note that an output on a terminal may not be assigned more than once.

Please note

- The specification regarding the status in which the machine lock is to be set is carried out in the context of the **Status assignment**.

- When the machine lock is activated using the available relay output of a DS 100, in general the value "1" is to be entered in the input field. In this case, the machine lock is set if a correspondingly configured status occurs <u>and</u> in the status <u>not assigned</u>.

Output batch change

Customer-specific assignment of an input with an automatic output batch change (MPL). As a default, the field should be left at 0.

PDE (Process Data Collection)

Collect process data

This parameter specifies whether or not process data are recorded for this machine. Process data cannot be recorded for this machine if this parameter is not set at a machine.

External connection

External connection

If this machine is assigned to a master terminal, the following selection of connection options is available:

Arburg control system	Arburg connection (only available if HYD-ALS is licensed)
Engel interfacing	Engel machine connection
No external device	No connection of an external device
DS100	DS100 connection
MT3	MT3 connection
PDE	Process data collection

If a DS100 or MT3 connection was activated, the device address field can be selected. If the option Engel interfacing is selected, the serial number field can be selected. The option Arburg control system enables the class field.

Note regarding the combination of connections on a master terminal:
"DS 100" and "No external device" allowed
"MT 3" and "No external device" allowed
"MT3" and "DS 100" not allowed!

Serial number (Engel interfacing)

The serial number of the connected Engel machine is entered here. A prerequisite for using Engel machines is the setting "EMS machine interface" in the HYDRA basic settings..

Device address

This field can be selected if a DS100 or MT3 connection was selected. The device address of the sub-bus participant is entered here.

Resource configuration tab

Resource master data

Type

Identifier regarding the type of resource:

Resource: A resource can be uniquely identified but is also actually present. It always has the number 1.

Anonymous resource: An anonymous resource cannot be uniquely identified. If the identifier is set, then the value in the field *Number* can be changed from 1 to another positive integer value. Anonymous resources cannot be posted because the actual reference to precisely one resource is not possible. Please note the information in the chapter *Anonymous resources*.

Required resource: A required resource is a substitute for one or more actual resources that can be identified. The resources that a required resource represents are specified in the configuration *WRM: Master data* > *Required resources*. The number results from the number of actual resources assigned to it.

Please note: If this field is empty, the resource is implicitly an ("actual") resource.

Equal type

Reserved for future use.

Version

Modification number; the program version can be stored here for resources of type DNC.

Number

This field can only be edited if it contains an anonymous resource and the identifier *Anonymous resource* is set (see above). A value > 1 indicates how many of these resources are available.

This field is automatically calculated for required resources.

Resource family

Assignment of a resource family. If the resource family is subsequently changed, an information dialog appears as a warning because user fields can possibly be assigned with the resource family.

Target utilization

Cycles

The target cycles serve as additional information regarding how long the resource is to be used.

Runtime

The target runtime serves as additional information regarding how long the resource is to be used.

Configuration

Target cycle

Target duration in seconds for 1000 machine cycles if this tool is used.

Please note: The target cycle stored in the OP is relevant for the planning in the HLS module and for the machine data collection.

Original partitioning

Partitioning of the tool (= multiplicity or number of cavities) when using this tool.

Current partitioning

Current partitioning of the tool. This partitioning can deviate from the original partitioning, e.g. if the original quantity can no longer be produced with a cycle due to a tool defect.

The posting of cycles to the tool is always carried out based on the current partitioning.

Please note: The partitioning stored in the OP is relevant for the planning in the HLS module and for the machine data collection.

Partitioning due to cavities

Setting the flag "partitioning due to cavities" causes the system to (re-)calculate the fields "current partitioning" and "original partitioning" using the assignments in cavity management. Then the fields can no longer be changed manually.

Log on with OP

This identifier is used to control whether or not the resource is logged on if it is assigned as a component in the list of production resources and tools for the operation. Possible values are:

None: The resource is not logged on.

Implicit: The resource assigned to the operation as a production resource and tool is automatically (implicitly) logged on with the operation; an explicit logon or change in logon status is not possible.

Explicit: The resource assigned to the operation as a production resource and tool can be logged on explicitly or another resource can be logged on instead. If the resource is not explicitly logged on or if no other resource is explicitly logged on, the current resource is implicitly logged on; in this way, the current resource serves as a "default".

Please note:
If another resource is explicitly logged on, it will be logged on for the resource that has the same **resource type** in the list of production resources and tools of the operation. For this reason, an explicit resource logon is only possible for resources that are included in the list of production resources and tools of the operation as a requirement. In this way, no resource can be logged on for which there is no requirement (= entry) in the list of production resources and tools.

In general, this identifier should be inactive for resource type DNC because a specific processing exists for it in the HYDRA module DNC (NC programs are logged on separately).

Resources that are defined in the BOM of the machine are also logged on.

Parallel logon/ planning possible

The tool can be logged on/ planned in parallel.

Caution: A resource can be logged on multiple times to only one machine. Consequently, the identifier "Parallel logon possible" refers to several different OPs on one machine.

Post to resource

Indicates whether or not the quantities and times are to be posted to the resource. Due to a high degree of complexity, this identifier should only be provided for those resources that will actually be evaluated later.

Planning

Setup time

Duration in hours for setting up the tool.

Please note: The setup time stored in the OP is relevant for the planning in the HLS module.

Retooling time (teardown)

Duration in hours for removing the tool.

Please note: The retooling time stored in the OP is relevant for the planning in the HLS module.

Assignment

No processing. The configuration option of the same name stored in the resource type is used for taking the resource allocation in the HYDRA shop floor planning into consideration.

Evaluation

Consider in evaluations

Reserved for future use.

File

File exists

Shows whether or not the file is stored in the specified path. The files are checked by a cyclic process and the corresponding flags are set subject to whether or not the file is available.

File name

File name; without file extension for DNC. The file extension is added based on the configuration in the resource type. The defined paths specify the storage location.

Comparison resources

Two comparison resources may be indicated here for energy consumption resources. They will then be shown in comparative evaluations/reports, e.g. the energy monitor.

Resource 1

Resource number of the resources to be compared

Resource type 1

Resource type of the resources to be compared

Resource 2

Resource number of the resources to be compared

Resource type 2

Resource type of the resources to be compared

Accuracy

More detailed information on measuring accuracy and measuring range may be entered here for test equipment resources.

User fields tab

User fields offer the possibility to store further customer-specific information to MES besides the available fields in MOC standard. The tab provides eight sub tabs each of which providing eight user fields. The so called user field key determines, which user fields are involved and which meaning they have.

Object type

User fields are configured for the relevant resource type, e.g. MNR for the workplace/machine.

User field key

Every user field key describes a combination of user fields. How the user field key is managed (and thus the meaning of the fields) varies for the individual objects. User field keys are defined together with the customer while customizing the system.

User fields

The following user fields are available after customizing the system:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6

Field data type	Number of fields
Text field, length 20	14
Text field, length 40	2

A maximum of 8 fields are shown for each page.

User field keys are not defined by default in the system. The system has to be customized respectively in order to be able to support this kind of user fields.

Comment tab

The Comment tab allows additional comments about the resource to be stored.

Main tab “resource attributes”

Additional resource attributes are shown using the user field definitions of the resource family. The "resource attributes" button is used for editing.

Resource list main tab

Shows the resource list for the selected resource. By clicking the "resource list" button you can directly go to the BOM application for editing purposes.

Toolbar

General tab

Insert

Opens the dialog for adding resource lists

Copy

Opens the dialog for copying resource lists.

Edit

Opens the dialog for editing resource lists.

Delete

Deletes one or several resources.

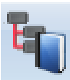
Resource tab

Configuration – resource status

Opens the "resource status" application

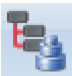
File - show file

Shows the file - only available for document resources configured as file-based resource without DNC processing in the resource type and if the corresponding license and function authorizations are available.



Go to - resource list

Opens the "resource list" application. The selected resource is entered as default value for the superior resource.



Go to – required resources

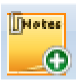
Opens the "required resources" application. The selected resource is entered as default value for the required resource.

Go to – cavity assignment

Opens the "cavity assignment" application. The selected resource is entered as default value.

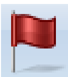
Go to - resource attributes

Opens the application "resource attributes". The selected resource is entered as default value.



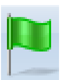
Functions – measures

Opens the "measures" application.



Functions – status change

Opens the dialog for changing the resource status.



Functions – release of resource

Opens the dialog for releasing a resource.

Functions – stock transfer

Opens the dialog for transferring/relocating a resource

Workplace tab


Configuration – status assignment

Opens the application "status assignment". The selected resource is taken over.


Configuration – counter configuration

Opens the application "counter configuration". The selected resource is taken over.


Configuration – terminal assignment

Opens the application "terminal assignment". The selected resource is taken over.


Entry – reasons

Opens the application "reasons". The selected resource is taken over.


Entry – Operator positions

Opens the application "operator positions". The selected resource is taken over.


Entry – premium indicator

Opens the application "premium indicator". The selected resource is taken over.


Groups - groups

Opens the application "groups". The group of the selected resource is taken over.


Groups – group assignment

Opens the application "group assignment". The selected resource is taken over.


Other – cycle parameter

Opens the application "cycle parameter". The selected resource is taken over.


Other - workforce requirements of workplaces

Opens the application "workforce requirements of workplaces". The selected resource is taken over.

DNC tab

The tab is only available if a DNC resource is selected. These are resources configured as resources with DNC processing in the resource type.


Configuration – resource status

Opens the "resource status" application


Configuration - assignment of DNC family to machine

Opens the application "assignment of DNC family to machine".


File - comparison editor

Opens the comparison editor for the selected resource or resources. Please see below for further information.


File - export

Exports the file entered for the resource. The target file is input using the file explorer.


File - import

Imports the file entered for the resource. The source file is selected using the file explorer.


File - viewer

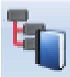
Opens the file entered for the resource for viewing using the defined viewer program.


File - editor

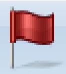
Opens the file entered for the resource for editing using the defined editing program.


Go to - resource attributes

Opens the application "resource attributes". The selected resource is entered as default value.

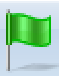

Go to - resource list

Opens the "resource list" application. The selected resource is entered as default value for the superior resource.



Functions – status change

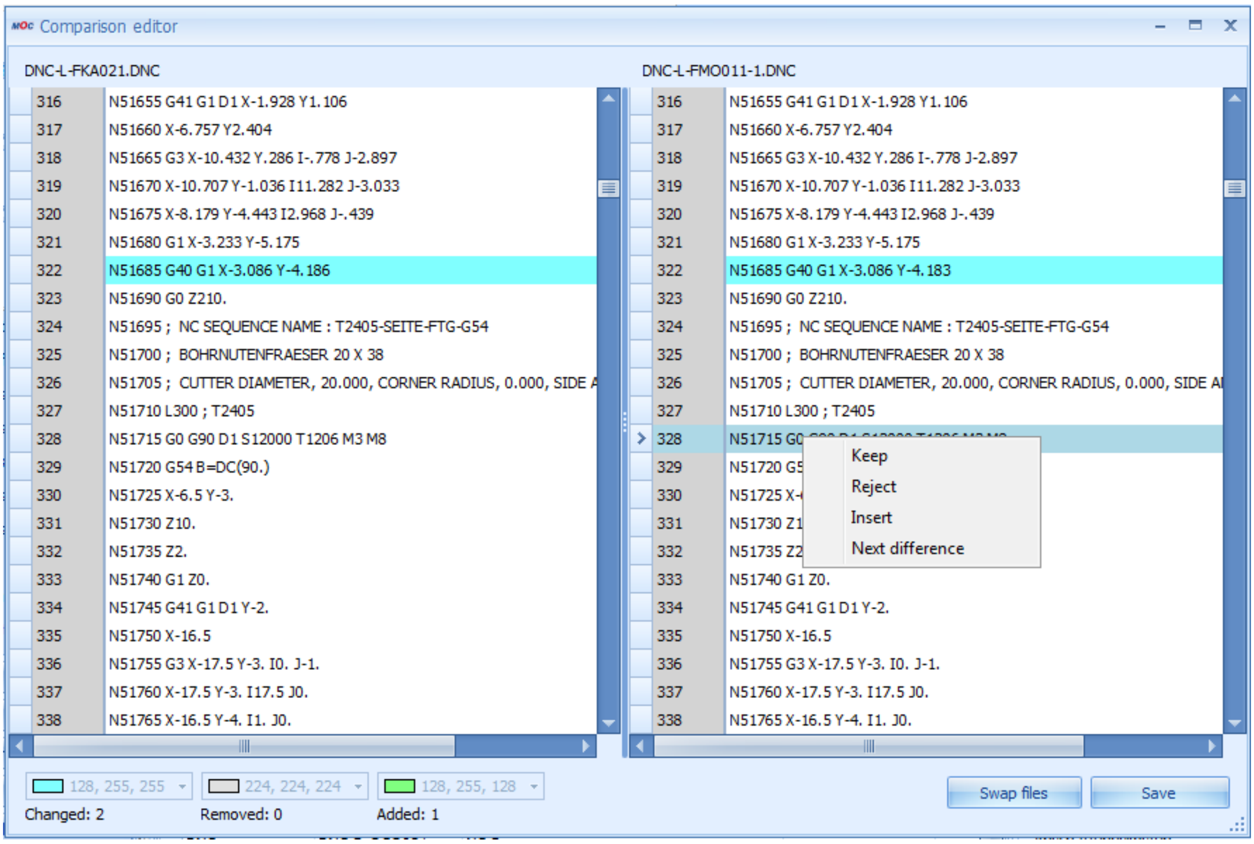
Opens the dialog for changing the resource status.



Functions – release of resource

Opens the dialog for releasing a resource.

Operation of the comparison editor



The comparison editor compares the files attached to the DNC resources. There are two operating modes:

Selection of one resource:

The released resource and the optimized version of the resource are displayed for comparison. The file entered on the right-hand side of the editor can also be changed. Once the changes have been made, they are taken over to the system, just as it is also the case for the simple editor. This mode can only be used for DNC types with the file processing type "optimized".

Selection of two resources:

If two resources are selected, the editor compares the two selected resources. The file type may be selected. The file entered to the right of the editor may also be changed. Once the changes have been made, they are taken over to the system, just as it is also the case for the simple editor.

The functions of the comparison editor can be opened using the relevant buttons or by clicking the right mouse button (context menu):

- Reject: The detected difference (on the right) is rejected. The value from the left file is accepted. The difference is no longer selected.
- Keep: The detected difference (on the right) is accepted. The difference is no longer selected.
- Next difference: goes to the next difference.
- Insert: Inserts a row at the current position.
- Contents of a row can always be changed by clicking the row and inputting values. The row can be left by clicking Esc without making any changes. The row is then highlighted as "changed".
- Swap windows: this button allows for windows to be swapped. This function is necessary if two resources are compared with each other. Their order results from the display order in the table and the system does not know which resource needs to be changed. This button is not available if only one resource is selected. For only the optimized program version can be changed.
- Save: Saves the changes made to the file on the left-hand side.

Processing notes for workplaces and machines

Configuration modifications

In order that the settings or modifications made can be interpreted by the terminal shop floor program, the terminal to which the workplace/ machine is assigned must be restarted.

Deleting a machine/ workplace

If data is already recorded and saved for a machine/ workplace, the relevant configuration data of the machine (including the data from the status assignment, the maintenance calendar and the process parameters in the MDE module, among others) is automatically deleted as well.

The previously existing movement data (events, postings) are not automatically deleted with this action; instead, it is deleted later in context of the specified deletion cycles (default: 35 days).

→←

9 Resource Families

Summary

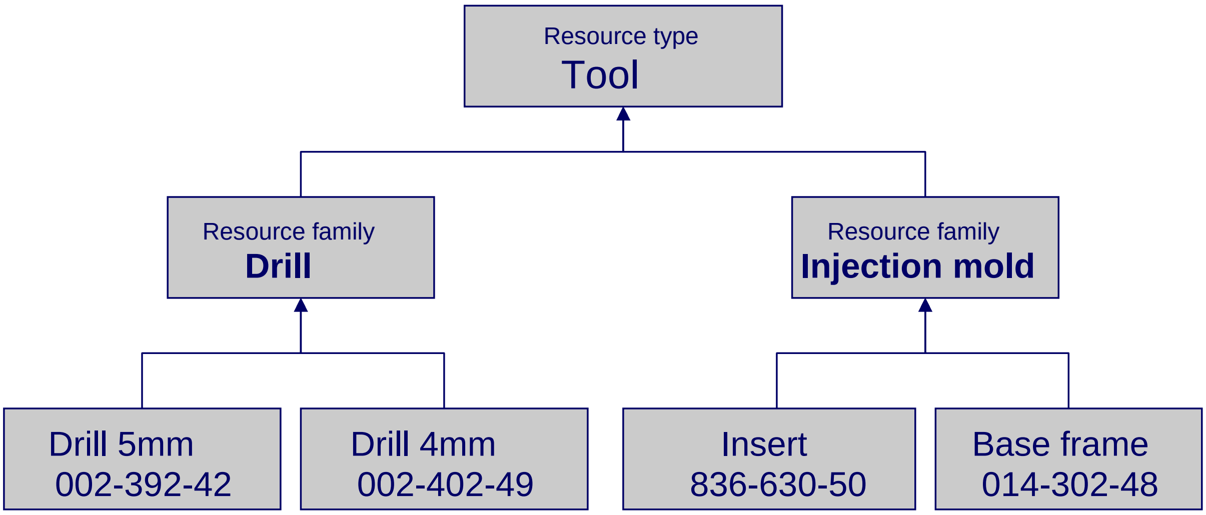
Menu	Master data → Resources → Resource families
Transaction code	resfam.*
Function authorization	mdrfam

This document describes the application "Resource families" within the Manufacturing Operation Center (MOC).

Usage

If the assignment between resource and resource type is assumed, it is clear that in one production company, different resources of the same resource type exist, which also are handled differently. That means that in general classification according to resource type is not sufficient for ordering the resources in a useful way.

By defining so-called resource families, a sub classification of resource types can be created. The following graphic shows an example of how the resource type "Tool" is divided into two resource families, "Drill" and "Injection mold". Each of the individual resources is assigned to one of the two resource families.



Integration

The resource families offer another structural level subordinate to the resource types. The function of the master-detail user fields of the resources that can be defined using the resource types can be refined by definition in the resource families. The resource families are used in particular in the DNC area as a main search criterion and as an assignment criterion on machines.

Selection parameters

In the selection panel, filters can be used for both superior and assigned resources. The following selection criteria are available in the respective application:

Resource type

Type of resource.

Resource family

Family to which the resource is assigned.

Field descriptions

Resource type

Resource type to which the resource families refers

Resource family

Unique "descriptive" designation of the resource family.

This value can be searched for (selected) in the various functions. Because a resource or its resource ID can only be uniquely identified within the resource type, in the evaluations the resource type of the resource is always displayed as well.

Description

Explanation of the resource family; functions as a comment.

Responsibility area

Definition of the responsibility area. By specifying a responsibility area for a family, the responsibility area for the assigned resource is also specified, which controls its visibility and editing options.

Field description for the *General* tab

User field key

Reference to a valid user field key. The specified user field key overwrites the specification in the resource type.

Note regarding DNC filtering using a DNC family and its search fields:

The definition of a suitable user field combination is important for using the flexible filter and search function in the DNC. The definition of such user field combinations is reserved for MPDV customizing. The assignment and use of this user field key is the user's responsibility. With the defined search fields, the DNC records are filtered on the terminal in addition to the DNC family of the machine and can be used as search criteria on the console.

Starting with release DNC 7.2, the following preconfigured user field keys will be delivered:

User field key	Description of the search fields
DNC_K	Plastic injection molding: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Tool, mandatory field, cannot be edited
DNC_K_V	Plastic injection molding: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Tool, mandatory field, cannot be edited 4. Version, mandatory field, can be edited
DNC_K_W	Plastic (tool reference only): 1. Tool, mandatory field, cannot be edited
DNC_K_WV	Plastic (tool reference and version): 1. Tool, mandatory field, cannot be edited 2. Version, mandatory field, can be edited
DNC_NC	NC programs: 1. Article, mandatory field, cannot be edited
DNC_NC_V	NC programs: 1. Article, mandatory field, cannot be edited 2. Version, mandatory field, can be edited
DNC_NC_M	NC programs 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited
DNC_NCMV	NC programs: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Version, mandatory field, can be edited
DNC_FREI	1. Search field 1, Text20, mandatory field, can be edited

	2. Search field 2, Text20, optional field, can be edited 3. Search field 3, Text20, optional field, can be edited 4. Search field 4, Text20, optional field, can be edited

Note regarding DNC administration

DNC records are used exclusively on machines. To prevent errors in input and assignments, a fixed assignment of a DNC resource family to every machine is carried out and saved in the data of the machine resource (field *Resource family DNC*). In this way, it can be ensured that only programs of one resource family and, indirectly, of one resource type can be loaded on the machine.

Furthermore, for DNC records administration criteria are needed that make selection and evaluation easier, so that identification and processing are also easier, and that enable checks. Because different machine types (e.g. injection molding machine, printer, NC machines) can be considered with the DNC administration, a strict specification of these criteria is not useful. For this reason, there is a resource family for which attributes are stored using the user fields; in turn, these attributes describe and specify the variable parameters.

The attributes are used for identification and can be provided with plausibility functions and assignments in order to create a context for the DNC programs with machines and operations (see the "User fields" chapter).

There are parameters, such as temperature or humidity that affect the behavior of the machine and can therefore influence production. These "environmental factors" can also be collected. To do this, further attributes just have to be defined in the user fields.

10 Inspection Stations

Summary

Menu	Master data → Quality management → Inspection stations
Transaction code	ista
Function authorization	ista

The inspection station catalog has been designed to categorize characteristics.

Utilization

The "inspection station number" field is the key field, i.e. while saving a new inspection station, the system checks whether there is already an inspection station with this key information.

The input of measures is easy to handle. Only an inspection station number and a corresponding designation have to be assigned.

By assigning an inspection station to a characteristic, it can be achieved that inspection orders are generated that only include the characteristics of an inspection station. This depends on a configuration option within the inspection plan header. In case inspection orders are generated for each inspection station, this information may, in turn, be used as filter for the inspection orders provided at the AIP inspection terminal. For this purpose, the terminal has to be configured accordingly.

Provided that the inspection plan header defines that inspection orders are not to be generated for each inspection station, the inspection station assigned to a characteristic is only a supplementary information.

By differentiating between active and inactive inspection stations, it can be defined whether or not they are still to be available in the selection lists for inspection stations in the later data acquisition process. Inactive inspection stations can be reactivated at any time.

Integration

Inspection stations are used in all applications dealing with characteristics. Moreover, the terminal configuration allows for inspection stations to be assigned.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately.

The "inactive" filter field allows for the data set to be restricted to active or inactive inspection stations.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies inspection stations that are no longer to be used in the active data acquisition process.

Toolbar

There are no other special function buttons in addition to the standard functions.

11 Customers

Summary

Menu	Master data → Quality management → Customers
Transaction code	cto
Function authorization	cto

The customer catalog has been designed to edit/keep customers. Provided that there is an interface to a higher-level system (e.g. ERP system), customers can be created automatically via interface. As soon as a new customer is created or changed in the ERP system, for example, the customer data record is automatically created or changed in the customer catalog including the defined information.

Utilization

The customer number uniquely identifies customers in all QM applications that refer to the customer catalog. The customer catalog is used as basis in particular for inspection requirements of production and and the goods issue as well as for the complaint management.

The “customer number” field is the key field, i.e. if a new customer is saved it is verified whether a customer with this key information exists already.

By distinguishing between active and inactive customers, it may be defined whether or not they are available in certain selection lists. Thus, no complaint can be created for an inactive customer, for example. However, inactive customers may be evaluated at any time. Moreover, inactive customers can be reactivated at any time.

Extensive address and contact data can be defined in addition to the customer number and the designation.

If a customer is designated as “party in charge” they will be included in the selection list for the parties in charge. Such selection lists are integrated in different detail applications. The list of the parties in charge is accessed mainly in the complaint management function and when measures are generated.

Integration

Customer data is a global catalog that is used in many QM applications. The below list shows the applications referring to the customer catalog.

- External people
- Departments

- Production inspection planning
- Production inspection requirements
- Complaint management
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2, and 3 are available in addition to the customer number and customer name.

The active or inactive customers can be restricted using the filter field “inactive”.

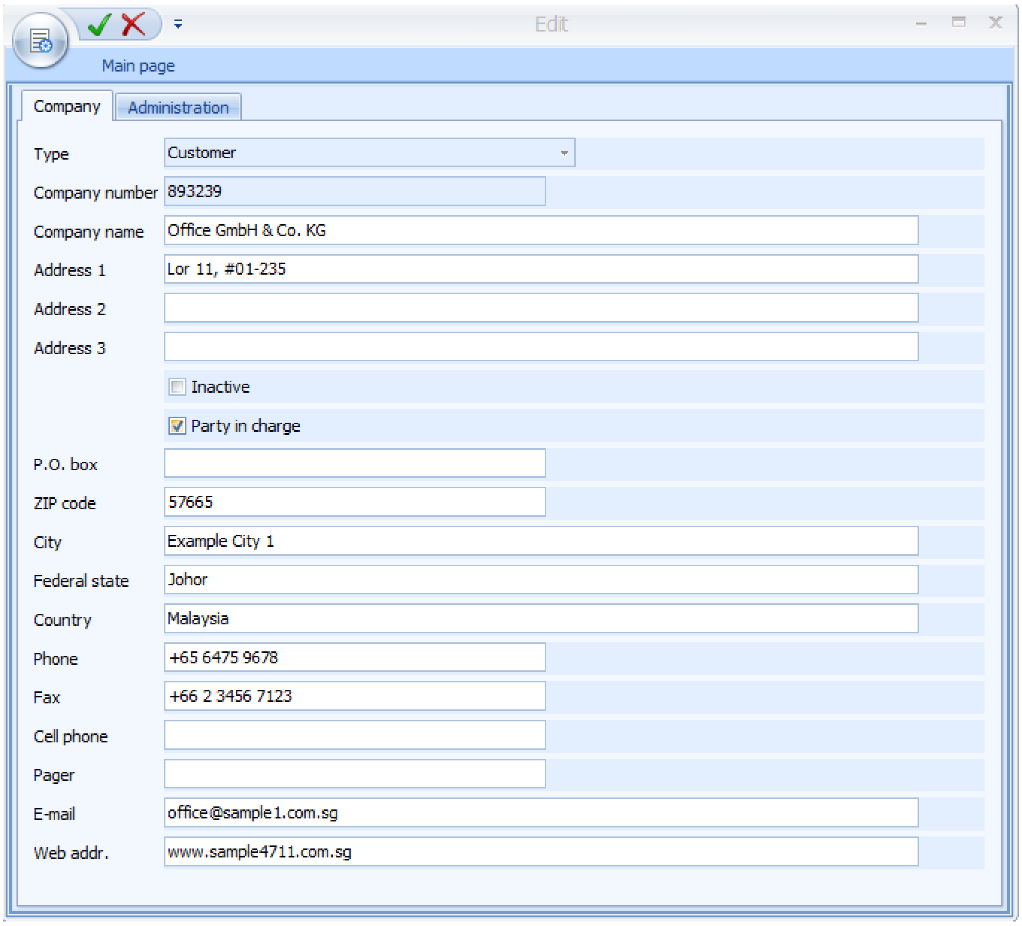
Field Descriptions

The available fields are self-explanatory and are not described separately, except for the address fields.

The content of the individual address fields is not specified by default and, as a result, may be defined by the user. Normally, address field 1 should include further details on the company, e.g. “site X”. As there is no field defined for the street, address field 2 or 3 (to be preferred) is to be used for the street including street number.

Editing functions

The below dialog opens to edit a data record.



Toolbar

There are no other special function buttons in addition to the standard functions.

12 Manufacturer

Summary

Menu	Master data → Quality management → Manufacturer
Transaction code	mft
Function authorization	mft

The catalog of manufacturers has been designed to edit and update the list of manufacturers. Provided that there is an interface to a higher level system (e.g. ERP system) the manufacturers can be created automatically by interface. As soon as a new manufacturer is created or an existing manufacturer is changed in the ERP system, the corresponding manufacturer record is automatically created or changed with the defined information in the catalog of manufacturers.

Utilization

The manufacturer number uniquely identifies the manufacturers in all QM applications that refer to the catalog of manufacturers. In particular, the inspection requests of the goods receipt refer to the catalog of manufacturers.

The "manufacturer number" field is the key field, i.e. while saving a new manufacturer, the system checks whether there is already a manufacturer with this key information.

By differentiating between active and inactive manufacturers, it can be defined whether or not they are still to be available in certain selection lists. Consequently, it is, for example, impossible to create a goods receipt inspection request for an inactive manufacturer. However, it is possible to evaluate inactive manufacturers at any time. Moreover, inactive manufacturers may be reactivated at any time.

Extensive address and contact data may be defined in addition to the manufacturer number and designation.

If the manufacturer is designated as being a "party in charge" by clicking the corresponding field, the manufacturer is included in the selection list for the parties responsible. Such selection lists are integrated in different detail applications. The list of the parties responsible is mainly used within complaint management and for the creation of measures.

Integration

Manufacturer data is a global catalog that is used in many QM applications. Please find below some applications that refer to the manufacturer catalog.

- External people

- Departments
- Goods receipt inspection planning
- Goods receipt inspection requests
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2 and 3 are available in addition to the manufacturer number and the designation.

The "inactive" filter field allows for the data set to be restricted to active or inactive manufacturers.

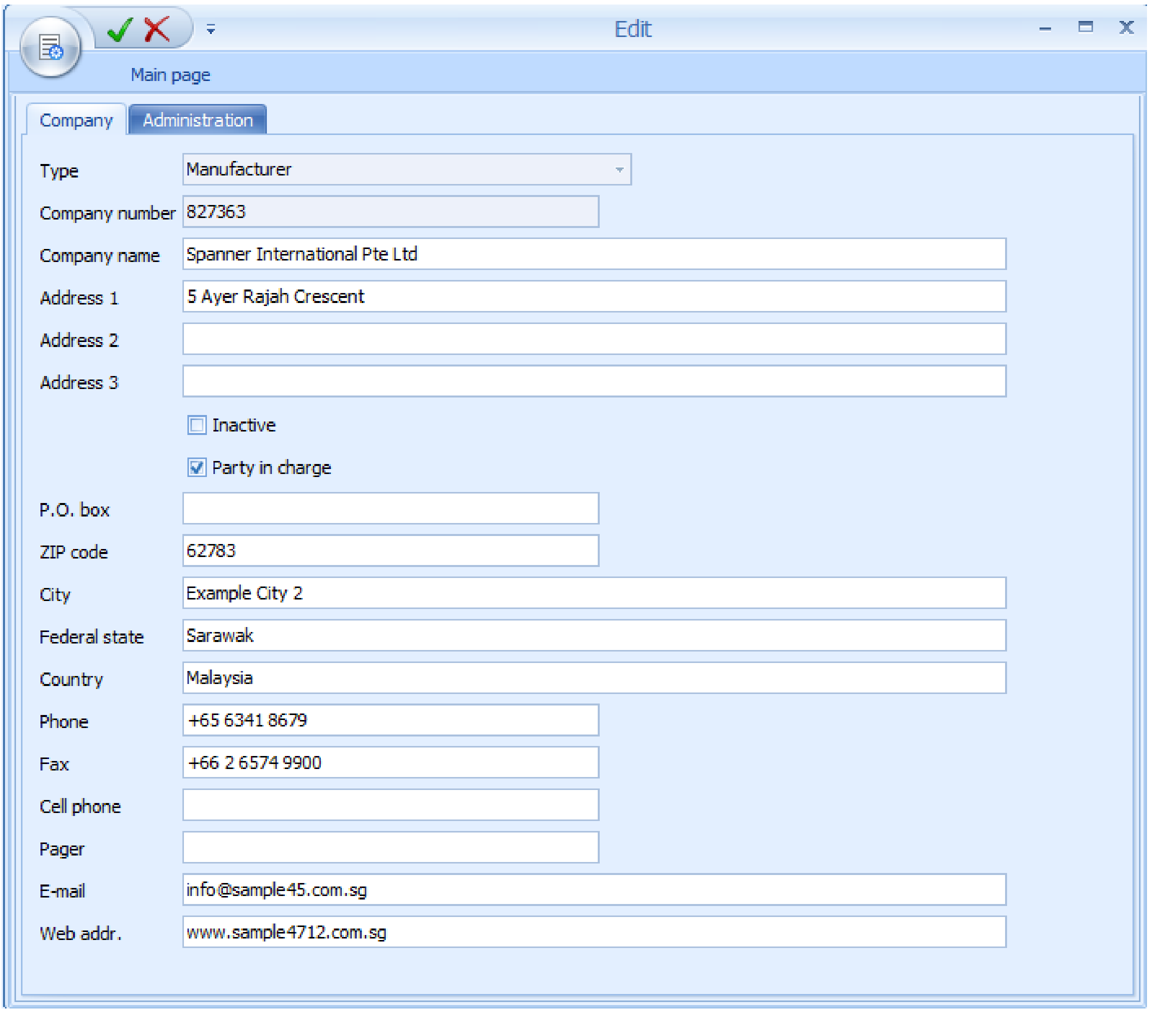
Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

The content of the individual address fields is not specified and, as a result, may be defined by the user. Normally, the address field 1 should include an addition to the company's name, e.g. "site X". As no field has explicitly been defined for the street, the address field 2 or 3 (to be preferred) is to be used for entering the street and street number.

Editing functions

The following dialog opens to edit a data record.



Toolbar

There are no other special function buttons in addition to the standard functions.

13 Suppliers

Summary

Menu	Master data → Quality management → Suppliers
Transaction code	sup
Function authorization	sup

The supplier catalog has been designed to edit and update the list of suppliers. Provided that there is an interface to a higher-level system (e.g. ERP system), suppliers may be created automatically by the interface. As soon as a new supplier is created or an existing supplier is changed in the ERP system, the data record for this supplier is automatically created or changed including the specified information within the HYDRA-CAQ supplier catalog.

Utilization

The supplier number uniquely identifies suppliers in all QM applications that access the supplier catalog. The supplier catalog is used, in particular, for inspection requests of goods receipt and the complaint management.

The "supplier number" field is the key field, i.e. while saving a new supplier, the system checks whether or not there is already a supplier with this key information.

By distinguishing between active and inactive suppliers, it may be defined whether or not the suppliers are available in certain selection lists. Consequently, it is, for example, impossible to create a goods receipt inspection request for an inactive supplier. However, inactive suppliers may be evaluated at any time. Moreover, inactive suppliers may also be reactivated at any time.

In addition to the supplier number and designation, it is possible to specify comprehensive address and contact details.

Being identified as "party in charge", this supplier is included in the selection list for responsible parties. Such selection lists are integrated in different detail applications. The list of the parties responsible is mainly used within complaint management and for the creation of measures.

Integration

Supplier data is a global catalog that is used in many QM applications. Please find below some applications that refer to the supplier catalog.

- External people
- Departments

- Goods receipt inspection planning
- Goods receipt inspection requests
- Complaints management
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2 and 3 are available in addition to the supplier number and the designation.

The "inactive" filter field allows for the data set to be restricted to active or inactive suppliers.

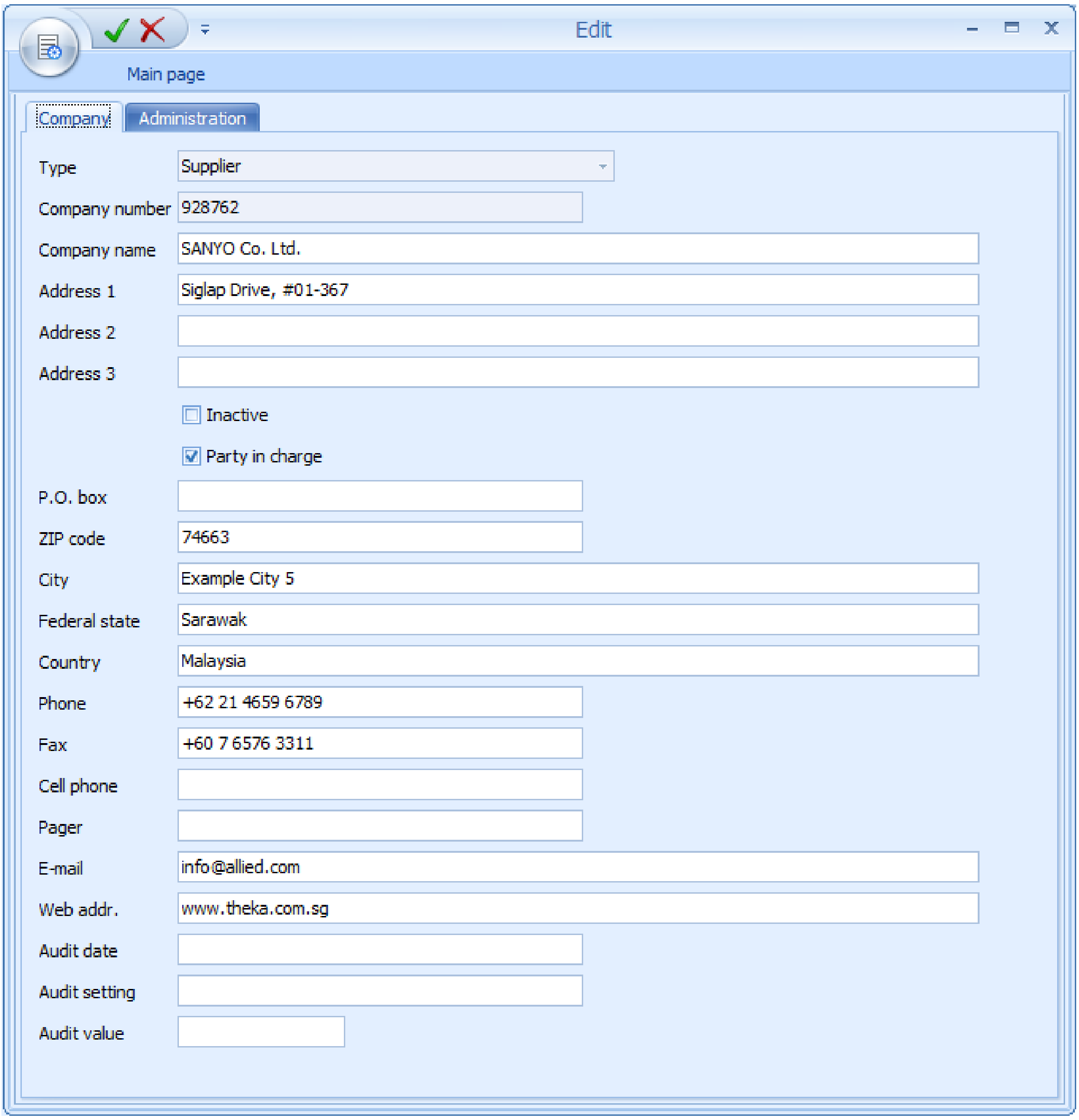
Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

The content of the individual address fields is not specified explicitly and, as a result, may be defined by the user. Normally, the address field 1 should include an addition to the company's name, e.g. "site X". As no field has explicitly been defined for the street, the address field 2 or 3 (to be preferred) is to be used for entering the street and street number.

Editing functions

The following dialog opens to edit a data record.



Toolbar

There are no other special function buttons in addition to the standard functions.

14 Failure Types

Summary

Menu	Master data → Quality management → Failure types
Transaction code	ftyp
Function authorization	ftyp

The catalog of failure types has been designed to describe the occurred deviations in more detail, e.g. deviations from specified limit values. In addition to the other failure catalogs (failure location, failure cause, causer), the failure type catalog is very important as it also includes inactive failure types by default. These inactive failure types are required for the generation of automatic failure types, e.g. in case a limit value has not been respected. For this reason, these inactive failure types must not be deleted. All inactive failure types that are important for the automatic generation of failures start with the ID number "AUTO:". There are, for example, automatic failure types for

- Non-observance of the upper tolerance limit
- Non-observance of the lower tolerance limit
- Non-observance of action and warning limits

Failures the designation of which includes the number sign (#) automatically trigger the generation of a complaint if they are assigned to a measured value of one of the areas in-production inspection, goods receipt inspection or goods issue inspection.

Utilization

The "failure analysis number" field is the key field, i.e. when saving a new failure type, the system checks whether there is already a failure type with this key information.

The input of failure types is easy to handle. Only a failure analysis number and a corresponding designation have to be assigned.

Failure type groups may optionally be defined beforehand. Consequently, the corresponding group can be assigned to the respective failure type. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The hierarchical tree entries of the group tree allow for the requested group to be selected and taken over. Then the "groups" field of the editing dialog for failure types shows the assigned group including the hierarchical group structure.

If failure types are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "failure type groups" application, which is described in the manual entitled "MOC_Groups.pdf".

Under certain circumstances, it might be reasonable and recommendable to use a self-explanatory structure for the failure type number as failure key.

By differentiating between active and inactive failure types, it can be defined whether or not they are still to be available in failure selection lists in the later data acquisition process. However, it is also possible to evaluate inactive failure types at any time. Moreover, inactive failure types can be reactivated at any time.

Integration

The failure types catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected in measurement recording the failure catalogs help describe the deviations in more detail and represent it in a way that allows for analyses/reports to be performed. Only in this way is it possible to determine failure mode analyses, take appropriate action (measures) and prevent the deviation from reoccurring.

In addition, this catalog is the basis for failure mode analyses relating to the failure types.

This catalog is also required for the creation of analysis selection catalogs as well as for using inspection chart characteristics within inspection planning and measurement recording. An analysis selection catalog includes a subset of all failures and restricts the list of failure (types) that can be selected during the collection process to those failures of the analysis selection catalog assigned to the characteristic. Consequently, the analysis selection catalog also determines the list of failure types for inspection chart characteristics.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Failure types of a group can be filtered in the "groups" tab using the icon and selecting a failure type group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive failure types.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies failure types that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

15 Failure Locations

Summary

Menu	Master data → Quality management → Failure locations
Transaction code	floc
Function authorization	floc

The catalog of failure locations has been designed to describe the deviations occurred in more detail, e.g. deviations from specified limit values.

Utilization

The "failure analysis number" field is the key field, i.e. when saving a new failure location, the system checks whether there is already a failure location with this key information.

The input of failure locations is easy to handle. A failure analysis number and a corresponding designation only have to be assigned.

Failure location groups may optionally be defined beforehand. Consequently, the corresponding group can be assigned to the respective failure location. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure then appears in the "groups" field of the editing dialog.

If failure locations are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "failure location groups" application, which is described in the manual entitled "MOC_Groups.pdf".

Under certain circumstances, it might be reasonable and recommendable to use a self-explanatory structure for the failure location number as failure key.

By differentiating between active and inactive failure locations, it can be defined whether or not they are still to be available in failure selection lists in the later data acquisition process. However, it is also possible to evaluate inactive failure locations at any time. Moreover, inactive failure locations can be reactivated at any time.

Integration

The failure location catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected in measurement recording the failure catalogs help describe the deviation in more detail and represent it in a way that allows for analyses/reports to be performed. Only in this way is it possible to determine failure mode analyses, take appropriate action (measures) and prevent the deviation from reoccurring.

In addition this catalog is the basis for failure mode analyses relating to the failure locations.

The failure locations of this catalog can also be integrated in analysis selection catalogs. An analysis selection catalog includes a subset of all failures and restricts the list of failure (locations) that can be selected during the collection process to those failures of the analysis selection catalog that is assigned to the characteristic.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Failure locations of a group can be filtered in the "groups" tab clicking the icon  and selecting a failure location group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive failure locations.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies failure locations that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of a tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

16 Failure Causes

Summary

Menu	Master data → Quality management → Failure causes
Transaction code	fcau
Function authorization	fcau

The catalog of failure causes has been designed to describe the occurred defects/failures in more detail, e.g. if limit values have been infringed.

Utilization

The "failure analysis number" field is the key field, i.e. when saving a new failure cause, the system checks whether there is already a failure cause with this key information.

The input of failure causes is easy to handle. A failure analysis number and a corresponding designation only have to be assigned.

Failure cause groups may optionally be defined beforehand. Consequently, the corresponding group can be assigned to the respective failure cause. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure then appears in the "groups" field of the editing dialog.

If failure causes are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "failure causes groups" application, which is described in the manual entitled "MOC_Groups.pdf".

Under certain circumstances it might be reasonable and recommendable to use a self-explanatory structure for the failure cause number as failure key.

By differentiating between active and inactive failure causes, it can be defined whether or not they are still to be available in failure selection lists in the later data acquisition process. However, it is also possible to evaluate inactive failure causes at any time. Moreover, inactive failure causes can be reactivated at any time.

Integration

The failure cause catalog is used, among other things, within measurement recording and in complaint management. If deviations with respect to tolerance limits are detected in measurement recording the failure cause catalog helps describe the actual failure cause in more detail and represents it in a way that allows for analyses/reports to be made. Only in this way is it possible to determine failure mode analyses, take appropriate action (measures) and prevent the deviation from reoccurring.

In addition this catalog is the basis for failure mode analyses relating to the failure cause.

The failure causes of this catalog can also be integrated in analysis selection catalogs. An analysis selection catalog includes a subset of all failures and restricts the list of failure (causes) that can be selected during the collection process to those of the analysis selection catalog assigned to the characteristic.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Failure causes of a group can be filtered in the "groups" tab using the  icon and selecting a failure cause group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive failure causes.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies failure causes that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of a tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

17 Causers

Summary

Menu	Master data → Quality management → Originator
Transaction code	ori
Function authorization	ori

The catalog of causers/originators has been designed to describe the cause for an initial situation. In this context, the causer or originator does not imperatively have to be a person or a group of people. A machine can also be the causer/originator.

Utilization

The "causer number" field is the key field, i.e. while saving a new causer/originator, the system checks whether there is already a causer with this key information.

The input of causers is easy to handle. Only an ID number and a corresponding designation have to be assigned.

Causer groups may optionally be defined in advance and the corresponding group can be assigned to the originator. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure can then be found in the "groups" field of the editing dialog for causers.

If causers are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "causer groups" application, which is described in the manual entitled MOC_Groups.pdf.

Under certain circumstances, it might be reasonable and recommendable to use a self-explanatory structure for the originator number as analysis key.

By distinguishing between active and inactive causers, it can be defined whether or not they are still to be available in selection lists for the causers in the later data acquisition process. However, it is possible to evaluate inactive originators at any time. Moreover, inactive causers can be reactivated at any time.

Integration

The catalog of originators is used, among other things, within measurement recording and in complaint management. A causer/originator may be added if defects are detected and documented within measurement recording. Only in this way is it possible to determine the basic causers, take appropriate action (measures) and prevent the defects from recurring during the failure mode analysis that follows.

The causers/originators of this catalog can also be integrated in analysis selection catalogs. An analysis selection catalog includes a subset of all failures and causers/originators and restricts the list of failures and causers that can be selected during the collection process to those of the analysis selection catalog assigned to the characteristic.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Causers/originators of a group can be filtered in the "groups" tab using the icon  and selecting a causer group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive causers.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies causers that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

18 Measures

Summary

Menu	Master data → Quality management → Measures
Transaction code	meas
Function authorization	meas

The catalog of measures has been designed to define the action to be taken, for example, to prevent the detected failures from recurring or to allow for them to be identified in time.

Utilization

The "measure number" field is the key field, i.e. while saving a new measure, the system checks whether there is already a measure with this key information.

The input of measures is easy to handle. Only a measure number and a corresponding designation have to be assigned.

Measure groups may optionally be defined beforehand and the corresponding group can be assigned to the measure. This option should not be missed out as it provides improved reports/evaluations or structures. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure can then be found in the "groups" field of the editing dialog for the measures.

If measures are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "measure groups" application, which is described in the manual entitled MOC_Groups.pdf.

By distinguishing between active and inactive measures, it can be defined whether or not they are still to be available in selection lists for the measures in the later data acquisition process. Inactive measures can be reactivated at any time.

Integration

The measures catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected within measurement recording actions can be triggered immediately on the basis of the catalog of measures. At first the assigned measures have the status "open". In complaint management a party responsible (e.g. a person) as well as a deadline may be defined. In addition to this, an actual date, the completion in % and effectiveness in % may be specified as well. Normally, these fields are only filled out, once a measure has been completed (measure status switches to "done").

Subject to whether the corresponding license for the escalation management module has been purchased, the measures can also be sent to the defined person by e-mail.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Measures of a group can be filtered in the "groups" tab using the icon and selecting a measure group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive measures.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies measures that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

19 Analysis Selection

Summary

Menu	Master data → Quality management → Analysis selection
Transaction code	asc
Function authorization	asc

Analysis selection catalogs allow for the set of failure types, failure locations, failure causes, originators, and measures that are available in measurement recording to be restricted specifically.

The overall application is divided into two hierarchical levels. A master-detail grid is used for presentation. The data records of the analysis selection are created on the first level. One level below, the corresponding failure types, locations, causes, originators and measures are assigned to these data records of the analysis selection. There is a separate tab for each assignment type on this second level.

Separate function keys to create, edit or delete data records are available on each level of the master-detail grid.

Utilization

The analysis selection number identifies the data records of the analysis selection uniquely in all QM applications in which they may be selected and assigned.

A data record of an analysis selection catalog only consists of a number and designation as well as of the flag to disable it ("inactive" field). The most crucial factor here is the possibility to assign failure types, locations, causes, originators and measures to a data record of the analysis selection.

If a characteristic is assigned an analysis selection catalog only the failure types, failure locations, failure causes, originators and measures listed in this catalog will be available for this characteristic when measured values are recorded. Consequently, the failure list, for example, can be designed in relation to characteristics.

Analysis selection catalogs are mainly used for the assignment to an inspection chart characteristic. An assignment is almost mandatory for inspection chart characteristics, as in any other case, the user may choose from the whole set of failure types of the entire master data catalog when recording measured values for this characteristic. This would make inspections confusing and too complex.

Integration

Analysis selection catalogs are used in all applications dealing with characteristics. By assigning analysis selection catalogs in these applications, the selection list for failures, originators and measures is restricted for measurement recording.

Prerequisite

The master data for failure types, failure locations, failure causes, originators and measures need to be maintained before this function can be used in a useful manner.

Selection criteria

Selection criteria are self-explanatory and are not described separately.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The check box "inactive" identifies data records of the analysis selection that are no longer to be used for the definition of characteristics (of inspection plans/inspection orders).

20 Inspection Planning

Summary

Menu	Quality management → In-production inspection → Inspection planning Quality management → Goods receipt → Inspection planning Quality management → Goods issue → Inspection planning Quality management → Initial sample → Inspection planning Quality management → Gage management → Inspection planning
Transaction code	iplp
Function authorization	iplp

Utilization

Subject to the field of application, inspection planning is the most important function, as inspection plans are the basis for the generation of inspection orders and the resulting inspections. This means inspections in the area of goods receipt, production and goods issue as well as initial sample inspections and the calibration of test equipment and measurement equipment. As the requirements as regards inspection planning are nearly identical in all areas, the HYDRA inspection planning functions are also nearly identical for these areas.

In creating inspection plans, the user is supported in most cases by the master data catalogs. For this reason it is very important to conscientiously maintain this master data.

Integration

Inspection planning is a prerequisite to generating inspection requirements, the corresponding inspection steps and, as a result, the relevant inspections/calibrations.

The inspection plan in use is referenced in the generated inspection requirements. Moreover, some evaluations/reports also refer to inspection planning. Consequently, data of the same inspection plan number can be added to the basis of a control chart for inspection order characteristics. The same applies to the global control chart analysis, where special inspection plan filter fields may be used. The global control chart analysis does not take into account calibration data and, as a result, no filtered calibration inspection plans.

Another essential relation has been established to the production control plan. As no production control plan can be created without having generated inspection plans beforehand.

Prerequisite

Relevant master data needs to be edited/maintained to be able to create inspection plans. Which master data has to be maintained depends on the respective field of application. However, it is a fundamental prerequisite that articles/items and characteristics are maintained beforehand. Article groups also need to be created, provided that inspection plans based on article groups are created.

Selection criteria

The paragraph that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt and goods issue. By default, the following areas are available.

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

Active

By checking the corresponding field, the inspection plan list can be restricted to active inspection plans. If this checkbox is not checked the list only shows inspection plans that are in the "in process" and "released" status. The third state of this checkbox (grayed out) shows all inspection plans. This is set by default.

Article number

Filters the article number with respect to inspection plans based on articles.

Article designation

Filters the article name with respect to inspection plans based on articles.

Article group

The article group tree can be opened by clicking the  icon, in case inspection plans for article groups are to be filtered. A function to cancel and accept the entries made is provided.

Operation

Operation number

Customer number

Direct input or opening of the customer catalog and taking over of the number from the entry selected there.

Supplier number

Direct input or opening of the supplier catalog and taking over of the number from the entry selected there.

Manufacturer number

Direct input or opening of the manufacturer catalog and taking over of the number from the entry selected there.

Field descriptions

Inspection plan header

Area, inspection plan number, inspection plan index

Uniqueness of all existing inspection plans is secured by the "area", "inspection plan number" and "inspection plan index". The area may be selected. The inspection plan number and inspection plan index may be entered using alphanumeric characters. All three fields are mandatory fields.

The input of these three pieces of information must be unique, i.e. no other inspection plan may exist that already includes this information. By assigning a structured inspection plan number, it is possible to provide specific information. This information might be useful later during sorting. As an alternative, the inspection plan number may also be used synonymously with the article number or gage group number. The inspection plan index corresponds to the inspection plan version. If an existing inspection plan version is to be modified, yet it cannot simply be edited because it has already been used to generate inspection orders, it is recommended to copy the original inspection plan and then modify the inspection plan index (e.g. incrementing it by 1). The inspection plan number should be maintained, if possible, to facilitate later evaluations/reports (e.g. for displaying control charts applying to several orders, in which case the inspection plan versions on which the evaluation is based differ while the inspection plan number is identical).

Inspection plan type

The inspection plan types "article inspection plan" and "group inspection plan" are distinguished. The "article inspection plan" type allows for an article to be assigned from the article master data for which this inspection plan is to be created. The "group inspection plan" option has to be selected to create an inspection plan for article groups. This allows for an article group to be selected. The selected article group is displayed in a separate field within the tree structure.

Article

Input of the article number. Provided that it is known, it may be entered directly. Otherwise, the article catalog can be opened and the requested article can be identified and taken over using the filter and sort criteria provided there. By selecting an article, the drawing issue number, article designation, customer article number and the drawing number are taken over from the master data record and displayed in the corresponding fields.

The customer article number is not displayed within the goods receipt inspection planning.

Nothing may be entered here if the field “article number” or “article group” is shown in gage management.

Drawing issue number

The drawing issue number can be entered directly, just as it is the case for the article number.

In case the article number and the drawing issue number are entered directly, the master data record is determined on the basis of this information while saving and the article designation, customer article number as well as the drawing number are displayed.

The article is uniquely identified by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.

Nothing may be entered here if the “drawing issue number” field is shown in gage management.

Operation assignment

One of the two assignment types "an inspection plan for each OP" and "an inspection plan for all OPs" may be selected.

By assigning operations, it is defined whether the inspection plan to be created should include the characteristics of different operations ("one inspection plan for all OPs") or whether the inspection plan should only include the characteristics of one operation. The fields "operation" and "operation designation" are shown if the "an inspection plan for each OP" option is selected. In any other case, operation details are assigned in the area of inspection plan characteristics. The option "one inspection plan for all OPs" should be selected as far as possible. This allows, for example, printing out of all inspection characteristics of an article, although it has several operations.

The option “one inspection plan for all OP“ should be used in areas that are not assigned to a production order. This provides the advantage that the “fictious” operation only has to be entered once in the inspection plan header, instead for every inspection plan characteristic.

However, the system has to be configured accordingly by MPDV consultants to be able to use this inspection plan variant.

Operation, operation designation

If operations are used, only one of the two fields "operation" or "operation designation" needs to be filled out. Only the operation number has to be entered in the "operation" field if an inspection requirement/inspection order is to be generated automatically by logging an operation of a production order on.

It has to be taken into consideration that these fields are used as search criteria when inspection orders are generated at a later point in time. Provided that only the designation is entered for the operation within the inspection plan, exclusively this information needs to be provided when inspection orders are generated.

A “fictious“ operation has to be assigned (e.g. 9999) even in areas that are not assigned to a production order (e.g. in goods receipt or in gage management). This is necessary to generate inspection steps at a later point in time.

Released/active

Shows whether the inspection plan is "released" and/or "active". If the inspection plan is released or active the corresponding checkboxes are checked. An inspection plan is released and activated, i.e. its status is changed, only by using the corresponding toolbar functions. An inspection plan has to be released before it can be activated.

Only active inspection plans are taken into consideration, when inspection orders are generated at a later point in time.

Released by / on

Shows the HYDRA user who has released it. The release date is displayed additionally.

Valid from / until

Instead of an "unrestricted" activation (using the toolbar), a validity period may be entered here. This period is then taken into account when the inspection order is generated. Yet activation for a certain period means that the user has no clear overview of currently valid inspection plans, and it is therefore recommended to use the "global/unrestricted" activation option using the toolbar if possible. If activated via the toolbar functions, the system carefully monitors whether an active inspection plan exists for the specified article and checks whether this plan includes the same drawing issue number, customers and suppliers. If this is indeed the case, the previously active inspection plan is automatically deactivated.

Type of inspection

Inspection based on characteristics or pieces

When it comes to an inspection based on characteristics and, say a sample size of 5, the characteristic is checked completely first. Whereas with the inspection based on pieces, the characteristic is changed every time a measured value is collected, as items are checked entirely.

Action

Create inspection step or create and immediately release inspection step

Only inspection steps that have been released can be checked.

Customer / supplier / manufacturer

If a customer / manufacturer or supplier is selected this inspection plan only applies for the selected company. Consequently, the customer / manufacturer / supplier becomes the key field of an active inspection plan. In this case, the inspection requirement must also include a company entry in order for this inspection plan to be used for the generation of the inspection requirement.

Dynamic modification type

Characteristic-related, batch-related or none

The authorization “iriscp.dynamic“ is required to show these fields.

Batch-related: In this case, it is defined on the level of inspection plan headers which transitional definition is to be used for the dynamic modification. In addition, the initial inspection severity is also defined on the level of inspection plan headers. Now only the reference to the dynamic modification norm is required on the level of inspection plan characteristics. In this context, however, the selection is restricted to those dynamic modification norms referring to the same inspection severity definition as the assigned transitional definition.

Characteristic-related: In this case, the transitional definition, initial inspection severity as well as the dynamic modification norm are referenced on the level of inspection plan characteristics. In this context, however, the inspection severity definition of the assigned transitional definition has to match the assigned dynamic modification norm.

None: The "dynamic modification" function has not been enabled for this inspection plan.

Transitional definition

The authorization “iriscp.dynamic“ is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is selected. It provides the possible inspection severities and controls switching between the inspection severities.

Initial inspection severity

The authorization “iriscp.dynamic“ is required to show these fields.
This field is only available if a dynamic modification relating to batches or characteristics is selected. It defines which inspection severity is used for checking the first goods received according to the basics for the dynamic modification history.

Initial sample form

The different form types available for initial sampling can be chosen here. The form type “VDA volume 2, 4th edition“ is only supported at the moment. Subject to the selected form type, a corresponding list of categories is shown in the inspection plan characteristics.

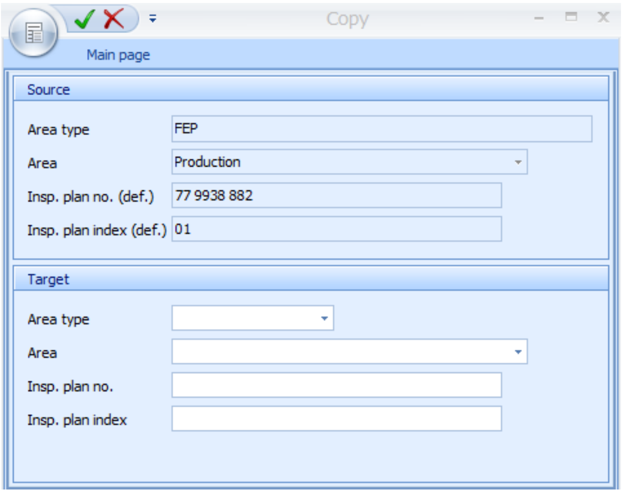
Editing functions

The key fields "area", "inspection plan" and "inspection plan index" cannot be changed in the editing mode.

Toolbar

Copy

The below dialog opens to copy an inspection plan:



The target area type and target area may be entered here. Normally, the user should select an area that is identical to that of the source inspection plan. The new inspection plan number and inspection plan index are to be entered afterwards. In case a new version is generated from an existing inspection plan, normally the same inspection plan number is used and only the inspection plan index is changed.

Activate

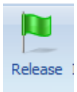
Function authorization: iplp.activate

Makes the inspection plan status "active“.

Deactivate

Function authorization: iplp.deactive

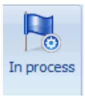
Puts an inspection plan that is in the "active" status back to the "released" status.



Release

Function authorization: iplp.release

Puts an inspection plan that is in the "in process" status to the "released" status.



In process

Function authorization: iplp.inprocess

Puts an inspection plan that is in the "released" status in the "in process" status.

"Print" detail application

Function authorization	iplp.print

The print dialog of the inspection plan header opens a list of available reports. These are Word forms. The potential content of these forms is determined by the Web services that are available in the respective context. The form entries, i.e. the content of the form list of the corresponding print dialog, are defined within the master data of quality management. This is where the basis for new forms is established and the corresponding form properties are defined. A corresponding license is required to be able to change the forms with respect to content and design.

Toolbar - print

There are no other special function buttons in addition to the standard functions.

"Inspection plan characteristics" detail application

The detail application of inspection plan characteristics is nearly identical to the master data of characteristics application. For this reason, reference is made only to additional features in the following.

Go to

For further information on the definition of characteristics, please refer to the functional description of the document entitled MOC_CharacteristicsQM.

On the level of inspection plan characteristics the corresponding inspection plan characteristics are assigned to the previously defined inspection plan header. Assignments are performed by way of initial data creation using the characteristics catalog and taking over the characteristic selected there. By accepting the characteristic, all master data entries are copied to the inspection plan characteristic. Every (copied) detail may be changed and / or complemented afterwards. The characteristic designation is often complemented to define it in more detail.

It is also possible to create a characteristic not included in the characteristics catalog. However, this is recommended only in exceptional cases, since all analyses (e.g. failure mode analysis) are based on characteristics found in the catalog. It is therefore preferable to properly maintain the characteristics catalog.

Different properties and settings can be defined, before specific characteristic data is complemented.

Field descriptions

Inspection plan characteristics

OP sequence

The operation sequence number (OP sequence/AFO) determines the order in which the inspection is later performed. Entries must be unique. In the ideal case, operation sequence numbering should be assigned in steps of 10. This provides the option of later inserting a new characteristic between two existing ones.

Characteristics number

Number of the characteristic selected from master data

Characteristic type

Decides whether the inspection is based on the collection of measured values (variable) or on the specification of the number of detected failures (attributive). When it comes to the attributive inspection, the decision is often only based on the "pass" or "fail" results. Further characteristic types are the inspection chart and the information characteristic. The information characteristic has only been designed to display a document while the inspection is running. Subject to the input type, the lower area of the dialog provides corresponding sampling schemes.

Inspection result base

Decides whether all samples or only the last sample is used to determine the inspection result (pass/fail).

Mandatory inspection

If this checkbox is activated, at least one measured value has to be entered for this characteristic, before an inspection order can be completed with this characteristic.

Calculate characteristic

Identifies a characteristic to be calculated

Formula

Further details can be taken from the manual dealing with the CAQ master data characteristics.

Initial sample creation

Shortlist of categories for initial sampling. The contents depend on the form type selected in the inspection plan header. Within the resulting initial sample report, the inspection plan characteristics are summarized by category and a separate page is printed for each group.

Copy characteristic

It may be defined here which characteristic is to be copied from an initial sample inspection plan to a production plan, for example. Only those characteristics are copied that have been marked relevant in this field.

Details

Defines whether - with respect to this characteristic - the contents of the "details" tab are defined within the inspection plan or whether they have to be determined from the available master data characteristic or an entry from the specification list when inspection requirements are later generated on the basis of this inspection plan. If the "from characteristic catalog" option is selected the "details" tab is hidden. The "from characteristics catalog" option is reasonable if it is a characteristic the specifications of which are to be defined identically for several articles (even applicable for several article groups). In this case, specifications can be defined centrally within the master data. This reduces the input and update efforts considerably.

Specifications

Defines whether - with respect to this characteristic - the contents of the "specification" tab are defined within the inspection plan or whether they have to be determined from the available master data characteristic or from a specification list entry, when inspection requirements are later generated on the basis of this inspection plan. The tabs "specifications", "chart 1", "default values chart 1", "chart 2" and "default values chart 2" are hidden if the option "from list" or "from characteristic catalog" is checked. The "from list" option is only reasonable if inspection plans for article groups are in use. The specification list is used for an inspection plan for article groups if the specifications vary from item to item for certain characteristics.

Dynamically modified

The authorization “iriscp.dynamic“ is required to show these fields.
This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. This option allows for definitions to be made that a characteristic is not to be modified dynamically, although the dynamic modification option is selected in the inspection plan header.

Transitional definition

The authorization “iriscp.dynamic“ is required to show these fields.
This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. The transitional definition that can be selected provides the possible inspection severities and controls switching between these different inspection severities. For the dynamic modification based on batches, the transitional definition is assigned in the inspection plan header and is not available on the level of inspection plan characteristics in this case.

Initial inspection severity

The authorization “iriscp.dynamic“ is required to show these fields.
It defines which inspection severity is used for checking the first goods received according to the basics for the dynamic modification history. Only the inspection severities of the previously selected transitional definition are available. For the dynamic modification based on batches, the transitional definition is assigned in the inspection plan header and is not available on the level of inspection plan characteristics in this case.

Norm

The authorization “iriscp.dynamic“ is required to show these fields.
This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. The dynamic modification norm to be used is assigned from master data.

Inspection level

The authorization “iriscp.dynamic“ is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. An inspection level can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

AQL

The authorization “iriscp.dynamic“ is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. An AQL value can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

Method

The authorization “iriscp.dynamic“ is required to show these fields.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually characterized by being relevant to dynamic modification. A method can only be selected if the assigned norm corresponds to DIN ISO 3951.

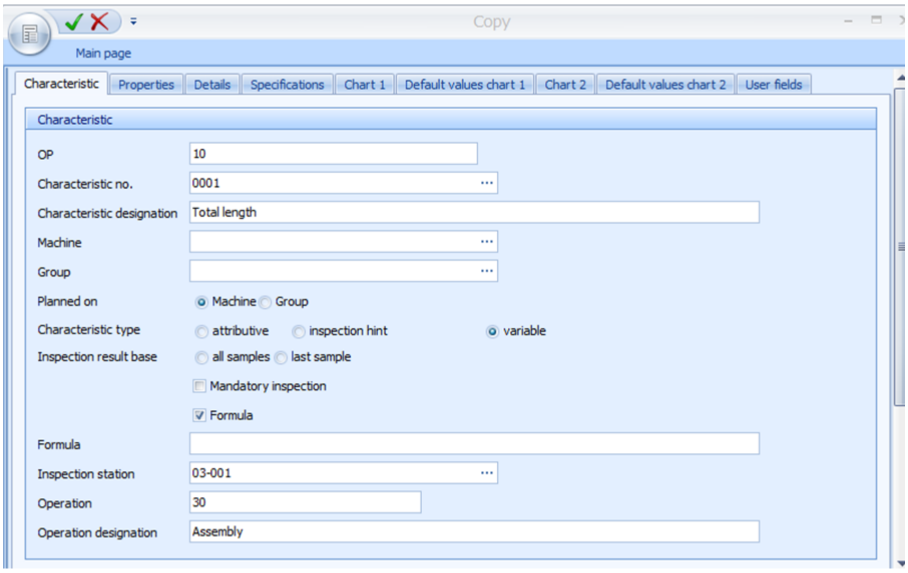
Go to

The fields of the respective tabs correspond to those of the characteristics master data. Further information on these fields can be gathered from the functional description of the document entitled MOC_CharacteristicsQM.

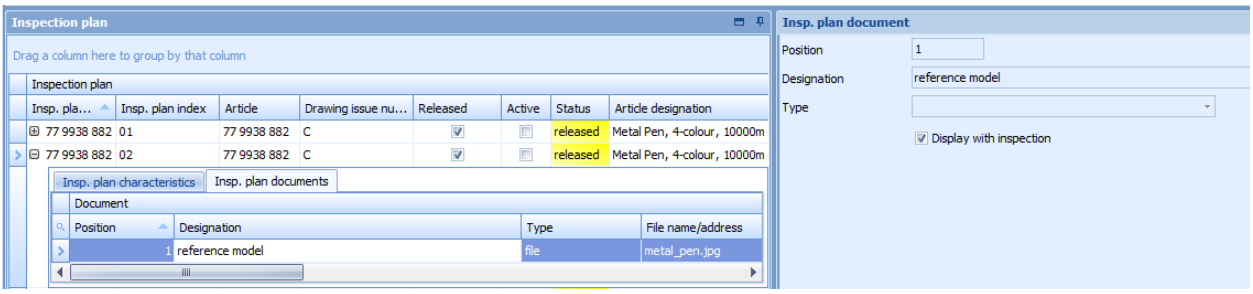
Inspection plan characteristics - toolbar

Copy

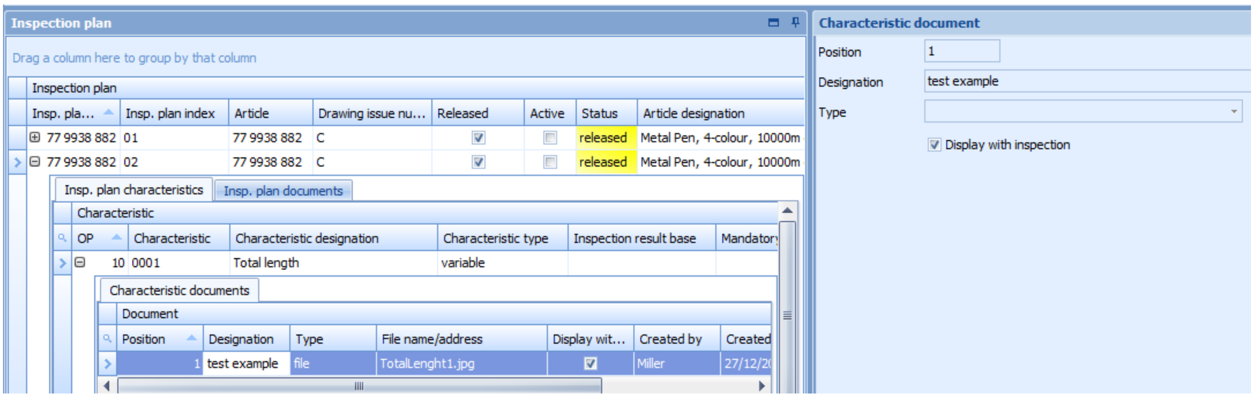
Data of the selected characteristic is opened in insert mode to be able to copy inspection plan characteristics. All fields can be changed. An operation sequence number that does not yet exist in this inspection plan has to be assigned to be able to save it.



"Inspection plan documents“ and "documents of inspection plan characteristics“ detail applications

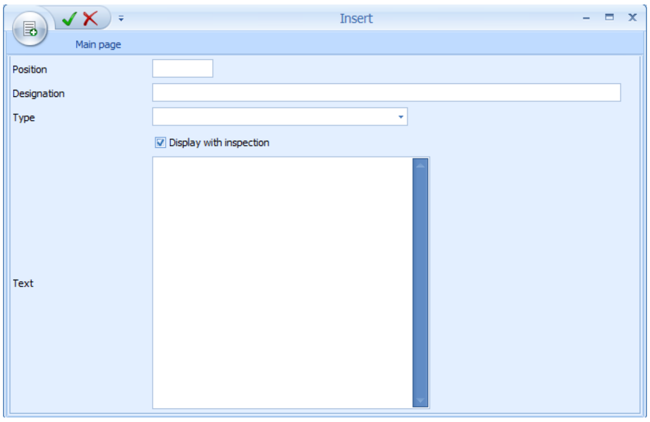


The above screenshot shows how an inspection plan document is assigned



The above screenshot shows how a document of an inspection plan characteristic is assigned

Provided that the "inspection plan documents" tab or the "characteristic documents" tab has been activated in the master detail grid, as many documents as required may be assigned to each inspection plan and each inspection plan characteristic. By enabling these tabs, the toolbar offers corresponding buttons to edit the documents.



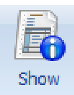
When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

The file types are: "FILE", "URL" and "Text". The file name including path may be entered manually with the "file" type. The "URL" file type enables access to the Internet or Intranet. The third file type "Text" allows for text to be entered directly.

A designation may be assigned to each defined document. Moreover, it may be determined in which order the documents are to be listed. The "position" field is used for this purpose (numeric input). The specifications made within this list must be unique. In addition, the checkbox "display with inspection" determines whether or not the document may be shown during the inspection process.

"Inspection plan documents“ and "documents of inspection plan characteristics" - toolbar

In addition to the standard features, there is also a button to display the documents.



Show documents

If a document link is defined this button opens and shows this document. However, a program, which is able to visualize the linked file type, has to be installed on the PC.

21 Inspection Requirements

Summary

Menu	Quality management → In-production inspection → Inspection requirement Quality management → Goods receipt → Inspection requirement Quality management → Goods issue → Inspection requirement Quality management → Initial sample → Inspection requirement Quality management → Gage management → Inspection requirement
Transaction code	irp[1]
Function authorization	irp

Utilization

The inspection requirement is as important as the production order is in the shop floor data collection module and includes one or several inspection steps (e.g. one inspection step for each operation). In this context, inspection steps result from the inspection plan search triggered by the generation of inspection requirements. For this reason, the inspection requirement rather includes general information, such as the order or article number. The inspection step that is subordinate to the inspection requirement finally includes the characteristics to be checked. Consequently, the inspection step or the operation of the production order (and, as a result, the inspection step as well) is selected and logged on to the terminal for measurement recording.

As the functional requirements are nearly identical in all areas (e.g. goods receipt, production, goods issue, gage management), the corresponding applications are structured nearly in the same way for these areas.

[1]New inspection requirements cannot be created if the "inspection requirement" application is opened by using transaction codes.

In creating inspection requirements, the user is generally supported by master data catalogs. For this reason, it is very important to conscientiously maintain master data.

Integration

Inspection requests/inspection steps are generated on the basis of inspection plans that have been created beforehand and refer to them. When being generated, the contents of inspection plans are copied into the inspection request/inspection steps. Thus, the relation is only based on referencing. The limited modification options that are enabled in the inspection plan do not affect already generated inspection requirements/inspection steps.

The inspection steps, in turn, are the "objects" that can be checked/inspected and the operation number connects them with the corresponding operation of the production order within production. Within gage management, inspection steps can be put on a level with the actual calibration order.

Nearly all reports/evaluations refer to the data of inspection requests/inspection steps and the available inspection step characteristics. Restricting filters are, e.g. the article number, order number, operation, and inspection step characteristic.

In gage management, inspection requirements are generated automatically from the activity calendar by a user action. In exceptional cases, inspection requirements can also be generated directly in the “inspection requirement” application. In this case however, the inspection plan number has to be entered manually.

Prerequisite

Inspection plans need to be created beforehand in the same area/context to be able to create inspection requests/inspection steps. Master data is also required. Which master data has to be maintained depends on the respective field of application.

Selection criteria

The list that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt or goods issue. By default, the following areas are available:
In-production inspection: Production
Goods receipt: Goods receipt
Goods issue: Goods issue
Initial sample: Initial sample inspection
Gage management: Gage management

Inspection request number

The inspection requirement number uniquely identifies the inspection request. HYDRA automatically generates the inspection requirement number, which is unique.

Status

By opening a status list, the different inspection request statuses may be filtered. In this way, it is possible to restrict the list of inspection requirements, e.g. to the completed inspection requests.

Skip lot (available for the goods receipt and goods issue only)

Due to dynamic modifications, an inspection requirement can get the classification "skip lot" and, as result, it will not be checked.

Order

Number of the corresponding production order or calibration order.

Article number

Direct input or opening of the article catalog and taking over of the number from the entry selected there.

Drawing issue number

The article is identified uniquely by the combination of "article number" and "drawing issue number".

Customer number

Direct input or opening of the customer catalog and taking over of the number from the entry selected there.

Supplier number

Direct input or opening of the supplier catalog and taking over of the number from the entry selected there.

Manufacturer number

Direct input or opening of the manufacturer catalog and taking over of the number from the entry selected there.

Actual date from / until

The time may be restricted here. Normally, in production this field is not filled out. In the goods receipt area the "actual date" is the date when the goods are delivered.

Planned date from / until

The time may be restricted here. Normally, in production this field is not filled out. In the goods receipt sector the "planned date" defines the date when the goods are (initially) planned to be delivered.

Manufacturing date from / until

The time may be restricted here.

"Inspection request" field descriptions

The elementary fields are described below.

Inspection requirement

Area

Area in which the inspection request is to be created. A selection list of the configured areas for in-production inspection, goods receipt, goods issue, initial sampling or gage management facilitates the creation process. By default, the following areas are available.

In-production inspection: Production
Goods receipt: Goods receipt
Goods issue: Goods issue
Initial sample: Initial sample inspection
Gage management: Gage management

This field is a key field for searching for the active inspection plan or the specified inspection plan including index on the basis of which the inspection request, inspection steps and inspection step characteristics are to be generated.

Inspection request number

The inspection request number is automatically generated upon saving. It is numeric, identifies an inspection request uniquely and cannot be changed.

Operation, operation designation

If operations are in use only one of the two fields "operation" or "operation designation" has to be filled out. Only the “operation number” field is to be filled out, provided that an inspection requirement and/or an inspection order is to be generated automatically by logging an operation of a production order on. The work plan structures cause the operation number to be referenced almost exclusively.

Subject to system configuration, this field can be a key field for searching for the active inspection plan on the basis of which the inspection requirement, inspection steps and inspection step characteristics are to be generated. If the system is configured accordingly, all inspection steps relating to the operations included in the relevant/detected inspection plan are generated already when the inspection requirement is generated.

Even areas that are not assigned to a production order (e.g. goods receipt and gage management), might require a “fictitious” operation (e.g. 9999) to be specified, subject to system configuration. This “fictitious” operation has to be included in the relevant/detected inspection plan.

Inspection plan number / inspection plan index (specified)

As an alternative to indicating key fields for searching for the active inspection plan to generate inspection requirements, the inspection plan that is to be used for the generation may also be entered in these fields.

Date

When a new data record is created, this field includes the system date, which may be changed afterwards. Once it has been saved, this field may no longer be changed.

Article

Input of the article number. Provided that it is known, it may be entered directly. Otherwise, the article catalog can be opened and the requested article can be identified and taken over using the filter and sort criteria provided there. By selecting an article, the drawing issue number, article designation, customer article number and the drawing number are taken over from the master data record and displayed in the corresponding fields.

This field is a key field for searching for the active inspection plan on the basis of which the inspection requirement, inspection steps and inspection step characteristics are to be generated, provided that no inspection plan including index has been specified.

This field is empty or should be left empty in gage management.

Drawing issue number

Like the article number, the drawing issue number can be entered directly.

This field is a key field for searching for the active inspection plan on the basis of which the inspection requirement, inspection steps and inspection step characteristics are to be generated, provided that no inspection plan including index has been specified.

In case the article number and the drawing issue number are entered directly, the master data record is determined on the basis of this information while saving and the article designation, customer article number as well as the drawing number are displayed.

The article is identified uniquely by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.

Provided that the "drawing issue number" field is shown in gage management, it is not allowed to assign any values to it here.

Companies

Customer / supplier / manufacturer

The corresponding company number may either directly be entered or by opening the corresponding master data catalog and selecting the corresponding entry. Once the inspection requirement has been selected and/or saved, the company name and address data are displayed additionally.

These fields are key fields for searching for the active inspection plan on the basis of which the inspection request, inspection steps and inspection step characteristics are to be generated, provided that no inspection plan including index has been specified.

Date - quantities

Actual quantity

The inspection scope is calculated subject to the selected sampling scheme and the actual quantity. This is especially true in goods receipt and goods issue if dynamic modification is enabled. Normally, in production this field remains empty.

Target quantity

In particular for the goods receipt, it is important to indicate the target quantity in order to determine the difference between the ordered quantity and the actually delivered quantity.

Scrap

This field is only relevant if HYDRA exclusively uses the CAQ functions and the shop floor data collection (BDE) module is not activated.

Rework

This field is only relevant if HYDRA exclusively uses the CAQ functions and the shop floor data collection (BDE) module is not activated.

Actually produced quantity (act. prod. qty.)

This field is only relevant if HYDRA exclusively uses the CAQ functions and the shop floor data collection (BDE) module is not activated.

Actual date

In particular for the goods receipt, it is important to indicate the actual date in order to determine the difference between the planned delivery date and the actual delivery date. Normally, in production this field remains empty.

Planned date

In particular for the goods receipt, it is important to indicate the planned date in order to determine the difference between the planned delivery date and the actual delivery date. Normally, in production this field remains empty.

Status - details

Status

The status describes the inspection requirement status, e.g. whether it is completed, unchecked ("created" status) or whether inspection results already exist ("partial result" status).

Overall result

The overall result defines whether the inspection request is classified as "pass" or "fail".

Usage decision

The usage decision classifies the overall result in more detail. The list of usage decisions can be complemented when customizing the system. The usage decision is preset subject to the overall result when inspection requirements are completed. A "pass" result leads to the usage decision "release" and a "fail" result leads to the usage decision "reject". The preset usage decision may be changed when completing the inspection requirement manually.

Skip lot

Due to dynamic modifications, an inspection requirement can be classified as "skip lot" and, as a result, it will not be checked.

PPS/ERP reference

If inspection requirements are automatically generated by higher-level systems using an interface, this reference number allows for unique assignments to be made for uploading the results.

PPS/ERP addition

If inspection requirements are automatically generated by higher-level systems using an interface, another piece of information may be added to the unique reference number for uploading results. This field is only filled out by interface.

Uploaded to PPS/ERP

If this field is enabled (by interface) it indicates that the upload to the higher-level system has been performed and the higher-level system has confirmed this via interface. This field is only populated by interface.

Inspection plan number/inspection plan index (used)

The inspection plan number and inspection plan index make clear which inspection plan was used as basis for the generation of the inspection requirement and subordinate data (inspection steps, inspection step characteristics).

Cavity assignment

The authorization "ipl_cav" is required for displaying these fields.

The options "none" or "sample" are available for cavity assignment. These options derive from the inspection plan.

Dynamic modification

Dynamic modification type

Shows the dynamic modification type of the respective inspection requirement. .
Batch-related: The available inspection plan has a dynamic modification based on batches.
Characteristic-related: The existing inspection plan has a dynamic modification based on characteristics. In this case, the transitional definition, the initial inspection severity as well as the dynamic modification norm are kept on the level of inspection step characteristics.
None: The available inspection plan does not include dynamic modification.

Transitional definition

These fields (number and designation) are only available if dynamic modification based on batches or characteristics is defined in the existing inspection plan. The transitional definition provides the possible inspection severities and controls switching between these different inspection severities. The content cannot be changed.

Determined inspection severity

It defines which inspection severity is/was used for checking the goods received according to the basics for the dynamic modification history. This field is only available if dynamic modification based on batches is defined in the existing inspection plan. The content cannot be changed.

Revised inspection severity

It defines which inspection severity is/was used for checking the goods received according to revising the inspection severity manually. This field is only available if dynamic modification based on batches is defined in the existing inspection plan.

Form

Party in charge type

Provides the shortlist of the different categories of responsible parties, e.g. department, customer, supplier, external persons

Party in charge

List of the responsible parties restricted to the type selected beforehand. This list includes all the master data that were marked accordingly in the “party in charge” field.

Party in charge name

Only shows the name/designation of the selected responsible party.

Form

Form type of the available inspection plan.

Calibration

Resource type

Shows the resource type of the resource to be calibrated. For gages, this is the type “PRM”.

Resource

Shows the resource number (gage number) that is assigned to this inspection requirement.

Maintenance

Shows the activity number that is assigned automatically by creating an activity in the activity calendar. The relevant activity is determined or referenced based on this unique ID number.

Calibration findings

Shows the result entered or determined by completing the calibration process, e.g. capable without restrictions

Calibration status

Shows the status entered or determined by completing the calibration process, e.g. released

Inspection requirement - editing functions

The key fields "area" and "inspection requirement number" as well as the fields "skip lot" and "uploaded to ERP/PPS" cannot be changed in the editing mode.

Inspection requirement toolbar

Release

Function authorization: irp.release

Transfers the inspection requirement to the status that existed at the time when it was completed, e.g. to the "partial result" status. This is required in cases where the inspection request has already been completed and is to be brought again into a condition that can be checked.

Complete

Function authorization: irp.complete

Opens a dialog to assign a usage decision, whereas a usage decision is already preset on the basis of the existing inspection results. The inspection request is finally transferred to the "completed" status, once it has been completed. The inspection steps pertaining to it can no longer be checked in this status.

Cancel

Function authorization: irp.cancel

Transfers the inspection requirement to the "canceled" status. This status no longer allows for the inspection steps pertaining to it to be checked.

"Print" detail application

Function authorization	irp.print

The print dialog of the inspection requirement header opens a list of available reports. These are Word forms. The potential content of these forms is determined by the Web services that are available in the respective context. The form entries, i.e. the content of the list of forms of the corresponding print dialog, are defined within the master data of quality management. This is also where the basis for new forms is established and the corresponding form properties are defined. A corresponding license is required to be able to change the forms with respect to content and design.

The report, 4th edition, 2nd volume VDA (German Association of the Automotive Industry), is available as special form for initial sampling. In addition to the cover sheet, the relevant annex sheets of the inspection characteristic results are printed as well. Each appendix includes the inspection characteristics including the relevant results for a category, e.g. data for dimensional checks (category: dimension).

Print - toolbar

There are no other special function buttons in addition to the standard functions/features.

"Complete inspection requirement“ detail application

Function authorization	irp.complete

The corresponding button has to be clicked in the toolbar to complete an inspection request. An editing dialog opens afterwards.

Information about the inspection steps pertaining to the inspection requirement is displayed in addition to information about the inspection requirement itself (i.e. order, batch, article, company data). An inspection requirement cannot be completed as long as any inspection steps are still open, i.e. the field “without result” includes a value greater than zero. Within this dialog box it is, however, possible to complete all inspection orders that are still open by using the button “complete all inspection orders”, provided that the conditions of a mandatory inspection that might exist are met. The result and further use of the inspection requirement may then be specified. If all of the inspection steps have been completed with “pass”, the system proposes “pass” as the result and “release” as the possible use. If any inspection step has been completed with “fail”, the system proposes "fail” as the result and “reject” as the possible use. Another result and use may still be selected though.

The list indicating the available usage decisions may be changed through MPDV customizing services.

"Inspection step" detail application

Proceeding from an inspection requirement in the master detail grid to an inspection requirement in the sub-level of "inspection steps" results in all inspection steps pertaining to the selected inspection requirement to be displayed. The detailed view of the selected inspection step additionally shows the inspection step data. The number or the combination of the inspection steps depends on the configurations made for the existing inspection plan. In case an inspection plan includes a configuration saying that the operation is defined on the level of inspection plan characteristics and, in addition to this, system configuration provides for the creation of all corresponding inspection steps along with the generation of the inspection requirement and the first inspection step, the inspection requirement includes an inspection step for every operation included in the inspection plan.

Moreover, it is possible to release, complete or cancel an inspection step using the toolbar functions.

Inspection steps may be deleted but not created or edited manually.

The current status of the inspection step can be taken from the "rating" category of the "inspection step" tab. The following statuses are possible.

Status	Description
No characteristic (KMM)	The inspection step has no characteristics able to be inspected. The inspection step is faulty and no inspection can be performed.
Created (ERS)	The inspection step has been created but not yet released for inspection. No measurement values have been recorded yet.
Inspection step released (FRE)	The inspection step has been released for inspection. No measurement values have been recorded yet.
Partial result (TER)	The inspection step has been released; some measurement values have been recorded.
Inspection in progress (IPR)	Measurement values for this inspection step are currently being recorded at another terminal.
Completed (ABG)	The inspection step has been completed.
Canceled (STO)	The inspection step has been canceled.
Skip lot (SKL)	The inspection step is in the "skip-lot" state.

"Inspection step" field descriptions

The essential inspection point fields are described in the paragraphs that follow. Self-explanatory fields are not explained.

Inspection step

Inspection step number

Unique number of the inspection step

Status

Status description of the inspection step, e.g. "completed", "partial result", etc.

Result

Qualitative classification of the inspection result based on the corresponding inspection results of the characteristics pertaining to it.

Workplace

Number of the workplace for which this inspection step has been generated.

Workplace designation

Name of the workplace for which this inspection step has been generated.

Inspection point identification

Up to nine different fields may be activated, which can or must be filled out later when inspection points are generated. By customizing the CAQ system options, it may be defined which fields are to be enabled, how they are labeled (for some fields) and whether or not it is a mandatory field.

If a field is activated and it is a mandatory field the corresponding checkbox will be "checked". The corresponding checkbox is "checked" and grayed out if it is not a mandatory field but enabled for the resulting data collection. If it is disabled, the checkbox of the field pertaining to it is not activated at all.

Once activated within the inspection point, the fields 1 to 3 are available for entering values with respect to "equipment", "functional location" and "physical sample". The fields 4 to 9 are user fields relating to inspection points. The fields 4 and 5 are "alphanumeric" fields, the fields 6 and 7 are numeric, the field 8 is a "date" field and field 9 is a "time" field.

Order data for the inspection step

Order

This is e.g. the production order number in production.

Operation

Number of the operation for which this inspection step has been generated. The operation number taken from the existing inspection plan is the basis in this context.

Operation designation

Operation designation pertaining to the operation number taken from the existing inspection plan. Normally, this field remains empty, as in nearly every case only the operation number is used in inspection planning. The operation designation is only used in exceptional cases.

Quantities

Rework

A rework quantity may be entered in this field when the inspection step is completed. This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, this field is useful in the goods receipt and goods issue area.

Scrap

A scrap quantity may be entered in this field when the inspection step is completed. This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, this field is useful in the goods receipt and goods issue area.

Actually produced quantity (act. prod. qty.)

The actually produced quantity may be entered in this field when the inspection step is completed. This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, this field is useful in the goods receipt and goods issue area.

Inspection step - editing functions

An inspection step cannot be edited. Only quantity specifications may be added while completing inspection steps.

Inspection step - toolbar

Release

Function authorization: irisp.release

Transfers an inspection step of the status "created" or "completed" to the "released" status or to the status that existed at the time when it was completed, e.g. to the "partial result" status.

Complete

Function authorization: irisp.complete

Opens a dialog where the (inspection) result is preset on the basis of the existing inspection results of the characteristics. By completing the inspection step, it is finally transferred to the "completed" status. Inspections can no longer be performed for inspection steps in this status.

Cancel

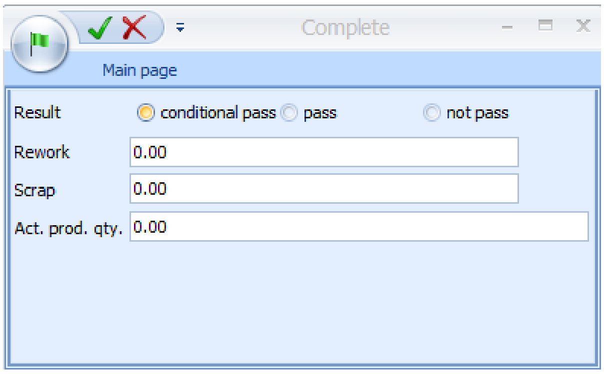
Function authorization: irisp.cancel

Transfers the inspection step to the "canceled" status. Inspections can no longer be performed for inspection steps in this status.

"Complete inspection step" detail application

Function authorization	irisp.complete

The corresponding button has to be clicked in the toolbar to complete an inspection step. The following editing dialog opens afterwards.



The system suggests the result "pass" if all corresponding inspection step characteristics have the inspection result "pass" or if they are not yet checked. The system suggests the result "fail" if there is at least one inspection step characteristic that has the inspection result "fail". However, the user is free to choose another result.

The "rework quantity", "scrap quantity" and the "actually produced quantity" may be entered when the inspection step is completed (irrelevant to gage management). This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, these fields are useful in the goods receipt and goods issue area. In particular in the goods receipt area, it is useful to record scrap and rework as these details may be uploaded for several inspection steps to a higher-level system at a later point in time when inspection requirements are completed.

"Inspection points" detail application

The standard configuration of the system specifies that inspection points are only used in the production area. Using inspection points only makes sense if several samples are drawn and checked within an order/inspection requirement. This is, for example, the case if inspections are to be performed subject to different events in production. This might, for example, be the case if a specified inspection interval based on quantities or time is reached. As the individual inspection plan characteristics might have different events that make inspections become due, the inspection points of an inspection step might indeed have different inspection characteristics.

Once an inspection has been performed, every inspection point is to be completed. Every inspection point has an inspection result (pass, fail) and a status (completed, open). Inspection points that have not yet been checked have a "pass" inspection result by default.

Please note

If text exceeding the field lengths existing in the relevant inspection point dialog is entered in inspection point fields that are also visible/used on AIP terminals, this field content will be cut off when saving the inspection point in AIP. Therefore, the field contents should not be longer than the field lengths configured for the AIP inspection point dialog. The field lengths of AIP dialogs can be adjusted as part of system customization.

"Inspection points" field descriptions

The essential inspection point fields are described in the paragraphs that follow. Self-explanatory fields are not explained.

Inspection point

Inspection point number

The inspection point number identifies the inspection point uniquely among the inspection points pertaining to the inspection step. The superordinate key fields "inspection step number" and "inspection request number" are shown in addition to the "inspection point" field.

Inspection result

The inspection result can be "pass" or "fail". Inspection points that have not yet been checked have the inspection result "pass" by default (standard system configuration).

Status

An inspection point can be completed or open. Only open inspection points may still be checked at the AIP terminal.

Inspection point identification

The fields, which have been identified as being activated in the superordinate inspection step, are shown here. The CAQ system options define which fields identify the inspection point and which ones require input. The following fields are displayed by default, whereas none of these fields requires input.

Date

Allows for a date to be entered. The system date is preset in case new inspection points are created. This date may directly be changed if they are created manually. Otherwise, the editing mode allows for changes to be made.

Time

Allows for a time to be entered. The system time is preset in case new inspection points are created. This time may directly be changed if they are created manually. Otherwise, the editing mode allows for changes to be made.

In addition to the configurable identification fields, there are still the following identification fields that are always available.

Workplace

If the inspection point is generated at an AIP terminal or the automatic generation is triggered by an inspection event of a specific terminal, this field shows the workplace of the triggering inspection station. If an inspection station is specified this inspection point will only be available at this inspection station.

Batch

Possibility to enter a corresponding batch.

Partial batch

Possibility to enter a corresponding partial batch.

Details

Generation of inspection points

If an inspection point is generated automatically on the basis of reaching a defined time or quantity interval, it has the flag "relating to time" or "relating to quantities". An inspection point that is assigned the flag "free" has been generated manually or on the basis of other events that make inspections become due.

Inspection points - editing functions

New inspection points cannot be created for completed or canceled inspection steps. The same applies to the modification of existing inspection points. To allow for an inspection point to be created or edited, the inspection step has to be released beforehand.

Inspection points - toolbar

Release

Function authorization: iripp.release

Transfers a completed inspection point to the "open" status.

Complete

Function authorization: iripp.complete

Opens a dialog window that allows for inspection points to be completed.

"Complete inspection point" detail application

Function authorization	iripp.complete

The corresponding button has to be clicked in the toolbar to complete an inspection point. Then the following editing dialog opens including an inspection result that cannot be changed and that is calculated on the basis of the inspection results for characteristics. In addition to this, in the tab “Evaluation” a usage decision for inspection points calculated on the basis of the inspection results for characteristics is preset, which may, however, be changed manually. To do so, there is a selection listof usage decisions for inspection points, filtered by the plant,, whereas every usage decision finally classifies the inspection point either as "pass" or "fail", irrespective of the inspection result. Additional information is displayed on the usage decision selected for the inspection point, once the selection has been confirmed. Additional information may, for example, be the code group, code number, catalog type and the factory/site.

MPDV services for customizing the system may provide further usage decisions for inspection points.

According to the configuration of identification fields, they can still be edited when inspection points are completed. If some identification fields require input (mandatory fields), they need to be filled out before inspection points can be completed. In addition to this, quantity fields may also be filled out.

"Inspection step/inspection point characteristics" detail application

Function authorization	iriscp.*

The detail application of inspection steps/inspection point characteristics is nearly identical to the master data of characteristics application. For this reason, reference is made only to additional features in the following.

Go to

For further information on the definition of characteristics, please refer to the functional description of the document entitled MOC_CharacteristicsQM.

The inspection step characteristics are displayed below the inspection step within the master-detail grid. Provided that inspection points are in use and inspection points have been generated, the list of inspection points may be opened below the inspection steps. This list again allows for the characteristics pertaining to the selected inspection point to be opened.

The existing inspection plan determines which characteristics are included in the inspection step. In case the inspection plan is configured in a way specifying that a separate inspection step is to be generated for each operation (applying to several workplaces), the inspection step only includes the characteristics of a specific operation. Provided that the inspection plan has additionally been configured to generate a separate inspection step for every workstation, an inspection step always includes the characteristics of a fixed combination of operation and workplace.

An inspection point, in turn, only includes characteristics of the same event causing inspections to become due. Even the manual generation of inspection points represents a due date event. An inspection point is generated automatically, once e.g. a quantity interval defined for the inspection step characteristics has been achieved. This inspection point includes all characteristics that have the same quantity interval or that have a quantity interval that has the achieved quantity interval as least common multiple.

Example:
Characteristic 1 has a quantity interval of 100 and characteristic 2 of 200. Once the first 100 items have been produced, an inspection point is generated, which only includes characteristic 1. After another 100 items that have been produced, another inspection point is generated that includes the characteristics 1 and 2.

If an inspection becomes due because of changing the machine status, the generated inspection point only includes characteristics that have to be checked on the basis of the same machine status change.

Only manually generated inspection points are not limited with respect to the corresponding characteristics. These inspection points always include all characteristics of the related inspection step.

Inspection step/inspection point characteristics field descriptions

The paragraphs that follow only describe the fields that are available in addition to the characteristic master data or inspection plan characteristics.

Characteristic

Initial sample creation

Shows the field content transferred from the basic inspection plan characteristic. A separate sheet/page is printed for each category in the printed initial sample report.

Properties

Dynamically modified

This checkbox is enabled, provided that a characteristic is dynamically modified.

Dynamic modification

Determined inspection severity/inspection severity designation

This field shows the inspection severity determined for this characteristic on the basis of the dynamic modification rules specified within the inspection plan.

Revised inspection severity/inspection severity designation

This field only exists if dynamic modification is based on characteristics. A new initial inspection severity is displayed here, provided that the user has intervened in the process of dynamic modification and changed the current inspection severity (new initial inspection severity).

Please note

A note is displayed in this field, in case an error has occurred with the dynamic modification of this characteristic while inspection steps are generated.

Details

No cavity

The authorization "ipl_cav" is required for displaying these fields.

This option allows for a characteristic to be highlighted as irrelevant to cavities, although the respective inspection requirement specifies that the relevant characteristics are (actually) to be recorded in relation to cavities.

This checkbox has always to be enabled for attributive characteristics and inspection charts.

Sample group

This field specifies which sample group this characteristic belongs to.

Sampling

If a characteristic is assigned a sample group, this checkbox specifies whether or not a sample is to be generated by way of this characteristic. If this is the case, it is a "sampling characteristic". Consequently, the checkbox is to be enabled. A sample group is to be assigned to the sampling characteristic.

If these fields are not available, you require a new program version of this application.

Specifications

Sample size

This field shows the determined sample size, if within an inspection plan characteristic a sampling scheme is defined that does not require the sample size to be indicated, as it is calculated on the basis of the actual quantity of the inspection requirement. This is, for example, the case for the sampling schemes "batch inspection" or "100% inspection".

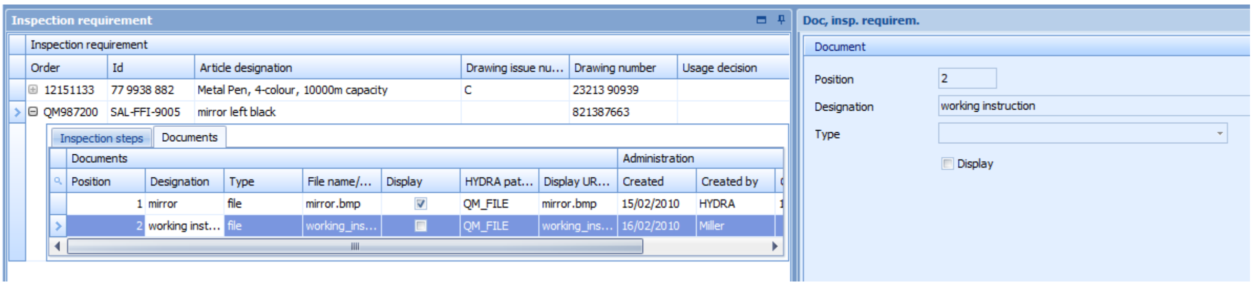
Go to

The other fields of the respective tabs correspond to those of the characteristic master data and are described in the documentation entitled "CAQ characteristic master data".

Inspection step/inspection point characteristics - editing functions

Inspection step or inspection point characteristics cannot be edited.

“Inspection request documents" and "characteristic documents" detail applications



The above screenshot shows how an inspection requirement document is assigned.

Provided that the "documents" tab has been activated, as many documents as required may be assigned to each inspection requirement and each inspection step characteristic within the master-detail grid. If this tab is activated in the master-detail grid corresponding editing buttons are enabled in the toolbar to edit the documents.

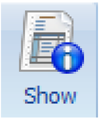
When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

The file types are "File", "URL" and "Text". The file name including the path may be entered manually for the "file" type. The "URL" file type enables access to the Internet or Intranet. The third file type "Text" allows for text to be entered directly.

A designation may be assigned to each defined document. Moreover, it may be determined in which order the documents are to be listed. The "position" field is used for this purpose (numeric input). The specifications made within this list must be unique. In addition, the checkbox "display with inspection" determines whether or not the document may be shown during the inspection process.

"Inspection plan documents“ and "documents for inspection plan characteristics" toolbar

In addition to the standard functions, there is also the button to show the documents.



Show documents

If a document link is defined this button opens and shows this document. However, a program, which can show the linked file type, has to be installed on the PC.

"Failure" detail application

A "failure list" detail application is provided on the level of inspection requirements. This list shows all failures of the type "failure type", "failure location", "failure cause" and "originator", which have been assigned to the characteristics, samples or measured values during the inspection process. The list even shows the failure types that have been generated automatically, e.g. "upper tolerance limit violated".

In addition to the respective failure, the following referenced data is also available.

- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number
- Failure type
- Characteristic number and designation
- Workplace number and designation
- Weighting (number, e.g. for an inspection chart characteristic)
- Comment,
- Failure date,
- Failure time

The list may be restricted in a flexible way by using the individual column filter. In addition, the grouping function allows for failures to be listed for every characteristic, for example.

Failure list - editing functions

Failure analysis entries can be created, edited or deleted for inspection requirements. If a new data record is created, further key information, such as

- Inspection step number,
- OP sequence number (uniquely identifies the characteristic),
- Sample number
- Value number

may be added to the assignment.

The target group can be found in the administrative area, as "administrative" skills are required for more precise assignment of this key information.

These authorizations are required.

Function authorization	failure.create => create new failure failure.edit => edit failure failure.delete => delete failure

Users must have profound knowledge of the inspection processes as they are not guided through the process of data collection/modification and validation checking is not performed.

The fields "area" and "inspection requirement number" can neither be modified nor assigned values during initial data creation. This data is taken from the previously selected inspection requirement when it comes to initial data creation.

The other fields can be assigned values as part of initial data creation. Only the fields "weighting" and "comment" may be changed in the editing mode.

"Measures" detail application

A "measure list" detail application is provided on the level of inspection requirements. This list shows all measures which have been assigned to the characteristics, samples or measured values during the inspection process.

In addition to the respective measure number and designation, the following data is also referenced.

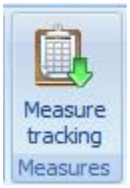
- Inspection requirement
- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number

- Measure type
- Party in charge incl. type
- Status
- Comment
- Text
- Effectiveness

The list may be restricted in a flexible manner by using the individual column filter. The grouping function provides further analysis options.

"Measures" toolbar

The button "measure tracking" can be used to open the measure tracking dialog to edit and create measures. Measure tracking is filtered by the internal and unique "serial number" of the measure in the "references" tab. Filtering is omitted if no measure is selected.



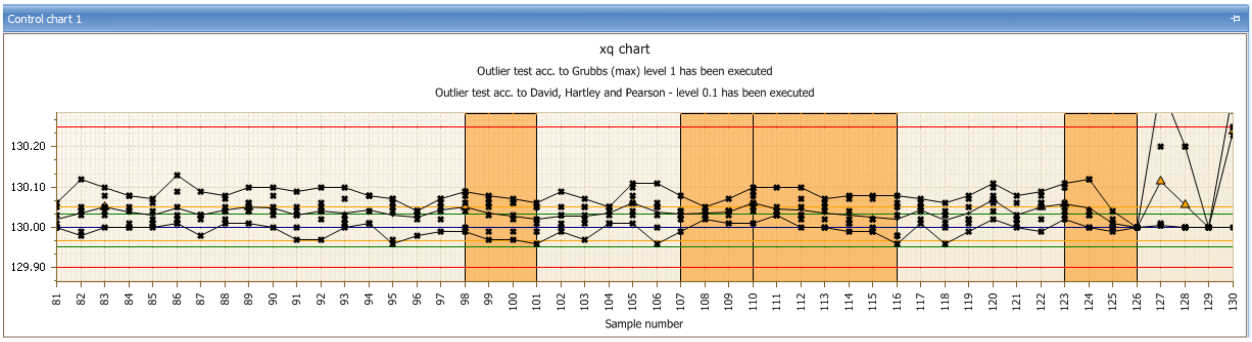
Go to measure tracking

Function authorization	failure.* => opens measure tracking

"Control chart 1 / 2" detail application

The control charts 1 and 2 defined for the inspection step characteristic are displayed by default. In case they are not defined for the inspection step characteristic, the xq and s charts are used for variable characteristics and the p and u chart for attributive characteristics.

All data of the selected inspection step characteristic is used as basis for graphic preparation.



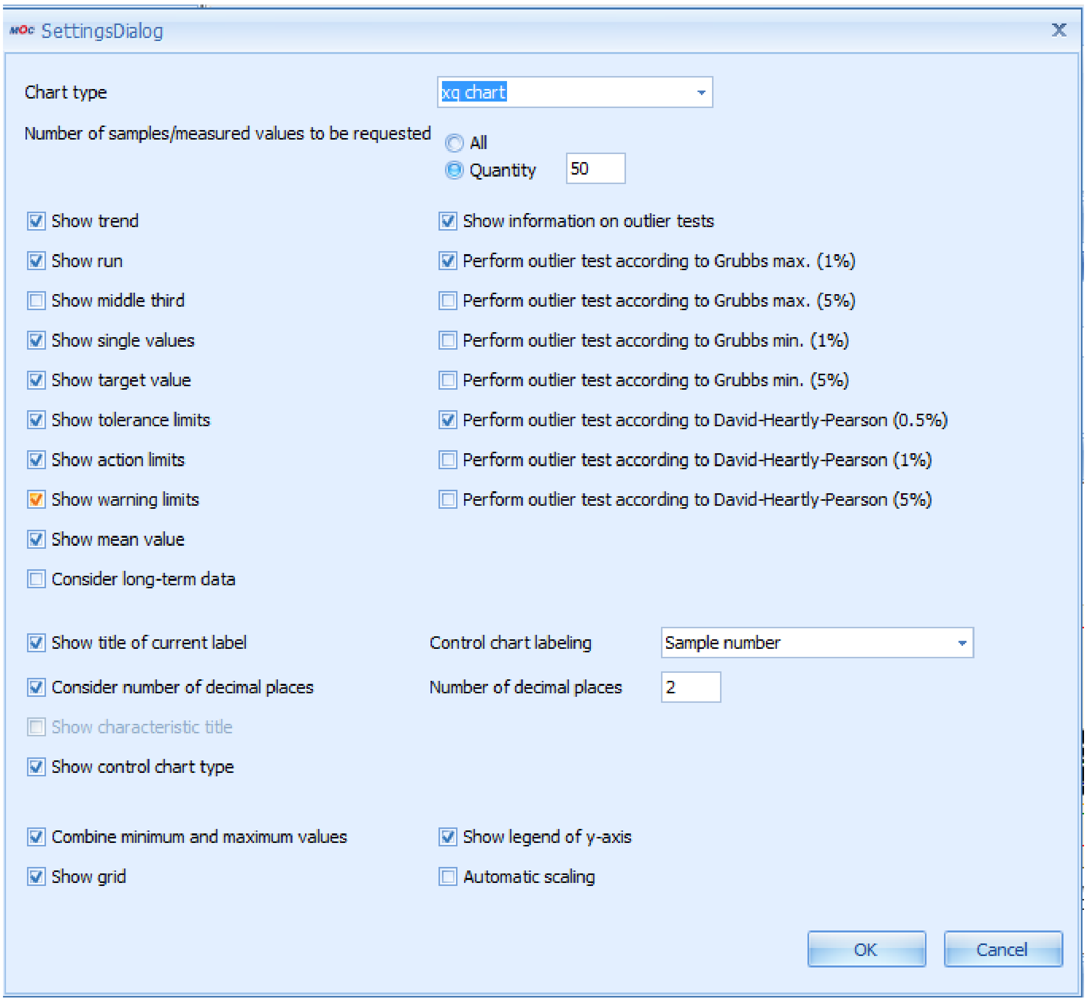
The contents of this application are defined by opening the dialog to configure "control chart 1" or "control chart 2". Changes made via this dialog are saved according to the user's requirements. Provided that no special settings have been made, the xq chart ("variable" characteristic type) or the p chart ("inspection chart" or "attributive" characteristic type) is displayed, by default. Subject to the characteristic type, the user may choose to display one of the following control charts by default.

Variable characteristic

- Xq chart
- s chart
- R chart
- Single value chart
- Median chart

Attributive characteristic

- p chart
- np chart
- c chart
- u chart



The paragraphs that follow explain the essential configuration options.

Number of the samples to be requested

Specifies how many samples or single measured values are to be displayed in the control chart.

Display …

The options that include the term "display ..." enable or disable the presentation of the corresponding data in the control chart.

Consider long-term data

This option has to be checked to integrate archived data from the medium-term data area.

Combine minimum and maximum values

It is recommendable to show the minimum and maximum values that are each connected by a line to improve the presentation of the range of dispersion of single values.

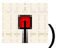
Automatic scaling

The "automatic scaling" function allows for all values to be displayed, irrespective of the existing limit values. This shows even extreme outliers in the control chart. The disadvantage is that there is less space for the other measured values and, it is very often the case that a changing value pattern can hardly be recognized.

Show trend / run / middle third

The monitoring functions "trend", "run" and "MiddleThird" allow better surveillance of a process than by using the control chart alone. However, they can only be used with control charts for xq and median. The trend allows visualization of an upward or downward tendency in the process over several samples. By default, these are seven subsequent rising or falling values. The run shows sections in which the process runs above or below the mean value (when displayed, otherwise the target value) over several samples. By default a run is recognized if seven subsequent values are above the mean value. The number of seven samples/values for recognizing a trend or run, which is set by default, may be changed by customizing the system. "MiddleThird" refers to an unusually high or low number of values, in the section of the control chart viewed, within the middle third of the area bounded by the action limits.

Respective analyses are performed automatically for "trend", "run" and "MiddleThird". The control chart shows in a graphic if a trend, run and / or MiddleThird is available/detected. The following events are altogether represented by icons or color codes in the control chart.

- Trend (can be recognized by a colored area)
- Run (can be recognized by a colored area))
- Middle Third (can be recognized by a colored area)
- Outlier (icon: )
- Xq violates action limit (icon )

The presentation of outliers is restricted to the xq chart and median chart and connected with the presentation of single values. There are different functions for performing and presenting outlier tests. The different outlier tests for the different levels may be activated separately. Provided that the presentation of outlier tests has been activated, the result of this test is displayed in text form above the corresponding control chart.

The below outlier tests are available for the different inspection levels, whereas the inspection level is indicated in parentheses.

- Grubbs max. (1 %)
- Grubbs max. (5 %)
- Grubbs min. (1 %)
- Grubbs min. (5 %)
- David-Hartley-Pearson (0.5 %)
- David-Hartley-Pearson (1 %)
- David-Hartley-Pearson (5 %)

Notes on outlier tests

- The outlier tests do not refer to the collectivity of all samples, i.e. every sample is considered individually. Consequently, the following phenomena may appear:
  - Despite a large range between minimum and maximum value no outlier can be identified. A reason for this may be the equal distribution of the single values within the sample.
  - Despite a low range between minimum and maximum value an outlier can be identified. A reason for this may be an accumulation of many values at one “point” so that an individual value having a certain distance to this agglomeration is identified as outlier within the sample.
- At least three values have to be in the sample in order to be able to perform an outlier test. The more values are available within a sample the more uniform the general view of the outliers becomes compared to all samples.
- The Grubbs outlier test is performed with a sample size of 2 < n < 148 (n = number of values within a sample). The outlier test according to David, Hartley und Pearson is performed at a sample size of 2 < n < 1251 (n = number of values within a sample).

X-axis labeling

The below information is available to label the x-axis of control charts.

- Sample number
- Order number
- PPS reference number
- Purchase order number
- Batch
- Article number
- Machine number
- Cavity number
- Date + time of the first measured value of a sample
- Date + time of the last measured value of the sample
- Date + time of sample completion
- Badge number of the first measured value of the sample
- Badge number of the last measured value of the sample
- Badge number of sample completion

If these fields are not available, you require a new program version of this application.

- Partial batch
- Workplace
- Production workplace

- Field 1
- Field 2
- Field 3
- Field 4
- Field 5
- Field 6
- Field 7
- Field 8

The fields "field 1" to "field 8" are enabled subject to their usage by MPDV customizing and assigned an individual designation. As these field names are flexible, they are only entitled "field 1" to "field 8" in this document. Field 1 includes, for example, the tool if cavity-related data collection is enabled.

Tool tips within the control chart

When a value is labeled being an outlier (red rhomb) detailed information on which test(s) was (were) the crucial factor for this determination is displayed when going with the mouse over this labeling (rhomb).

For mean values the tool tip shows the value, date and time for the first and last measured value of this sample as well as the corresponding inspection request number and inspection step number.

The exact value is shown for single measured values.

A special note indicating why an area is colored is also displayed for the colored areas when a trend, run or middlethird is recognized.

Sorting of measured values

The type of control chart sorting is determined by the server and depends on how the control chart filters are configured. If the filters "operation sequence" and "inspection request number" or "operation sequence" and "inspection step number" are indicated the characteristic is identified uniquely and sorting is based on the sample number.

Sorting is based on date and time if either the filter field "operation sequence" is left empty or it is filled out and the fields "inspection request number" and "inspection step" number are left empty.

The optical presentation of trend or run always refers to how the measured values are sorted. However, the automatic generation of the failure type "trend" is always based on the measured values being sorted by the sample number, as the numbers for the operation sequence, inspection request and the inspection step are always known at the time data is recorded.

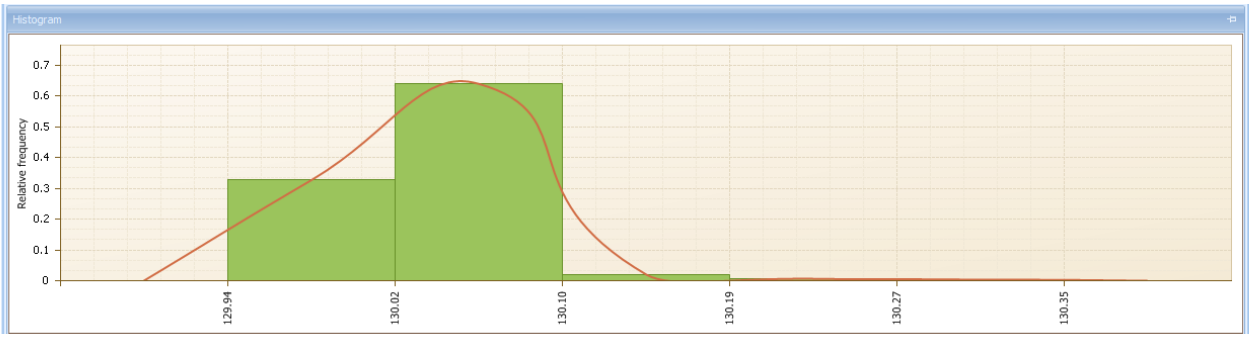
Consequently, the following scenario might be possible.

- The control chart is sorted by date and time and a trend is shown, although the automatic failure "trend" has not been generated.

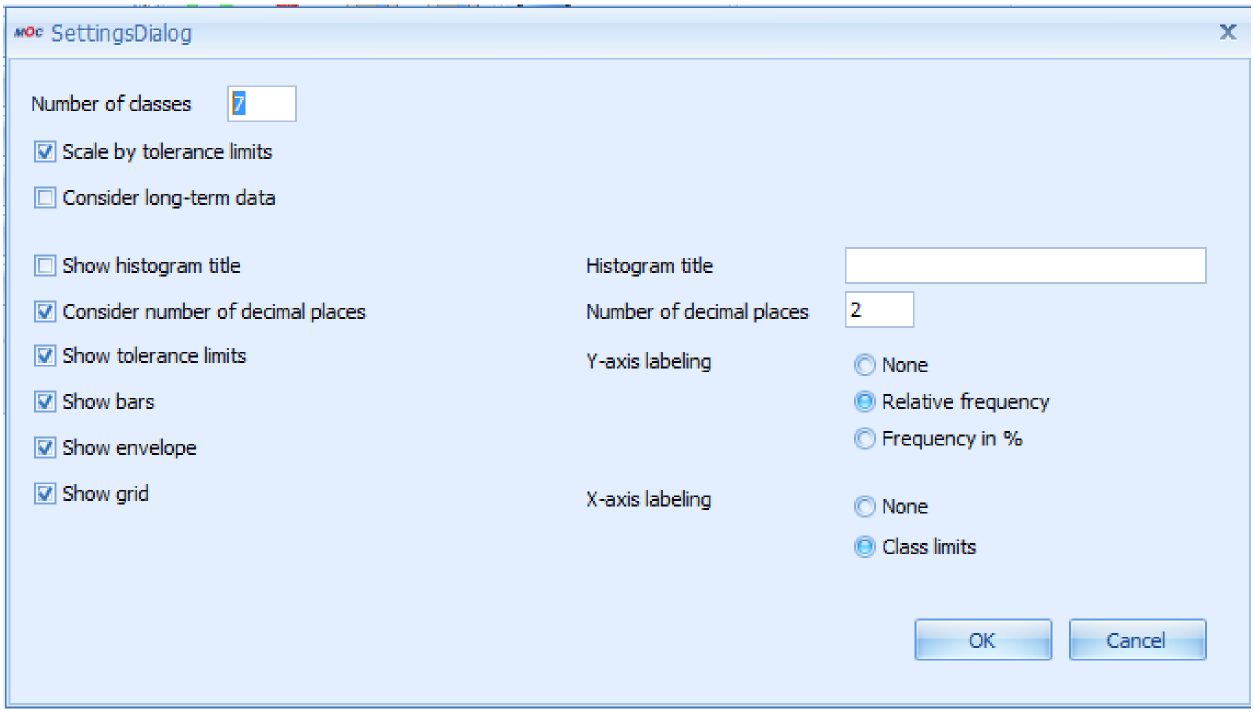
The control chart is sorted by the sample number and no trend is shown, although the automatic failure "trend" has been generated.

"Histogram" detail application

All data of the selected inspection step characteristic is used as basis for graphic preparation.



Unlike the control charts, which display a specified excerpt from the sample set, the histogram is always based on the entire set of available samples matching the selection filter criteria. The appearance of the histogram is determined by the number of classes and by any elements additionally displayed. The contents of this application are defined by opening the dialog to configure the "histogram". Changes made via this dialog are saved according to the user's requirements.



The paragraphs that follow explain the essential configuration options.

Number of classes

Determines the number of histogram classes according to which the measured values are to be distributed. If represented within the tolerance limits (option "scale by tolerance limits") one histogram class each is outside of the tolerance limits.

Scale by tolerance limits

Enabled: The classes are in between the tolerance limits with each one "outlier class" to the left and to the right.

Disabled: The classes include the range of all measured values (no separate classes for values outside of the tolerance limits).

Consider long-term data

Includes the archived data from the medium-term data area.

Show histogram title

If a special title is to be displayed it may be entered by enabling this option.

x-axis labeling

The x-axis shows the corresponding values of the class limits if the "class limits" option is set. The number of decimal places that are to be displayed may be set by the two configuration options for decimal places.

Consider the number of decimal places

Takes into account the defined number of decimal places in the x-axis labeling.

Control chart 1/2 and histogram

Control chart 1 settings

Opens a dialog to configure the settings of control chart 1. The corresponding details are described in the respective detail application.

Control chart 2 settings

Opens a dialog to configure the settings of control chart 2. The corresponding details are described in the respective detail application.

Histogram settings

Opens a dialog to configure the histogram settings. The corresponding details are described in the respective detail application.

"Samples" detail application

The statistical key figures about every single sample can be found in the "samples" detail application on the level of inspection step characteristics. In addition to the referenced key fields, such as the inspection requirement number, inspection step number and sample number, provided that they are available or can be calculated, the key figures

- Xq
- Xq floating
- R floating
- Standard deviation s
- s floating
- Minimum
- Maximum
- Range
- Median
- P
- U
- Number of defects

are listed. The list additionally shows the characteristic specifications (tolerance, action and warning limits) as well as the referenced machine.

"Samples" toolbar

Complete sample

It might be necessary to complete samples manually, as measured values can be entered on "administration" level or as there might be inspections without inspection points including the creation of new samples and/or inspections without having reached the sample size. Samples can even be completed if they have not been completed by AIP inspection data collection.

Release sample

This function enables samples to be released once more and the collection of additional measured values for this sample number, provided the defined sample size has not yet been reached.

"Single values" detail application

The single measured values for all variable inspection step characteristics can be found in the "single values" detail application on the level of inspection step characteristics. For attributive characteristics the following information

- Number of defects
- Number of NCU (non-conforming units) and
- Defects

is displayed. Moreover, it can be recognized whether the measured value or the attributive assessment is valid or invalid.

The characteristic specifications (tolerance, action and warning limits) are listed in addition to the referenced key fields, such as the inspection requirement number, inspection step number, sample number, and value number. Every entry also shows the date and time when it was recorded and edited as well as the responsible user.

Single values - editing functions

Measured values can be created, edited or deleted for a variable inspection step characteristic and/or inspection point characteristic and the number of non-conforming units may be created, edited or deleted for attributive characteristics on "administration" level. If a new data record is created, further key information, such as

- the sample number
- and value number

may be added to specify the assignment.

The target group can be found in the administrative area, as "administrative" skills are required to provide for more precise assignment of this key information. Users must have profound knowledge of the inspection processes, as they are not guided through the process of data collection/modification and validation checking is not performed.

These authorizations are required.

Function authorization	value.create => create new measured value value.edit => edit measured value value.delete => delete measured value

Users must have profound knowledge of inspection processes, as they are not guided through the process of data collection/modification and validation checking is not performed.

The fields

- Area,
- Inspection request number,
- Inspection step no.,
- Inspection step number (only available for inspection point characteristics),
- OP sequence (uniquely identifies the characteristic),
- Upper tolerance limit,
- Target value,
- Lower tolerance limit and
- Number of decimal places

can neither be modified nor assigned values during initial data creation. When it comes to initial data creation, this data is taken from the previously selected characteristic and the referenced inspection requirement as well as from the inspection point.

The other fields can be assigned values and/or changed during initial data creation.

The field "single value of cavity" is only visible if

- the inspection requirement shows an ID indicating that data is generally collected in relation to cavities and
- the relevant characteristic is not assigned an ID indicating that it still has to be checked without relation to cavities.

If a new data record is created, the field for the measured value will be restricted to the number of decimal places. The other decimal places are filled up with zeros. For technical reasons, these "additional" decimal places are shown during processing.

The field "measured value" is shown for variable characteristics. Instead of the measured value field, the fields "sample size (checked)", and "number of DU" are shown for attributive characteristics, i.e. this also includes inspection chart characteristics.

"Statistics" detail application

The common statistical key figures about the selected characteristic can be found in the "statistic" detail application on the level of inspection step characteristics.

For variable characteristics these are

- Number of samples
- Number of measured values
- Xqq
- Minimum
- Maximum
- R
- S
- S relative
- Sigma
- Cp and
- Cpk

For attributive characteristics these are

- Number of non-conforming units

- Number of defects
- p and
- u.